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DEPARTMENT OF DEFENSE HANDBOOK

RANGE LASER SAFETY

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This handbook provides uniform guidance for the safe use of military lasers and laser systems on Department of Defense (DoD) ranges. Each military service has previously established normal procedures for approving laser ranges and has the responsibility to organize, train, and equip their forces. Services will establish a range laser safety program as part of their overall range safety program. This guidance is intended to supplement these procedures and training requirements. It does not replace those procedures nor release individuals from compliance with the requirements of their particular service. The authority for this handbook is the DoD Laser System Safety Working Group (LSSWG) established by DoDI 6055.15, DoD Laser Protection Program for Military Lasers. Guidance for laser systems not addressed here should be obtained from the LSSWG.

3. The aim of this DoD handbook is to establish range safety for the evaluation and control of lasers under military control to reduce to a minimum the hazards to personnel, property, and the environment.

4. The installation commander is responsible for laser range operations while day-to-day operations are managed through the Installation Laser Range Authority.

5. This handbook applies to laser systems with optical radiation emission in the wavelength range of 100 nanometers to 1 millimeter, pulsed and continuous wave systems.

6. This handbook does not apply to medical or laboratory uses, where additional professional review may be needed to establish health and safety controls; other outdoor use is described in ANSI Z136.6, American National Standard for Safe Use of Lasers Outdoors.

7. This handbook applies to all hazard categories or hazard classifications defined by ANSI Z136.1, American National Standard for Safe Use of Lasers, and IEC 60825-1, Safety of Laser Products – Part 1: Equipment Classification and Requirements.

8. The intent is for this document to provide sufficient guidance for laser system use on a laser range.

9. Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This handbook contains general and detailed guidance to be followed in evaluating and recommending range laser safety procedures, which are intended to serve as a guide to the safe use of laser systems on military operational training and test ranges. This handbook is for guidance only and cannot be cited as a requirement.

1.2 Application. This handbook applies to all ranges where service approved laser systems are employed, or research and development lasers are being evaluated. The handbook addresses the roles of several levels of authority including institutional, installation, and unit, and subsequent positions of laser safety responsibility within the services.

This document is applicable to all Department of Defense (DoD) member ranges and all DoD laser operations conducted on non-DoD controlled ranges. The guidance in this document does not replace other procedures or release individuals from compliance with the requirements of their particular service.

This document implements information and methodologies in accordance with the North Atlantic Treaty Organization (NATO) Standardization Agreement (STANAG) 3606.

Suggestions for any change, revision, or cancellation of this handbook are to be submitted through the DoD Laser Systems Safety Working Group (LSSWG).

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed below are not necessarily all of the publications referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

INTERNATIONAL STANDARDIZATION AGREEMENTS

ARSP-4 - Laser Safety for Military Use

STANAG 3606 - Laser Safety for Military Use

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1425 - Safety Design Requirements for Military Lasers and Associated Support Equipment

(Copies of these publications are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein.

BUREAU OF MEDICINE AND SURGERY (BUMED)

BUMEDINST 6470.23 - Operational Non-Ionizing Radiation Health Protection

(Copies of this publication are available online at www.med.navy.mil.)

CODE OF FEDERAL REGULATIONS (CFR)

21 CFR 1040 - Performance Standards for Light-Emitting Products

(Copies of this publication are available online at www.ecfr.gov.)

DEPARTMENT OF DEFENSE ISSUANCES

DoDI 6055.15 - DoD Laser Protection Program for Military Lasers

(Copies of this publication are available online at www.esd.whs.mil/DD/.)

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DEPARTMENT OF THE NAVY ISSUANCES

OPNAVINST 5100.27/MCO 5104.1 - Navy Laser Hazards Control Program

SECNAVINST 5100.14 - Department of the Navy Laser Protection Program

(Copies of these publications are available online at https://www.secnav.navy.mil/doni/default.aspx.)

FEDERAL AVIATION ADMINISTRATION (FAA)

FAA Order 7930.2 - Notice to Airmen (NOTAM)

(Copies of this publication are available online at www.faa.gov.)

JOINT CHIEFS OF STAFF

Joint Publication (JP) 3-09.3 - Close Air Support

(Copies of this publication are available online at https://www.jcs.mil/.)

US AIR FORCE PUBLICATIONS

AFMAN 13-212V1 - Range Planning and Operations

AFI 48-139 - Laser and Optical Radiation Protection Program

AFI 91-401 - Directed Energy System Safety

(Copies of these publications are available online at http://www.e-publishing.af.mil.)

US ARMY PUBLICATIONS

AR 40-5 - Army Public Health Program

AR 385-10 - The Army Safety and Occupational Health Program

AR 385-63 - Range Safety

DA PAM 385-10 - Army Safety and Occupational Health Program Procedures

DA PAM 385-40 - Army Mishap Investigations and Reporting

TB MED 524 - Occupational and Environmental Health: Control of Hazards to Health from Laser Radiation

(Copies of these publications are available online at https://armypubs.army.mil/.)

US NAVY/MARINE CORPS PUBLICATIONS

MCO 3550.9 - Operational Training Range Certification/Recertification Program

MCO P3550.10 - Policies and Procedures for Range and Training Area (RTA) Management

MCO 3570.1 - Range Safety

(Copies of these publications are available online at https://www.marines.mil/News/Publications/.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z136.1 - American National Standard for Safe Use of Lasers

ANSI Z136.6 - American National Standard for Safe Use of Lasers Outdoors

(Copies of these publications are available online at http://webstore.ansi.org/.)

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3. DEFINITIONS

The following definitions and terms are used in this handbook. For other definitions associated with laser safety, refer to ANSI Z136.1, American National Standard for Safe Use of Lasers.

3.1 Administrative controls. A category of control measures used to eliminate hazards or reduce the degree of risk to personnel and equipment. Administrative controls reduce risks through specific actions. Administrative controls can be implemented by either local or higher authority.

3.2 Atmospheric attenuation. The reduction of energy of a laser beam due to absorption and scattering along its path by atmospheric conditions, such as smoke, haze, fog, or precipitation.

3.3 Attenuation. The decrease in the energy of any optical radiation beam as it passes through an absorbing or scattering medium, or both.

3.4 Buffer angle. An angle added to the beam divergence or intended laser projection field to ensure a protection zone.

3.5 Buffer zone. The volume of space around a propagating laser beam defined by the buffer angle.

3.6 Controlled area. An area of occupancy or activity, where personnel/equipment are subject to control and supervision for the purpose of protection from radiation hazards.

3.7 Danger zone. An area determined by analysis of weapons characteristics and historical patterns to present risk to personnel or equipment within a designated three-dimensional space.

3.8 Diffuse reflection. Reflection from a surface in which the beam is scattered in all directions (for example, a reflection from a rough surface).

3.9 Engineering controls. A category of control measure used to eliminate hazards or reduce the degree of risk to personnel and equipment. Engineering controls use engineering methods to reduce risks by design, material selection, or substitution.

3.10 Exclusion zone. A designated area which aircraft headings should avoid because of the possibility of false target indications caused by atmospheric scatter from the laser beam within short distances from the laser exit port. An exclusion zone is established as a sector where the apex is at the target and extends equidistant (10 degrees) either side of the target-to-laser designator line. This zone extends vertically to infinity and has a horizontal limit of 20 degrees.

3.11 Flash blindness. A visual impairment during and following exposure to light of extremely high intensity, caused by blooming (saturation) of the retinal pigment, that may last for a few seconds to a few minutes.

3.12 Fratricide. The employment of friendly weapons with the intent to kill the enemy or destroy his equipment and facilities and which results in unforeseen and unintentional death, injury, or damage to friendly personnel or equipment.

3.13 Installation laser range authority. The organization that provides oversight for range laser activities at the installation level for the Navy, Air Force, and Marines is the Installation Range Laser Safety Officer. For the Army it is the Installation Range Control Officer.

3.14 Institutional laser range authority. The organization that provides oversight to range laser activities at the institutional level is the Institutional Range Laser Safety Specialist.

3.15 Laser. A device that emits light (electromagnetic radiation) through a process called stimulated emission. The term laser is an acronym for Light Amplification by Stimulated Emission of Radiation.

3.16 Laser beam divergence. The full angle increase in diameter of the laser beam with distance from the exit aperture of the laser.

3.17 Laser footprint. The projection of the laser beam on the ground or target area, to include the buffer angle.

3.18 Laser-guided weapon. A weapon that electro-optically seeks reflected laser from a separate designator source to strike a target. The laser is used to guide the munitions to a target. The acquisition device, which is a seeker and guidance kit mounted on the laser-guided weapon, seeks coded laser energy reflected from the target.

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3.19 Laser range. A range designated for laser systems use.

3.20 Laser range certification. An approval based on an evaluation process conducted to ensure safety margins are determined to allow for the diverse application of the many lasers that may be used on a range, defining the degree of laser radiation hazard possible and to recommend control measures.

3.21 Laser spot size. The area of a laser beam's projection on a target. The laser spot size is a function of beam divergence and the distance from the laser designator to the target.

3.22 Laser spot tracker. A type of laser acquisition device, normally mounted on fixed-wing or rotary wing aircraft, used to aid visual acquisition of the target to be engaged by another weapon.

3.23 Laser surface danger zone (LSDZ). Designated area where laser radiation levels may exceed maximum permissible exposure levels, thereby requiring control during laser use. Some organizations use this term interchangeably with Nominal Hazard Zone (NHZ), which includes airspace. To minimize confusion in this document, we will refer to LSDZ generally and use NHZ when not specifically referring to surface dangers. The LSDZ is an area where unauthorized personnel and equipment are not permitted, and laser Personal Protective Equipment (PPE) (eye and skin) is required for personnel who may be exposed within this area.

3.24 Laser target designator. A laser used to mark a target or to guide munitions to a target.

3.25 Laser training area (LTA). A training area in which laser systems are used.

3.26 Maximum permissible exposure (MPE). The level of laser radiation to which a person may be exposed without hazardous effect or adverse biological changes in the eye or skin.

3.27 Milliradian (mrad). Unit of angular measure. One mrad equals one thousandth of a radian. One degree equals 17.45 milliradians.

3.28 Night vision goggles/devices. A nonthermal image intensification device used to enhance night vision.

3.29 Nominal hazard zone (NHZ). Describes the space within which the level of the direct, reflected, or scattered radiation may exceed the applicable MPE during normal operation. Exposure levels outside the NHZ are below the applicable MPE level. The NHZ consists of the target area plus the buffer zones. The NHZ is often confined by the application of backstops. For un-terminated direct viewing this zone would extend to the Nominal Ocular Hazard Distance (NOHD).

3.30 Nominal ocular hazard distance (NOHD). NOHD is the distance from an operating laser to the point where the laser irradiance or radiant exposure is not expected to exceed the appropriate MPE. For the purpose of this document, the term NOHD is used generically such that it does not imply the use of or exclusion of magnifying optics. Terms for the hazard distance specifically associated with viewing lasers with magnifying optics include NOHD-M (magnified) and Extended Nominal Ocular Hazard Distance (ENOHD).

3.31 Ocular interruption (OI) devices. Laser systems that are used to intentionally illuminate the eyes of an individual(s) to provide a warning to them. Fielded as a non-lethal device to be used as part of the escalation of force procedures with vehicle checkpoints, entry control points, perimeter security, and convoys.

3.32 Operator-controlled laser. A laser with sufficiently short NOHD (approximately 100 meters unaided viewing and 700 meters for optically aided viewing) that the operator can effectively mitigate the hazard by controlling the use of the laser without the requirements of other control measures typically associated with higher classed laser systems. The operator is responsible for monitoring the controlled area.

3.33 Optical density. The amount of laser protection afforded by a particular eye protection for a particular laser wavelength.

3.34 Optically aided viewing. Viewing of lasers using an optical magnifier, such as binoculars or a scope.

3.35 Personal protective equipment (PPE). Control measures used to protect personnel from the hazardous effects (eye and skin hazards) of lasers. PPE include goggles and spectacles to protect the eyes, and clothing and gloves to protect the skin.

3.36 Scatter. An effect caused by a laser beam reflecting off of particles in the air. Scatter is most pronounced at small distances from the laser source.

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3.37 Seeker. Laser acquisition device: it identifies laser designated targets so they can be engaged with a ground, airborne, or naval weapon.

3.38 Specular reflection. A reflection of a laser beam off smooth mirror-like surface causing the beam to remain coherent upon leaving the surface.

3.39 Spillover. Spillover occurs when part of the laser spot misses the intended target. This can be also termed underspill (in front of target) or overspill (beyond the target).

3.40 Standard operating procedure (SOP). Formal written description of the safety and administrative procedures to be followed in performing a specific task.

3.41 Surface danger zone (SDZ). The ground and airspace designated within the training complex (to include associated safety areas) for vertical and lateral containment of projectiles, fragments, debris, and components resulting from the firing, launching, or detonation of weapon systems, to include explosives and demolitions.

3.42 Unit laser range authority. The organization that provides oversight to range laser activities at the unit level for the Navy, Air Force, and Marines is the Unit Laser Safety Officer. For the Army it is the Laser Range Safety Officer.

3.43 Weapon danger zone (WDZ). The ground and airspace for lateral and vertical containment of projectiles, fragments, debris, and components resulting from the firing, launching, and detonation of aviation-delivered ordnance. It reflects the minimum land and air requirement, to include terrain mitigation, needed to safely employ a given weapon. This zone accounts for inaccuracy, failures, ricochets, and broaching/purposing of a specific weapon/munitions type delivered by a specific aircraft type. The WDZ “footprint” is based on the specific weapon characteristics, type of delivery being executed, the type of platform (aircraft) delivering the ordnance, and level of containment acceptable to the installation commander.

4. ROLES AND RESPONSIBILITIES

4.1 Range operating authority/range commander/institutional laser range authority.

4.1.1 Institutional guidance regarding range laser safety. The Institutional Laser Range Authority provides institutional guidance to installations via service-specific regulations. The Institutional Laser Range Authority is responsible for maintaining publications relating to laser use on ranges and establishing, or recommending, the requirements for training programs. If there are questions regarding range laser safety and certification, they should be directed to the Institutional Laser Range Authority.

4.1.2 Laser range certification. When empowered, the Institutional Laser Range Authority is responsible for certifying laser ranges. A laser range certification provides recommendations for the safe use of specific lasers and laser activities. The Institutional Laser Range Authority should:

a. Review the laser systems to be employed.

b. Verify range boundaries.

c. Verify laser firing and target area, line, or points.

d. Verify laser target condition.

e. Verify Laser Surface Danger Zones (LSDZ).

f. Determine if safe laser activities can be supported.

g. Ensure the LSDZ is contained within an area of control.

h. Ensure geospatial (GIS) data sets supporting the Certification are in DoD compliant format.

i. Determine encroachment or conflict issues.

j. Review access control measures.

k. Verify compliance of Standard Operating Procedures (SOP) and proposed operations.

l. Validate Personal Protective Equipment (PPE) Program.

4.1.3 Laser range hazard analysis. The Institutional Laser Range Authority is responsible for conducting laser range hazard analysis and for generating laser danger zones. The Institutional Laser Range Authority is responsible

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for maintaining its methods and, if necessary, automated software tools for generating danger zones. The Institutional Laser Range Authority should:

a. Determine buffer angle.

b. Validate laser firing area/line/points.

c. Validate laser target area/line/points/area offset.

d. Validate location of laser target area/line/points.

e. Validate magnetic declination.

f. Generate calculations for LSDZ.

g. Generate flight profiles, if applicable.

h. Capture area with digital pictures or video for report.

i Create GIS data holding of Laser Range Certification features.

j. Summarize hazard analysis findings and recommendations.

4.2 Installation laser range authority/installation laser safety officer.

4.2.1 Operating procedures. The Installation Laser Range Authority is responsible for maintaining operating procedures pertaining to installation range laser safety. To ensure proper control of hazardous laser radiation, the Installation Laser Range Authority should institute the following:

a. Publish and enforce safety instructions for the safe use of lasers at their range(s). Ensure that written procedures exist to minimize laser radiation hazards and other laser related range hazards, such as engaging on the laser target designator and wrong targets.

b. Regulations should be developed, or updated as necessary, to account for new laser systems and scenarios. The laser range certification should be used to review and to ensure overall laser range safety regulations are current.

c. Ensure operating procedures for specific laser systems inform laser users of the potential hazards from the laser systems under their control during laser operation. Checklists for evaluating SOPs are provided in Appendix B. The Installation Laser Range Authority should require users to adhere to range SOPs for all laser range training operations.

d. Keep records of the date, start and stop time for lasing periods, and type of laser or other appropriate information for each laser operation in the laser firing log.

e. Ensure a range Laser Safety Officer (LSO) is assigned to be responsible for ensuring appropriate safety control measures are followed.

f. Through proper management of operating procedures, the Installation Laser Range Authority should ensure the following are acknowledged to ensure currency and to mitigate risk:

(1) Range-approved or Laser Systems Review Board (LSRB) (service specific approved) laser systems.

(2) Laser range certification criteria.

(3) Range laser operations.

(4) Range laser and target maintenance procedures.

(5) Laser hazard control measures.

(6) Emergency response procedures.

(7) PPE.

(8) Communications.

(9) Notices (Airmen, Mariners, Federal Aviation Administration [FAA]).

(10) Agency Notifications - Memorandums of Agreement (MOA) or Memorandums of Understanding (MOU).

(11) Request for authorization for above the horizon lasing from the Laser Clearinghouse (LCH).

(12) Target maintenance program.

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4.2.2 Laser training plan. The Installation Laser Range Authority reviews laser training plans as part of the approval process. This focus is on verifying the information to ensure the range supports the execution of the training plan in a safe manner. In addition, the Installation Laser Range Authority reviews the risk management proposal, as necessary, and identifies revisions to the training plan to ensure compliance with safety and operational procedures and policies. The Installation Laser Range Authority should assist units, as necessary, in developing a plan that will meet training objectives. As part of this review, the Installation Laser Range Authority should review/determine the following data points within the laser training plan. Note that service-specific guidance will dictate whether the Installation Laser Range Authority "reviews" or "determines" the following data points:

a. Laser system is approved.

b. Laser use is constrained within the limits of the LSDZ.

c. Ground personnel locations to ensure those requiring PPE are identified.

d. Hazard area clearance procedures for unprotected personnel and equipment.

e. Control measures for ensuring personnel and equipment remain clear of the LSDZ.

f. Laser output power to ensure it is in compliance with local SOP.

g. Targets, laser firing area/line/points, orbit points, and laser to target orientation to ensure they can be supported by the laser system and the range.

h. PPE requirements to ensure they support the laser.

i. Training mode or filter requirements.

j. Emergency response procedures.

k. Public and agency notifications.

4.2.3 LSDZ.

a. Installation Laser Range Authority personnel may generate LSDZs, as necessary, to support training plans, map data, and information requests.

b. The generated LSDZ will be contained within the certified laser range. This effort considers the laser system data and range data, and performing risk mitigation to ensure the laser training can be safely conducted on a given range.

4.2.4 Laser systems. The Installation Laser Range Authority is responsible for reviewing laser systems for use on a certified laser range. This includes verifying the system falls within the current certification for that range. The Installation Laser Range Authority should:

a. Review range specifications and certification.

b. Review training plans including tactics, laser firing area/line/points, LSDZ, and laser target area.

c. Review laser specifications.

d. Compare characteristics of laser systems already authorized for use on the range with the new laser system.

e. Compare laser specifications with maximum supportable buffer angles on the range.

f. Compare the nominal ocular hazard distance (NOHD) of the laser with range specifications and intended training activities.

g. Determine if the laser system exceeds or falls within current range certification.

h. Recommend approval or disapproval of the use of the laser system on the range.

4.2.5 Laser range design. When new ranges requiring laser operations are considered, the Installation Laser Range Authority provides input to the design of the range cognizant of technical requirements. The Installation Laser Range Authority should review training requirements and tactics and should:

a. Conduct site analysis to determine range design requirements.

b. Recommend whether an existing range can be modified or a new range needs to be established to meet the training/testing requirement.

c. Perform risk assessment to determine potential hazards.

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d. Provide technical guidance on range design to support safety, mission, and environmental requirements.

e. Provide technical guidance on construction requirements.

f. Consult the Institutional Laser Range Authority.

4.2.6 Range laser safety compliance. The Installation Laser Range Authority should perform range laser safety compliance inspection or audits:

a. Confirm laser use is compliant with applicable SOP and laser range safety certification report.

b. Verify the existence and placement of range boundary warning signs.

c. Inspect targets for serviceable condition and placement/orientation and ensure proper target maintenance.

d. Verify accessibility, condition, height, and coordinates of ground laser system firing area/line/points.

e. Inspect line of sight between laser firing position and targets.

f. Verify the condition and height of the backstop, as applicable.

g. Verify there are no specular reflectors in the Laser Target Area (LTA).

h. Inspect the range for state of maintenance.

i. Provide inspection/audit results to the range laser certifying authority

j. Verify authorized laser system listing is indicative of actual systems used and is current.

4.2.7 Laser range certification. The Installation Laser Range Authority accepts responsibility for the laser range certification as performed by the Institutional Laser Range Authority. The Installation Laser Range Authority assists the range certifying authority in performing range certification for the safe use of lasers. Certifications should consider, but not be limited to, the extent of range boundaries, required warning signs, number and location of non-removable specular reflectors, access control, airspace restrictions, local operating procedures, and environmental conditions or restrictions. The Installation Laser Range Authority should:

a. Review the laser systems to be employed.

b. Identify range boundaries.

c. Identify airspace restrictions.

d. Identify laser firing area/line points.

e. Identify laser target area/line points.

f. Identify LSDZ limitations.

g. Provide a digital range map.

h. Identify points of interest (towers, structures, etc.).

i. Identify encroachment/conflict issues.

j. Provide target replacement and maintenance.

4.2.8 Laser briefings and indoctrination.

a. Public notices - The Installation Laser Range Authority advises the affected public of the presence of laser operations, where deemed appropriate.

b. Affected users - The Installation Laser Range Authority provides laser briefings and indoctrinations in accordance with service-specific guidance. Laser indoctrination should be provided at the same time as the basic weapons systems instruction. The indoctrination should be at the user level avoiding complex scientific data or terminology. Proper channels for obtaining professional safety and medical assistance should be addressed during indoctrination. In addition to instructions on relevant devices, indoctrination should include:

(1) Principles of laser reflection.

(2) Laser hazards.

(3) Safety standards or operational control procedures of laser systems.

(4) Range laser operations.

(5) Encroachment issue.

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(6) Address the National Environmental Protection Act concerns.

4.2.9 Laser incident investigations.

a. Review incident in accordance with local SOP and training plan.

b. Verify that the DoD Laser Safety Event Hotline has been notified of the incident.

c. Provide technical advice on laser capabilities and laser hazard effects.

d. Gather information about the incident.

e. Prepare and submit data for the investigation report.

4.3 Unit laser safety authority or unit LSO.

4.3.1 Laser training plan. The Unit Laser Range Authority is responsible for preparing a laser training plan to submit to the Installation Laser Range Authority for approval to perform laser activities on a specific LTA. As part of preparing a laser training plan, the Unit Laser Range Authority should:

a. Determine laser operations in support of training requirements.

b. Review training to be accomplished against local operating procedures.

c. Select a range whose laser range certification supports the laser system(s) to be used and training exercise to be accomplished.

d. Review laser mode and tactics to be employed to ensure it supports the laser system and range.

e. Identify targets, laser firing area/line/points, laser to target orientation, and orbit points that can be supported by the LSDZ.

f. Identify personnel/equipment locations.

g. Identify range hazard clearing requirements.

h. Identify PPE requirements.

i. Identify communications requirements.

j. Identify emergency response procedures.

k. Present the unit's laser usage recording practices.

4.3.2 Conduct of range laser safety inspection. The Unit Laser Range Authority should conduct a laser safety inspection of the range and its operations prior to its use to include:

a. Ensure laser warning signs have been posted.

b. Check range conditions (targets/backstop, range boundaries, laser firing area/line/points).

4.3.3 Safety brief/pre-mission brief.

a. Laser systems to be used.

b. Authorized tactics, laser firing positions, laser target orientation, weapons release points, and weapons performance as applicable.

c. Drawings, photographs, descriptions, or grid points of authorized targets.

d. Communication procedures that include specific frequencies (or channels), controlling authorities, and standardized terminology.

e. Acquisitions, identification, and tracking procedures for targets.

f. Missile/ordnance mode of operations.

g. Requirements for beam termination.

h. Control measures to minimize the risk of unauthorized personnel or equipment entering the LSDZ.

i. Run-in headings and flight profiles to be used for airborne laser operations or permissible LSDZ for laser operations.

j. Review of mission profiles to prevent misdirection of laser guided weapons.

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k. Type of PPE to be worn and how it should be worn.

l. Potential hazards posed by the laser system, the target area, maintenance area, types of warning signs to be posted, and specific procedures to avoid these hazards.

m. Risk considerations for location of personnel and equipment within the WDZ for observing/lasing the target area to weapons impact.

n. Review the applicable SOP.

4.3.4 Laser operations.

a. Laser systems/targets.

(1) Ensure only service-approved lasers are employed on the range.

(2) Ensure laser systems are at the approved operating position or firing points and always pointed toward the target; verify laser firing area/line/points and laser to target orientation.

(3) Ensure that laser systems are only fired at authorized targets.

(4) Ensure that the target is positively identified in accordance with appropriate safety procedures before operation of a laser system.

b. LSDZ.

(1) Ensure all unprotected personnel and equipment in the immediate area of the laser firing position are outside the LSDZ while the laser is in use.

(2) Ensure LSDZ is clear of all non-essential personnel and equipment.

c. Pre-fire checks - Supervise pre-fire checks. Pre-fire checks that require operation of the laser system may be made in a controlled area with the laser beam terminated by an approved backstop. Pre-fire checks that do not require operation of the laser, but require use of the optics, may be safely made in any area. To operate the optics without firing the laser, follow SOP to ensure power to the laser is turned off.

d. Cease fire operations.

(1) A specular reflection is presented in the target area.

(2) Poor target tracking is observed.

(3) Non-essential personnel and equipment enter the LSDZ.

e. General supervision.

(1) Maintain communication with the laser system operators, Installation Laser Range Authority, and all affected range personnel.

(2) Ensure that the laser system NOHD does not extend beyond controlled boundaries.

(3) Ensure personnel follow safety procedures in accordance with local SOPs.

(4) Ensure training filters or modes are used, as required.

(5) Ensure PPE are being used, as required.

(6) Ensure the approved training plan is followed.

(7) Coordinate emergency response as necessary.

f. Laser incident investigations - The Unit Laser Range Authority should participate in laser incident investigations that occur during their training. The Unit Laser Range Authority should:

(1) Ensure the laser system involved is quarantined.

(2) Report the incident to the Installation Laser Range Authority.

(3) Provide information on training activity/exercise.

(4) Provide information on what happened.

(5) Provide information on personnel who may have been exposed to hazardous laser energy.

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4.4 Laser safety range review assessor.

a. The Laser Safety Range Review Assessor (LSRRA) is an independent range review authority with the specialized technical expertise to assess the safe use of lasers on military testing and training ranges.

b. The LSSRA will make assessments based on:

(1) Guidance from DoDI 6055.15, this document, and other laser range evaluation and assessment tools in accordance with military department-specific laser safety program policy and procedures.

(2) Advising the cognizant range use certifying or approving authority regarding hazards, risks, and possible remediation, including appropriate protections, mitigations, and precautions.

5. GENERAL LASER SAFETY INFORMATION

5.1 Laser/laser system.

a. Lasers emit light in a narrow, coherent beam of energy. Due to the increased directional intensity of optical radiation generated by a laser, a concentrated optical beam is present at considerable distances which may present a hazard to personnel and equipment.

b. A laser system is an assembly of electrical, mechanical, optical, and software components which includes a laser. The terms “laser”, “laser system”, and “laser device” are considered equivalent within this document.

5.2 Laser classification. Laser systems are classified according to their relative hazards from Class 1 (least hazardous) to Class 4 (most hazardous), summarized as follows:

a. Class 1 - laser systems pose no hazard under any normal viewing conditions, aided or unaided.

b. Class 1M - laser systems are only hazardous when viewed by magnifying optics.

c. Class 2 - laser systems are low-power visible wavelength lasers which are not considered hazardous for momentary (0.25 second) unintentional exposure because the normal observer will blink or look away before eye damage can occur.

d. Class 2M - laser systems are low-power visible wavelength lasers similar to Class 2 but are hazardous when viewed with magnifying optics even for a momentary exposure.

e. Class 3R - laser systems are considered safe, if handled carefully, with restricted intra-beam viewing.

f. Class 3B - laser systems are powerful and can cause serious eye injury for exposures of very short duration. They can be hazardous for long distances downrange from the laser system.

g. Class 4 - laser systems are very powerful and the most dangerous laser systems. They can be hazardous for extremely long distances downrange from the laser system. They can also present a potential diffuse reflection viewing hazard as well as a possible skin or fire hazard.

5.3 Laser system applications. Developments in laser technology have resulted in an increase in the use of laser systems for military application. Military lasers are used primarily for target acquisition, target designation, range finding, fire control, Ocular Interruption (OI), directed energy weaponry, and communications.

a. Laser ranging and designation - Laser target ranging, and target designation systems can provide accurate range, azimuth, and elevation information to locate enemy targets. In combination with global positioning system (GPS), lasers can provide accurate enemy target locations. Laser target designators mark targets for engagement. When within range, the laser designator can be aimed so the energy precisely designates a chosen spot on the target.

b. Laser spot tracker - The laser spot tracker identifies the reflected laser energy from a laser designator off the target and displays the target’s position on a display panel. Unless this is used with a visual verification, an erroneous target could be detected (the laser designator).

c. Laser guided weapons - A laser-guided weapon uses a laser to guide the munition to the target by illuminating the target with coded pulses of laser energy that is diffusely reflected from the target. Some laser-guided weapons require laser target designation before launch / release, during the entire time of flight, or only during the terminal portion of flight.

d. Laser marking - Laser marking involves using a laser to get an individual’s attention (from a distance on the ground or in an aircraft) to point out the location of a target using the laser.

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e. OI devices - OI devices should meet stringent safety criteria and be able to deliver a warning effect to targeted personnel by obscuring their vision. The devices currently in use may be Class 3R or greater laser systems that can be employed in training safely, but due to the intensity of the laser beam, it poses an eye hazard within the NOHD if not employed correctly. The laser system should be terminated if exposure distances approach the NOHD of the system (known and briefed prior to use).

5.4 Laser operations. Laser operations involve employment of a laser system to use a weapon, acquire a target for illumination, pointing, target designation, weapon guidance, range-finding, communication, or to provide OI. These laser systems may be mounted on a platform, a weapon, or they may be hand-held. Laser operations fall into nine different engagement scenarios that are detailed below in table I.

TABLE I. Laser engagement scenarios.

Ground to Ground	Air to Ground
Ground to Air	Air to Air
Ground to Surface	Air to Subsurface
Subsurface to Ground	Subsurface to Air
Subsurface to Subsurface	

5.5 Laser reflections. A laser beam reflects off surfaces it encounters. The magnitude of the reflection is dependent upon the power level of that laser, the material surface, and the angle of incidence at which the laser energy strikes the surface. In practice, this is similar to how light striking a surface will reflect at the same angle that it interacts with the surface.

a. Specular - Reflection of a laser beam off smooth surfaces such as a mirror, a still body of water, clean ice, plate glass, or chrome-plated metal produces a type of reflection known as specular reflection. Laser beams reflected off specular surfaces (especially flat specular surfaces) retain much of their collimation and energy, and thus still may be a hazard for a considerable distance.

b. Diffuse - Reflection of a laser beam off rough surfaces. Examples of diffuse reflectors include dry foliage, rock, camouflage, soil, matte paint, aluminum cans, paper, terrain, roadways, and old ordnance. Diffuse targets reflect light and laser energy in a hemispherical pattern centered at the point of impact (see figure 1). A diffuse surface is one that distorts (or diffuses) the beam shape, normally resulting in a safe-to-view reflection from outside the target area. The radiation will scatter in many different directions depending on the angle of incidence. Diffuse surfaces reflect energy in all possible directions including toward the laser. A hazardous diffuse reflection can only be produced by a Class 4 laser system.

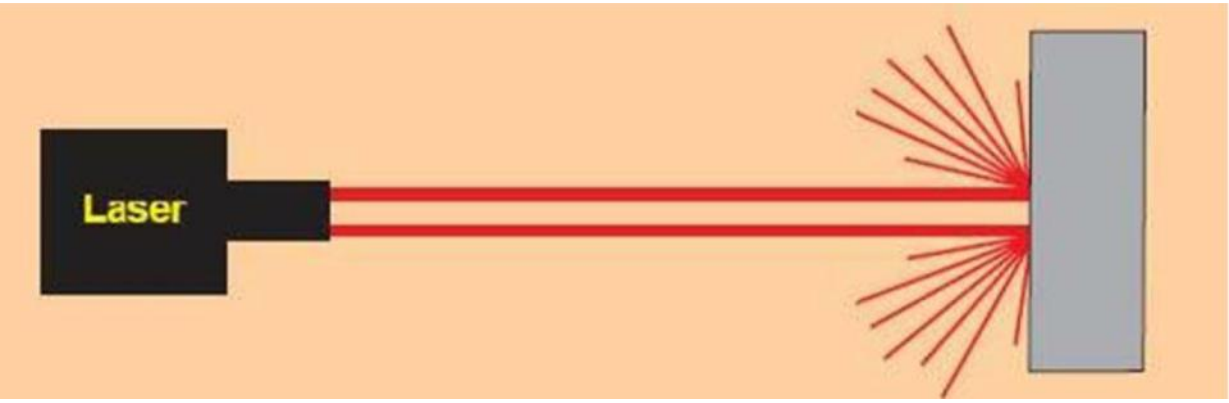


FIGURE 1. Diffuse reflection.

5.6 Control measures. The purpose of range laser safety is to prevent injury to both military personnel and the general public from laser radiation, and to ensure that only intended target areas are engaged by the laser. The goal

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is to accomplish this objective without placing unnecessary restrictions on laser system use. Therefore, control measures should be implemented to manage these risks. Different control measures are required depending on the class of the laser, the operational environment, and the training of personnel. Most control measures employ common sense practices aimed at limiting laser exposure, thus reducing the risk. Generally, controls are required for all laser systems. There are three categories of control measures in order of preference: Engineering, Administrative, and PPE. These measures can be used to eliminate hazards or reduce the degree of risk.

a. Engineering Controls reduce risks by design, material selection, or substitution. Examples of engineering controls include:

(1) Engineering controls associated with the laser system and built into the design of the laser system are considered engineering controls. These include software, switches, interlocks, protective housing or guards, training filters, key control, and beam stop. For example, a training filter can attenuate the radiation level of a laser beam below the MPE limit. Laser systems should not be altered by either removing or modifying these controls (see additional information in MIL-STD-1425 for military lasers and 21 CFR 1040 for commercial lasers).

(2) Backstops are the most important aspect to range laser safety and are the assurance that the laser hazard is terminated within a controlled area. A man-made or natural backstop on the range can be used to terminate the laser beam. If no backstop is available, the laser hazard is required to be controlled out to at least the NOHD, and possibly to the distance at which visual interference effects may produce secondary hazards.

b. Administrative controls reduce risks through specific documented policies, procedures, and actions, and include those controls that can be established and managed by range personnel, as well as the documentation and training that is required of participating personnel.

(1) SOPs are formal written descriptions of the safety and administrative procedures to be followed in performing a specific task and are the most important administrative control.

(2) Training and education for personnel (institutional, installation, and unit laser range authorities) to recognize laser hazards and take appropriate precautionary actions are control measures. Training requirements are dictated by service policy.

(3) Designation of installation and unit laser range authorities in writing is an administrative control to ensure range operations run safely.

(4) Warning signs and notices should be posted, when possible, around the installation laser training complex to warn and prohibit entry by unauthorized persons, and to alert authorized personnel entering a hazard area where lasers are employed (see figure 2). Warning signs or markers should be placed to ensure they are visible to individuals attempting to enter training complex areas at any point around its perimeter. The local SOP should provide for the placement and inspection of laser warning signs at the boundaries of the controlled areas and all access points in a way that will ensure that a person cannot enter the range without seeing at least one sign within a legible distance.

(5) Access control limits exposure to a laser hazard. Access to an LTA should be controlled. Access controls should prevent unprotected personnel and equipment from entering the LSDZ during laser operations. This objective can be met by determining where the laser hazard is expected to be and using the same control measures that are in place for live weapons fire, including roadblocks, gates, Local Notice to Mariners, Notice to Airmen, and cameras where the range is unmanned.

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FIGURE 2. Warning sign.

(6) Coordination of special use airspace (SUA) activities considered hazardous to nonparticipating aircraft or requiring SUA to segregate it from other users of the National Airspace System, or in airspace of host countries, should not be conducted until appropriate SUA has been designated and activated for that purpose. SUA is required to be designated and activated prior to conducting any activity over 45 meters above ground level that would be hazardous to aircraft. Types of SUAs that may be established include, but are not limited to:

(a) Restricted areas identified by an area on the surface of the earth within which the flight of aircraft, while not wholly prohibited, is subject to restrictions. Restricted areas should be designated when determined necessary to confine or segregate activities considered to be hazardous to nonparticipating aircraft (see figure 3).

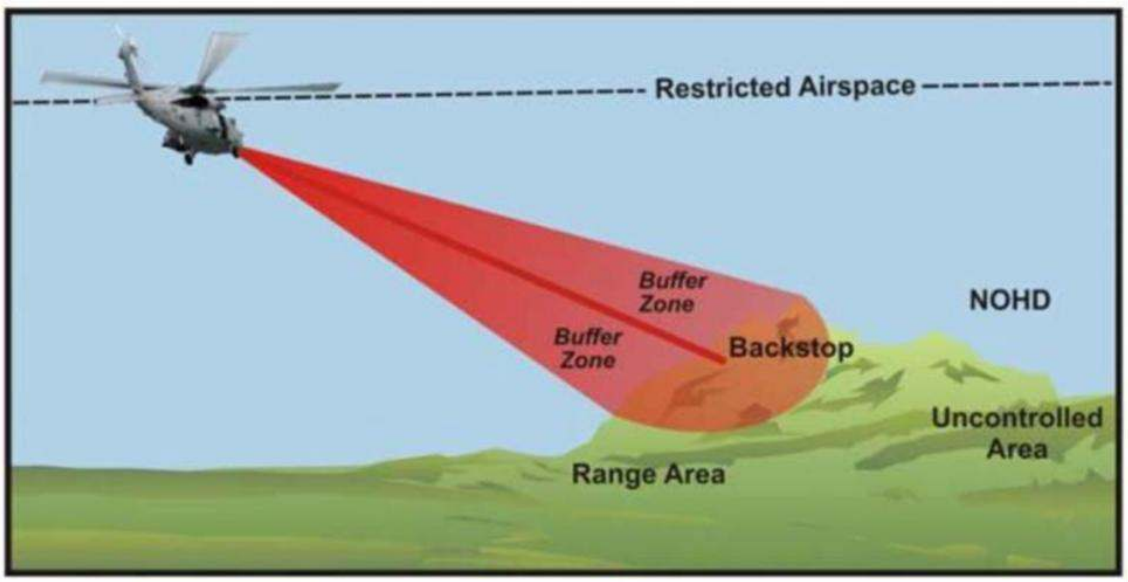


FIGURE 3. Restricted airspace.

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(b) Warning areas are defined dimensions that contain activity that may be hazardous to nonparticipating aircraft. The purpose of such warning areas is to warn non-participating pilots of the potential danger. A warning area may be located over domestic or international waters, or both.

(c) Military Operations Areas (MOA) are defined as vertical and lateral limits established for the purpose of containing certain military training activities that include, but are not limited to, air combat tactics, air intercepts, acrobatics, formation flying, and low-altitude tactics in airspace as free as possible from nonparticipating aircraft.

(d) Controlled Firing area (CFA) is established to contain activities that, if not conducted in a controlled environment, would be hazardous to nonparticipating aircraft. The distinguishing feature of a CFA, as compared to other SUAs, is that its activities are suspended immediately when spotter aircraft, radar, or ground lookout positions indicate an aircraft might be approaching the area.

(e) Coordination of navigable waterway requirements that apply to lasing over navigable waters, to include intracoastal waterways, can only be waived by U.S. Army Corps of Engineers (USACE). Installation commanders should notify the USACE Division or District Commanders, and the U.S. Coast Guard District Office of the waterway involved, operations to be conducted, and sector of waterway needed for closure.

c. PPE is considered a control measure because it protects personnel from the hazardous effects (eye and skin hazards) of lasers. PPE should be used when engineering and administrative controls do not reduce the hazard to acceptable levels. PPE includes goggles and spectacles to protect the eyes, and clothing and gloves to protect the skin.

(1) On some ranges, personnel and moving targets may be required to be on the range during laser operations for instrumentation operations, munitions impact spotting, and other required activities. The type of PPE required, if any, should be determined for each occupied location.

(2) Personnel within the LSDZ should wear Laser Eye Protection (LEP) and protective clothing, if applicable, during laser operations. LEP will match the required optical density at the wavelength of the laser system being used. LEP provides a measure of protection for a particular wavelength associated with a laser. A laser filter designed to protect against one wavelength may not provide protection against lasers operating at different wavelengths. If more than one type of laser is used, LEP should provide full spectrum protection for all wavelengths involved; the highest optical density required for all systems of the same wavelength should be used. All LEPs should have protection parameters permanently labeled for each set.

(3) The wavelength, optical density, and LEP requirements for currently fielded laser systems may be obtained from the Installation/Institutional Laser Range Authority.

6. GENERAL RANGE LASER SAFETY

6.1 Fundamentals. The fundamental concept of range laser safety is to prevent direct and collateral injury or damage resulting from laser use. Personnel using, or supervising the use of lasers, should be thoroughly familiar with all aspects of laser operations and associated dangers. The following guidelines should be used in conjunction with the guidance provided in referenced publications when employing lasers. Lasers should be treated as direct-fire weapons. Precautions associated with direct-fire weapons should be applied to all lasers operated on military ranges.

a. General service range laser safety requirements are outlined in AR 385-10, AR 385-63/MCO 3570.1, OPNAVINST 5100.27/MCO 5104.1, and AFMAN 13-212V1. Prior to laser use, the Unit Laser Range Authority should brief personnel on the use of lasers.

b. The use of Class 3B, Class 4, or DoD exempt lasers on test and training ranges should be conducted only at installations that have been certified for the safe use of lasers in accordance with service-specific requirements. Special consideration may be given to ranges using operator-controlled lasers.

c. A survey of the proposed lasing and target area should be accomplished to determine laser elevation and azimuth limits within the laser footprint. Restrictions should be applied to prevent lasing above the horizon unless authorized.

6.2 Laser systems.

a. Laser systems should only be directed at approved targets/target areas and only from approved operating positions/areas or on designated headings and altitudes, or angles and azimuths.

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b. Laser systems should only be used on ranges approved for such use.

c. Stationary Continuously Operating Lasers. Uses of lasers, such as light detection and ranging (LiDAR), space probes, or laser warning lights operating continuously in airspace may require additional controls. These emissions may require coordination with the applicable service airspace representative, the FAA, and LCH if the laser is directed above the horizon.

d. When lasers are not in use, hazardous laser output should be prevented by removing batteries or implementing engineering controls such as a software control, output covers, or rotating the laser into the stowed position unless otherwise specifically authorized by the local SOP.

e. Non-laser activities, such as viewing through common optics, can be conducted outside of a laser-controlled area by instituting procedures that ensure power to the laser is turned off.

f. When laser systems have both training and combat operating modes, the combat operating mode should be employed only when the SOP or the cognizant authority authorizes use.

g. Laser systems should not be employed on a range that is outside the scope of an applicable SOP (see Appendix A).

6.3 Unprotected personnel. Unprotected personnel will not be exposed to laser radiation within the LSDZ of the laser system.

6.4 Protected personnel.

a. Personnel within the LSDZ should wear PPE during laser operations.

b. Eye wear will be approved for the wavelength and corresponding optical density of the laser system being used.

c. Skin protection should be worn when appropriate.

6.5 Aided viewing.

a. Aided viewing involves the use of optical devices including binoculars, scopes, and rangefinders.

b. The magnification of laser energy can significantly increase the probability of eye injury. The use of magnifying optical devices to observe the target during laser operation is permitted if specular surfaces have been removed from the target area, appropriate filters are used, or it is being viewed beyond the NOHD with magnifying optics (NOHD-M).

c. Optical devices not marked with the level of protection should be assumed to offer no protection unless verified. Personnel should not view direct laser radiation with optical instruments within associated NOHD unless such instruments have appropriate laser filters in place.

6.6 Night vision devices (NVD).

a. NVDs may be used to detect lasers. Night vision devices should not be used for LEP.

b. These devices are not “cover-all” goggles and laser energy may enter the eye from offset angles where protection is not afforded.

c. The damage threshold for NVDs may be as low as, or lower than, the damage threshold for the human eye.

d. These devices can be bloomed (saturated), damaged, or destroyed from exposure to laser radiation.

6.7 Range maintenance.

a. A maintenance program for vegetation control should be established to ensure a clear line of sight can be maintained between the laser firing point and the target. Ensure all vegetation is trimmed or eliminated to maintain line of sight as environmental policy allows.

b. A maintenance program for the removal of specular reflectors should be established. This can be accomplished via range area cleanup, draining ponds, or painting over targets.

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7. LASER TARGET AND LASER TARGET AREA CONSIDERATIONS

7.1 Target types.

a. The targets employed in conjunction with a laser operation are selected based on the weapons employed and the testing and training requirements.

b. Operators and crews should conduct laser operations only on approved targets and in approved target areas.

7.2 Target material and diffuse reflections. In a laser operation involving a laser-guided weapon, the target is designated using a laser. A laser-guided weapon acquires and follows the coded diffuse reflected laser energy to strike the target. This means that the target needs to reflect sufficient laser energy for the laser seeker to lock-on. Certain materials reflect laser energy better than others. For targets with higher reflectivity, the probability of a laser seeker picking up the laser spot is increased. Table II presents the amount of reflectivity for various types of materials.

TABLE II. Amount of reflectivity.

Material	Amount (%) of Reflectivity
Olive Drab Metal (dirty)	2 to 30%
Concrete	10 to 15%
Asphalt	10 to 25%
Unpolished Aluminum	55%
Vegetation	30 to 70%
Brick	55 to 90%

7.3 Specular reflection.

a. Approved laser targets should not contain specular reflective (mirror-like) surfaces unless testing or training dictates.

b. Specular reflections can redirect the beam out of the controlled area and can result in eye or skin injury and may also damage equipment.

c. A flat specular reflector is a mirror-like surface perpendicular to a laser beam and can reflect the beam toward the laser position (see figure 4).

d. Examples of flat specular reflecting targets may include:

(1) Flat glass.

(2) Flat window.

(3) Still water.

(4) Instrument gauge.

(5) Vehicle rear view mirror.

(6) Vision view blocks.

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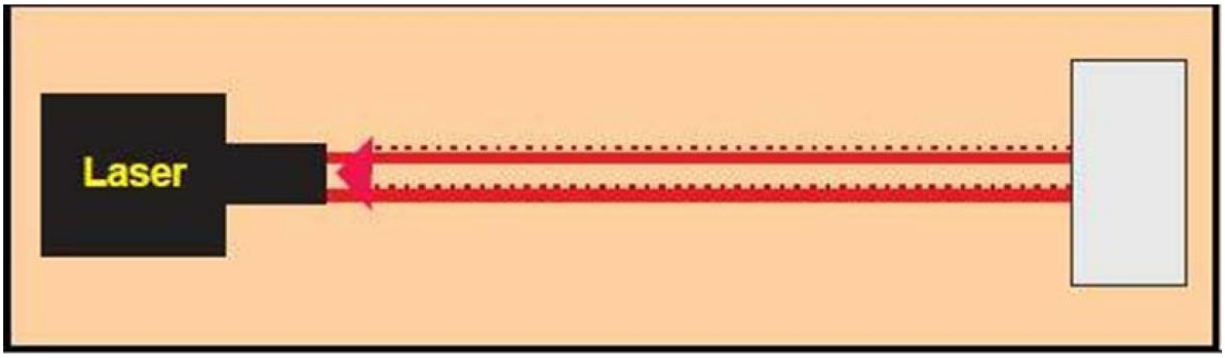


FIGURE 4. Specular (mirror-like) reflection from perpendicular surface.

e. Concave reflective surfaces can focus the reflected beam and cause the reflection to be more hazardous than the incident beam (figure 5). Normally, these reflections are of little concern because it is improbable that the surface is perfectly concave (focuses the beam to a single point) or perfectly reflective. Examples of curved specular reflectors include optical sights, curved windows, vehicle bumpers, headlight assemblies, or bottles. Glossy foliage, raindrops, fog, and most other natural objects are not considered to be specular surfaces that would create ocular hazards. This is because their convex curved reflective surfaces cause the beam to spread, and the reflected energy decreases quickly with distance.

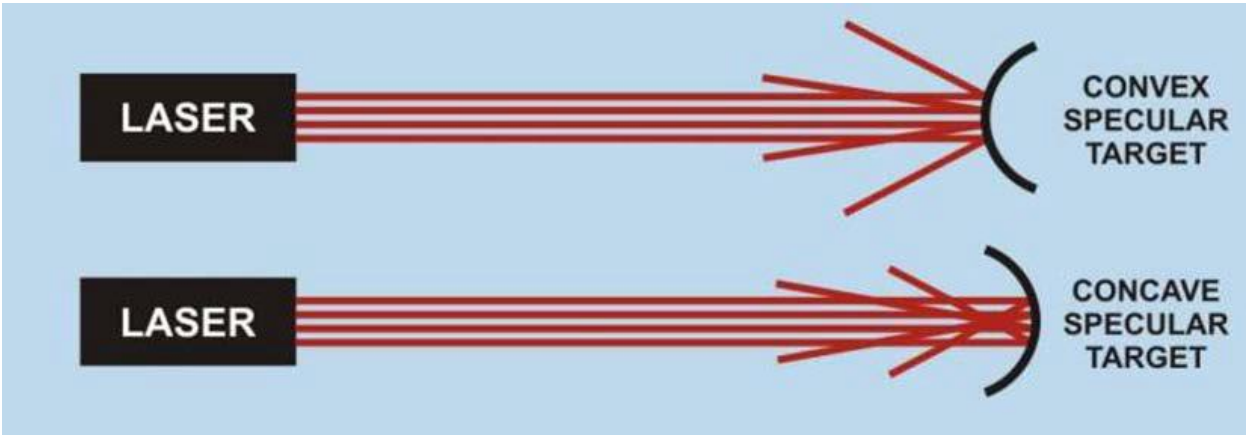


FIGURE 5. Reflection off curved specular reflector.

7.4 Target size. Target size impacts the effectiveness of laser operations by:

a. Laser spot size is the diameter of the beam at any given distance.

b. Laser beam divergence is the spread of the laser beam over distance. Laser spot size is a function of beam divergence and the distance from the laser system to the target. If a designator has a beam spread or divergence of 0.25 milliradian, its spot will have a diameter of approximately 0.25 meter at 1,000 meters in front of the designator. At 5,000 meters, the beam would spread to 1.25 meters; at 10,000 meters, the beam would spread to 2.5 meters (see figure 6).

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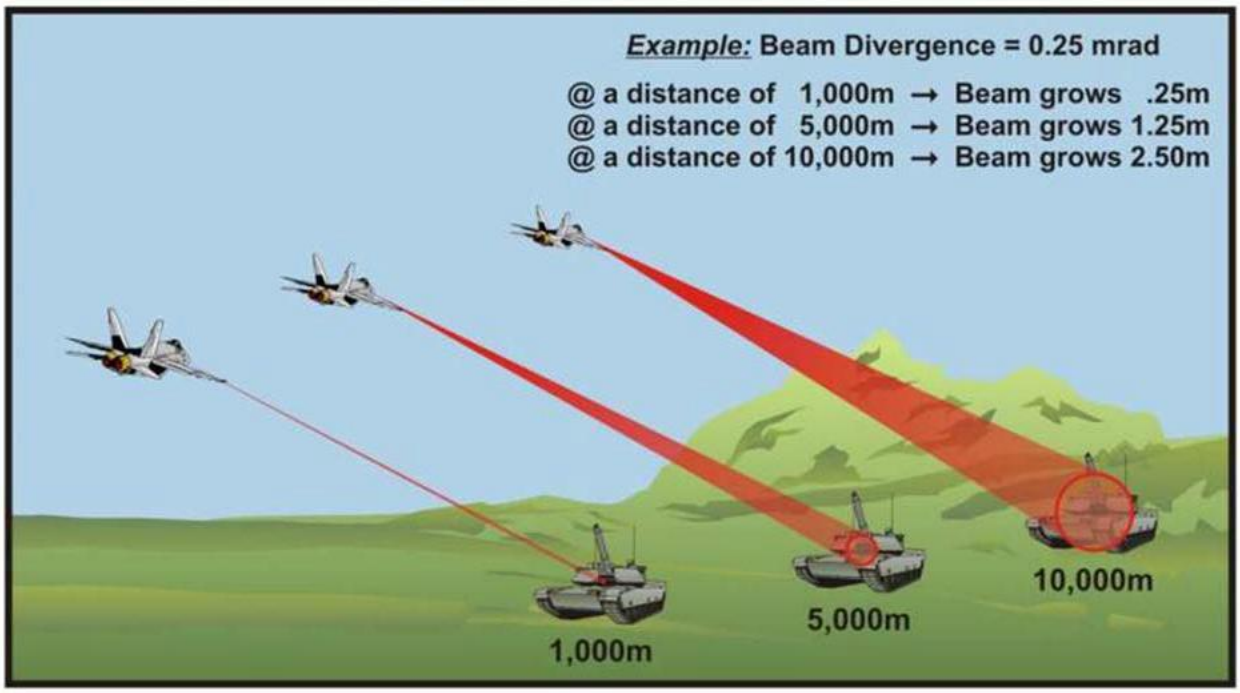


FIGURE 6. Beam divergence.

7.5 Target placement. On laser training areas, targets are placed to accommodate the training and test requirements. However, there are additional considerations for placement when employing lasers:

a. Targets should be positioned so the laser hazard can be contained within range boundaries or other designated constraints. This can be accomplished by elevating the lasing platform, using a backstop, or adjusting the distance from laser to target.

b. Targets should be located so that there is an uninterrupted line of sight from the lasing position to the target.

c. Targets should be oriented so the diffused reflected energy can be detected. This includes adjusting their position laterally as well as vertically.

d. Targets should be staged or positioned so that specular reflections are eliminated or mitigated.

7.6 Target maintenance.

a. Specular surfaces should be removed from all targets designated as laser targets prior to placement or if already in place, prior to engagement. If it is not feasible to remove all specular surfaces from the target, these surfaces should be covered with a diffuse material prior to use as a target.

b. Targets should be regularly inspected by Installation Laser Range Authority personnel to ensure they do not produce a specular reflection. Target maintenance includes the guidance of (1) and (2) below.

(1) Target condition - Careful attention should be paid to the condition of the target. A specular hazard can present itself upon impacts or explosions, which cause target materials to break apart, or from high energy laser/material interaction. Weather can deteriorate material as well. Broken or bent specular surfaces can have a flat surface remaining to generate a specular reflection. Target condition should be periodically checked by Installation Laser Range Authority personnel.

(a) Specular reflectors should be covered, removed, or rendered diffuse by painting with a matte (non-specular reflecting) paint, sandblasting, or scuffing up so they do not produce a reflection.

(b) Concave (bowl-shaped) surfaces with a large radius of curvature which could focus the reflected beam at longer distances outside the controlled area should be removed.

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(2) Target area condition - Terrain, unexploded ordnance, glossy painted surfaces, trash, and standing water can produce diffuse and specular reflectors within the target area. Snow is not a specular surface, but if thawed and refrozen, hazardous reflections can be found especially at low angles of incidence. Still water and clean ice can also reflect laser beams, especially at low angles of incidence. Figure 7 illustrates the reflection that can take place when a laser beam strikes still water. Target area condition should be reviewed periodically as determined necessary by the Installation Laser Range Authority to eliminate or mitigate specular hazards.

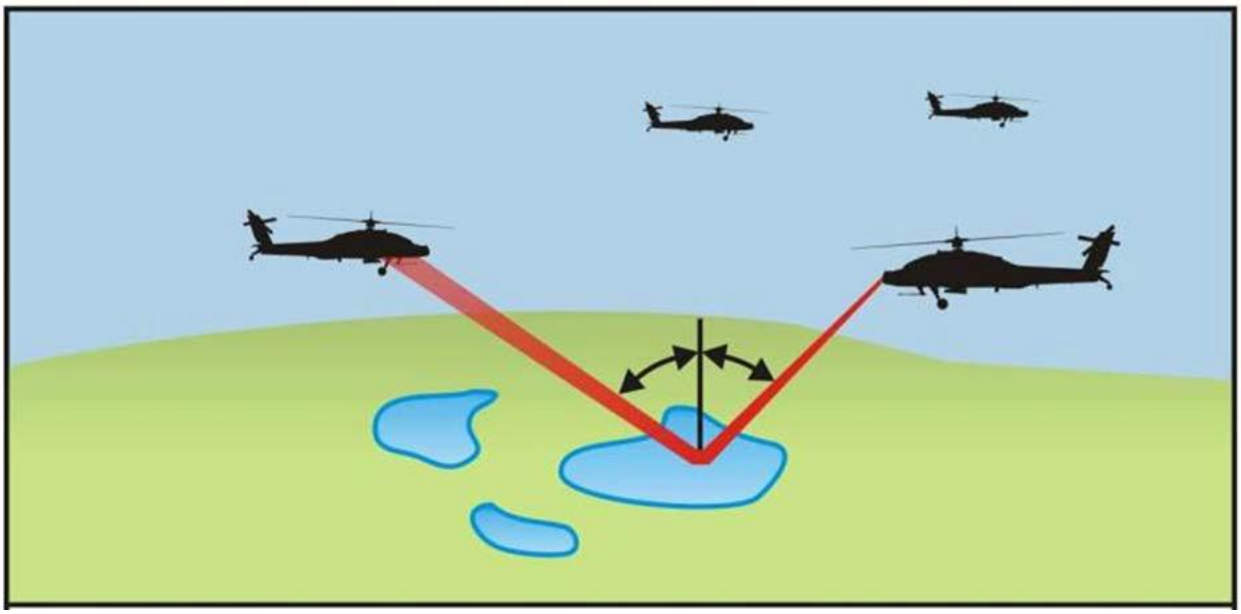


FIGURE 7. Example of airborne laser beam reflection.

(a) All specular hazards should be removed or mitigated from the laser training area. If potential reflections have not been considered for the approved target area, ranges will be closed (for example, when still water is present on the ground).

(b) Target location should be checked periodically by Installation Laser Range Authority personnel to ensure the location has not changed due to munitions impact. A change in target location or orientation can affect the angle of the laser beam to the target, thereby affecting the LSDZ for that laser system.

(c) The position and orientation of any specular reflectors that cannot be removed or rendered diffuse should be noted so they can be considered during the laser range certification. It may be possible to position the specular reflector away from the laser impact so that it will not be a hazard.

8. AIRBORNE LASER OPERATIONS

8.1 Considerations for laser operation. General following considerations and tenets apply to all types of laser operations whether they involve range finding, target marking (pointing), guidance, illumination, identification, or designation for guided munitions.

a. An unrestricted line of sight should exist between the laser system and the target. Since lasers are line of sight, their energy will reflect off objects in their path. If the laser beam reflects off any object other than the target, the laser may not designate the target correctly. Obstructions such as trees, limbs, leaves, grass, hills, and buildings between the laser system and target may prevent a clear, unobstructed view. Seasonal considerations come into play as well since the terrain in summer months may have more vegetation than in winter months. Jungle operations and weather conditions such as fog or low clouds could preclude the use of laser systems when a clear line of sight cannot be obtained.

b. Atmospheric attenuation is the reduction of laser power as the laser beam interacts with suspended matter which absorbs and scatters a fraction of the beam. This atmospheric absorption can reduce the effectiveness of the beam on the target.

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(1) The degree of atmospheric attenuation depends on the path length through and the optical density of the atmosphere. Environments that contain dust, smog, battlefield obscurants, and water vapor may absorb more light than a particle-free environment thus attenuating the laser to a greater degree.

(2) Smoke, haze, cloud, precipitation, and fog conditions can cause a laser beam to diffuse and thereby degrade delivery accuracy (see figure 8). Positioning of the laser system is a key to reducing the obscurants that can degrade laser performance. Possible considerations include positioning lasers on high ground where smoke is likely to be less heavy along the line of sight and repositioning from an obscured to a non-obscured position.

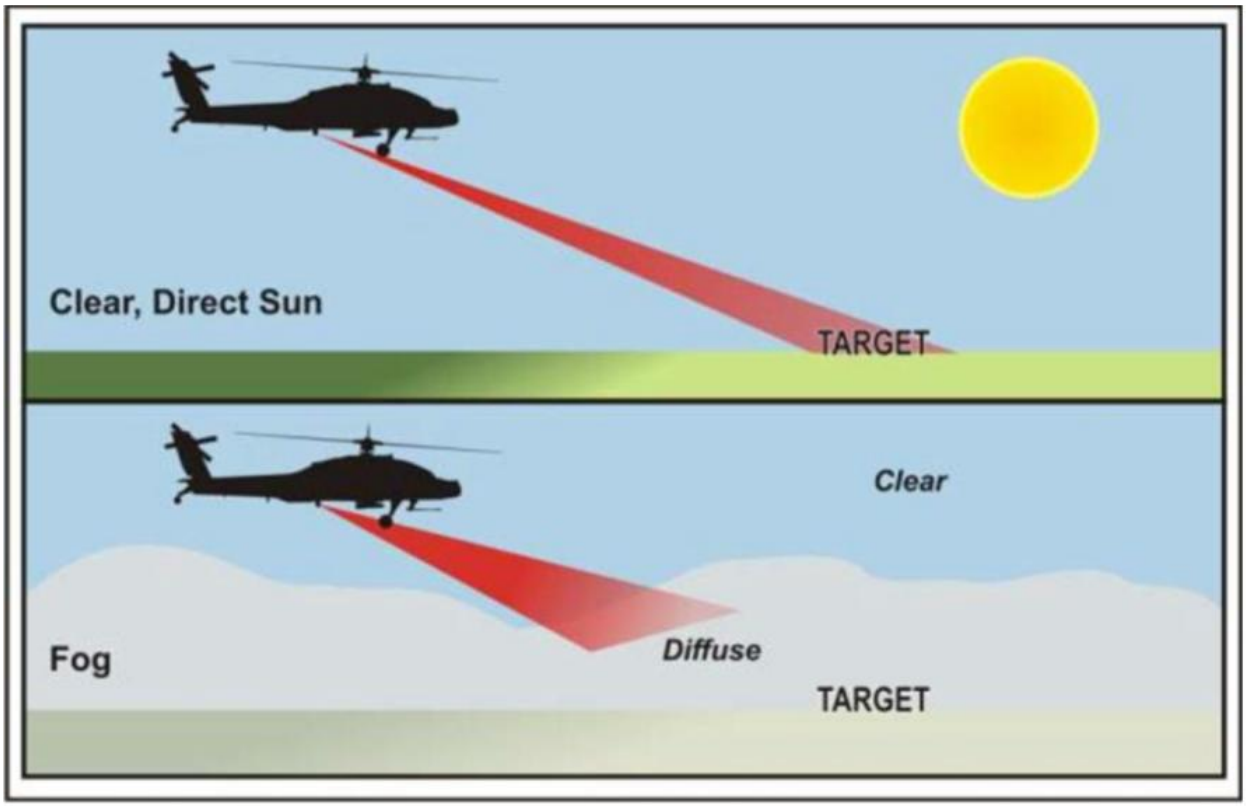


FIGURE 8. Atmospheric attenuation.

c. Laser operations planning is essential for both the unit and installation laser range authorities to understand the details of proposed laser employment tactics, as well as the technical aspects of all laser systems used on their ranges. Laser operation planning will vary with the type of laser operation. Laser marking, illuminating, and range finding do not require as much detailed planning with regards to coordination with other operators as laser designation (see 8.2). As laser systems are often used to identify, illuminate, or range a target in support of munitions delivery, laser operators should ensure their position is located outside the weapons danger zone to be safe from weapons effects. If it is necessary to be within the perimeter, ensure the position is outside minimum safe distances for the weapon system per service range safety regulations.

d. Minimum lasing altitudes for the aircraft are to ensure the LSDZs are contained within constraints such as roads, observation points, or range boundaries.

8.2 Laser target designator/seeker. Installation Laser Range Authority and unit range personnel should employ precautions to ensure the safety of personnel and equipment during a laser operation involving laser guided munitions. The following considerations should be given to ensure munitions are delivered on target and not directed toward the laser designator:

a. A seeker is a laser acquisition device. It detects laser designated targets so they can be engaged with ground, airborne, or naval weapons. In a laser operation involving a laser-guided weapon, the target is designated

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using a laser. A seeker acquires diffuse reflected laser energy off the target to direct the munition to that signal. The target cue provided by a seeker may be required because of the difficulty in seeing camouflaged targets at long ranges and high aircraft speeds. Seekers are set to identify the specific laser code (or Pulse Repetition Frequency [PRF]) used by the designator. A seeker can either be part of a laser guided munition or a standalone device such as a laser spot tracker.

b. Laser designators are used to illuminate targets to create a diffuse reflection for a seeker. Laser designators have selectable laser codes that are matched to the laser guided weapon's sensing system to achieve target lock-on.

c. Laser guided munitions and other laser detectors may unintentionally acquire radiation sources within the field of view of detection other than the target if certain precautions are not taken. This can result in fratricide if the munitions are guided to the laser rather than the target. Exclusion areas and tactics should be planned to ensure the angle between the laser designator line of sight and laser detectors will not mistakenly identify scattered radiation from the laser platform. The Installation Laser Range Authority should be familiar with the tactics and technical aspects of all laser systems used on their ranges.

(1) Laser codes are used for laser designation operations. Designators and seekers work together as a team on a specific code because seekers will not detect designators set on other codes. This allows for multiple targets to be identified and targeted by different systems / operators at the same time. To ensure the correct targets are prosecuted by the correct weapons, the PRF code of the laser target designator and seeker will be the same.

(2) Line of sight (unrestricted) needs to exist between designator and the target as well as between target and the seeker (see figure 9). For laser-guided weapons, line of sight is required prior to launch or after launch. If the laser beam reflects off any object, other than the target, the laser may not designate the target correctly, causing the seeker not to identify the target correctly.

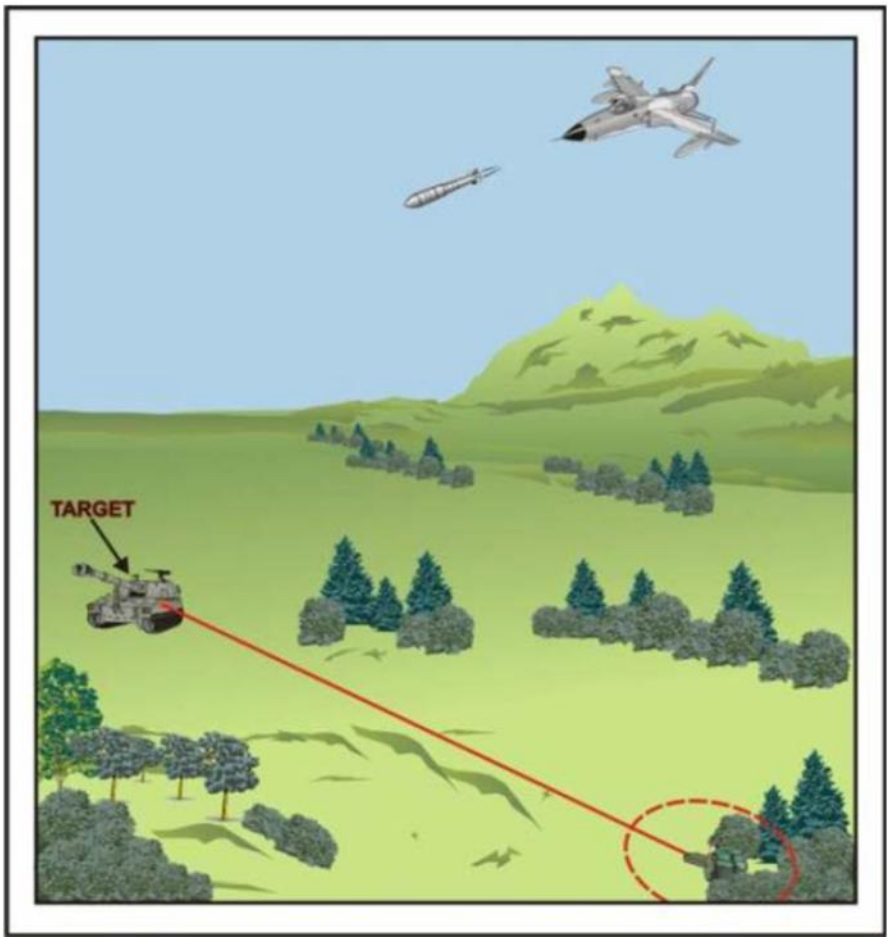


FIGURE 9. Line of sight.

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(3) Field of view limitations apply to all seekers and therefore should be oriented so that the target falls within that field of view, so the seeker is able to acquire the laser energy reflecting from the target.

(4) Wind effects are an important consideration for laser designator operator positioning for target areas where multiple weapon releases are anticipated. Laser designators should be positioned so that successive targets will not be obscured by smoke, dust, and debris from previous weapons impacts. Due to these factors, the laser target designator should be set upwind of targets and the targets should be designated from the farthest downwind to the most upwind.

d. Laser designation and weapons delivery tactics may involve the same platform or multiple platforms. The platform delivering the ordnance may not be the same platform that is designating the target, and subsequently, may impact the target from a different direction than the lasing platform. The following is a list of other factors that should be considered for laser designation operations:

(1) Attack heading should be provided by Terminal controllers to aircrews and, if providing the laser designation, the laser-to-target line. The direction of attack allows the seeker to sense sufficient diffuse laser energy reflecting from the designated target, minimize false target indications, and preclude the laser-guided weapon from tracking the laser target designator (see figure 10). These efforts should be coordinated by the exercise controller to ensure designators for other targets on the range are not using the same laser codes.

(2) Optimal attack zone is inside a 120-degree cone whose apex is at the target and extends to 60 degrees on either side of the target-to-laser designator line and is outside an exclusion zone, both laterally and vertically (see figure 10 and figure 11). This leaves an ideal attack zone of 50 degrees on either side of the exclusion zone and at an elevation that will ensure correct target acquisition and weapons function.

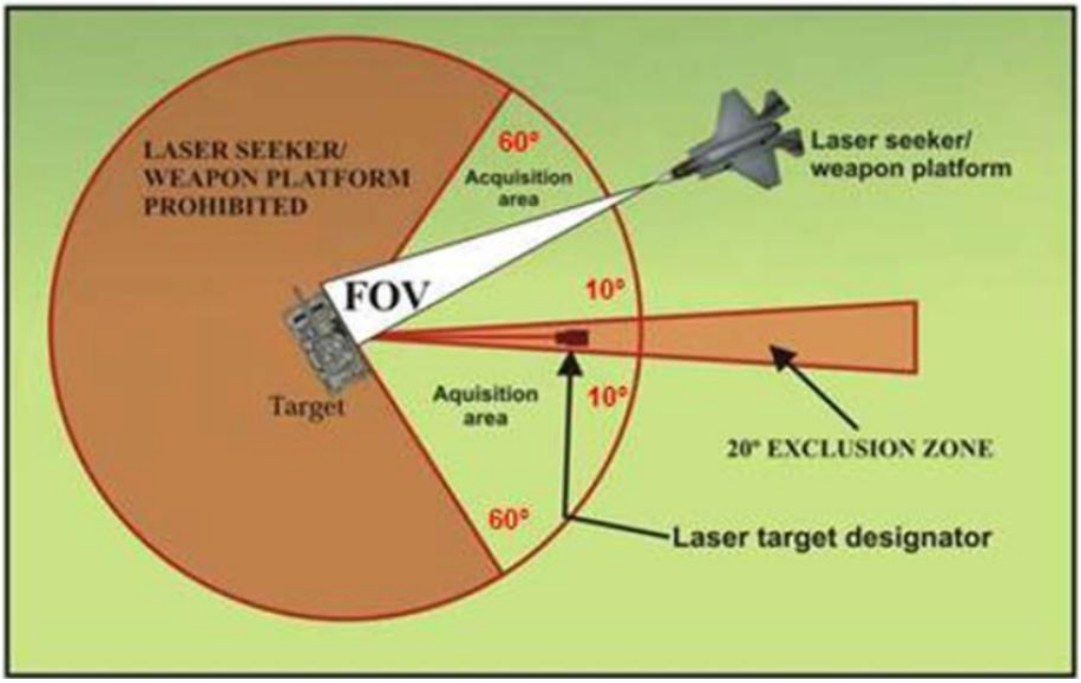


FIGURE 10. Optimal attack zone (top view).

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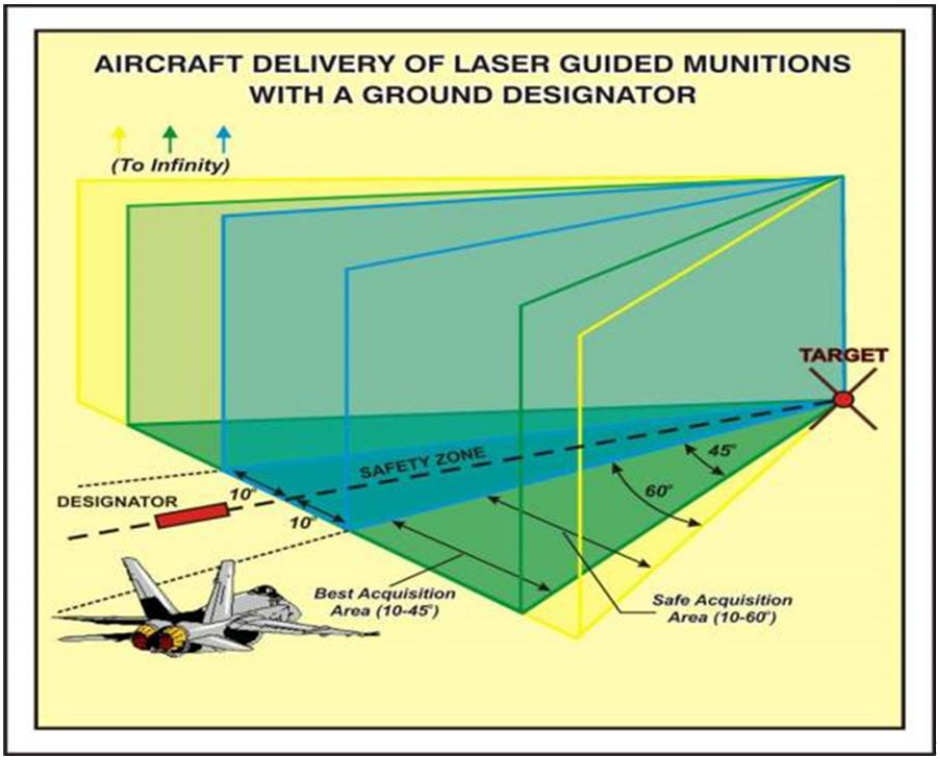


FIGURE 11. Optimal attack zone (side view).

(3) Exclusion zones are established as a sector whose apex is at the target and extends equidistant (10 degrees) on either side of the target-to-laser designator line. This zone extends vertically to infinity and has a horizontal limit of 20 degrees (see figure 12). Weapons release may occur in any area within weapons limits outside the exclusion zone that does not put the designator in jeopardy. Due to the possibility of false target acquisitions caused by atmospheric scatter from the laser beam within short distances from the laser exit port, attack headings should avoid designator exclusion zones unless the tactical situation can safely dictate otherwise.

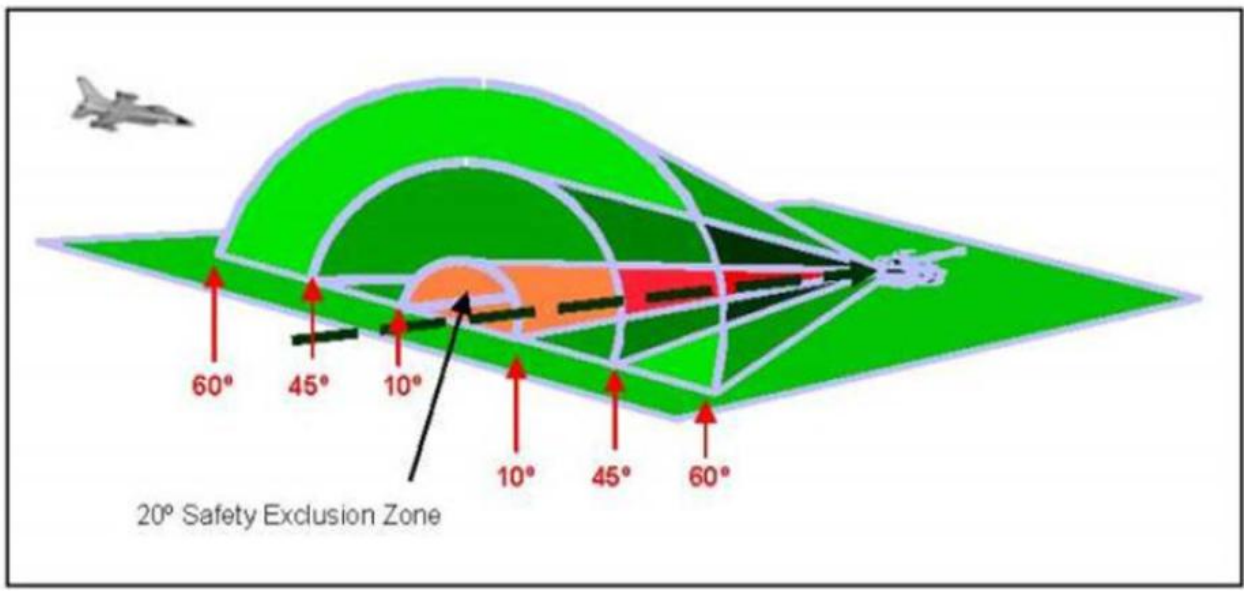


FIGURE 12. Optimal attack zone (vertical perspective).

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(4) Angle of attack or the degree of hazard to ground personnel and equipment operating the laser target designator varies with the attack angle of laser-guided munitions from the laser line of sight. The risk of acquiring the laser designator instead of the target in the optimal attack zone varies from high to low as the angle increases. The risk to the laser designator operator may be reduced by increasing the delivery aircraft altitude, offset angle, or the designator-to-target distance. While increasing the delivery offset angle improves safety, it may degrade the seeker's ability to acquire the laser spot.

(5) Laser munitions have a limited amount of time to detect the laser spot, lock-on, and guide to the target. Laser-guided weapons require a minimum amount of time to acquire and track a target. The laser designator should engage the target at the correct time and for the proper duration. The delivery system releases the weapon within the specific weapon's delivery envelope. Laser spot tracker-equipped aircraft may not have enough time to acquire the target under short designation time conditions. The required acquisition time is mission-specific and should be pre-briefed and coordinated by operators. Communications among aircrew and ground participants are crucial.

(6) Backup method for target acquisition should also be implemented. During training, aircrews should not use a laser spot tracker as the sole source for target verification since the seeker may lock-on to atmospheric backscatter or the laser target designator. To ensure proper target identification, the laser spot cue provided in the cockpit will align with the expected target location. Aircrews should verify they are attacking the target through additional means such as visual description, terrain features, or non-laser target marks.

e. Seeker lock-on errors can occur during laser designation operations. It is essential for the munition to correctly lock-on to the laser spot so the target can be accurately engaged. This applies to aerial, ground, and naval launched projectiles. Laser seekers may occasionally lock-on to other reflected energy instead of the target. Under some conditions, the seeker may incorrectly lock-on to the laser target designator. In this case, a seeker is most likely to detect stray energy only in the immediate vicinity of the designator. Errors include:

(1) Backscatter is of sufficient intensity in a portion of the laser beam to cause the seeker to lock-on. Backscatter refers to a portion of the laser energy that is scattered back in the direction of the seeker by an obscurant, such as dust, smoke, or clouds. Since backscatter energy competes with the diffuse reflected energy from the target, a seeker may attempt to lock-on to the obscurant rather than the target (see figure 13).

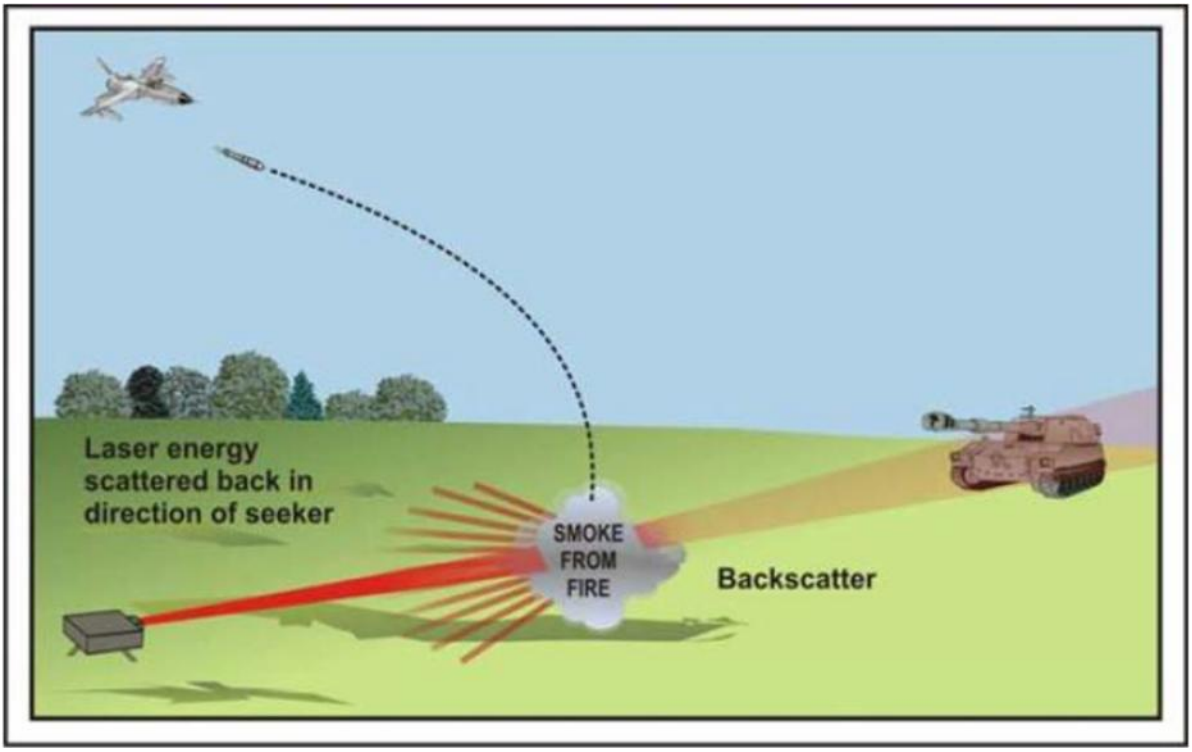


FIGURE 13. Backscatter.

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(2) If spillover energy (overspill or underspill) is of sufficient intensity to cause the seeker to lock-on, a potential for fratricide or collateral damage may exist. Spillover reflection occurs when the laser spot is larger than the intended target or when there is unsteady tracking of the target from the designator. Laser spot size depends on the laser system's beam divergence, the relative elevation of the designator compared to the target, distance of the designator to the target, and backstop/terrain around the target. This laser spillover can provide scattered reflections off objects around the target (see figure 14). Laser designator operators should generally aim at the center of mass of the portion of the target that produces the best reflectivity or in such a way as to avoid overspill and accomplish the intent of the weapon strike. In general, the closer the designator is to the target, the smaller the laser spot. Additionally, for airborne lasing, as the lasing aircraft gets closer to the target, the laser footprint will lessen reducing the spillover.

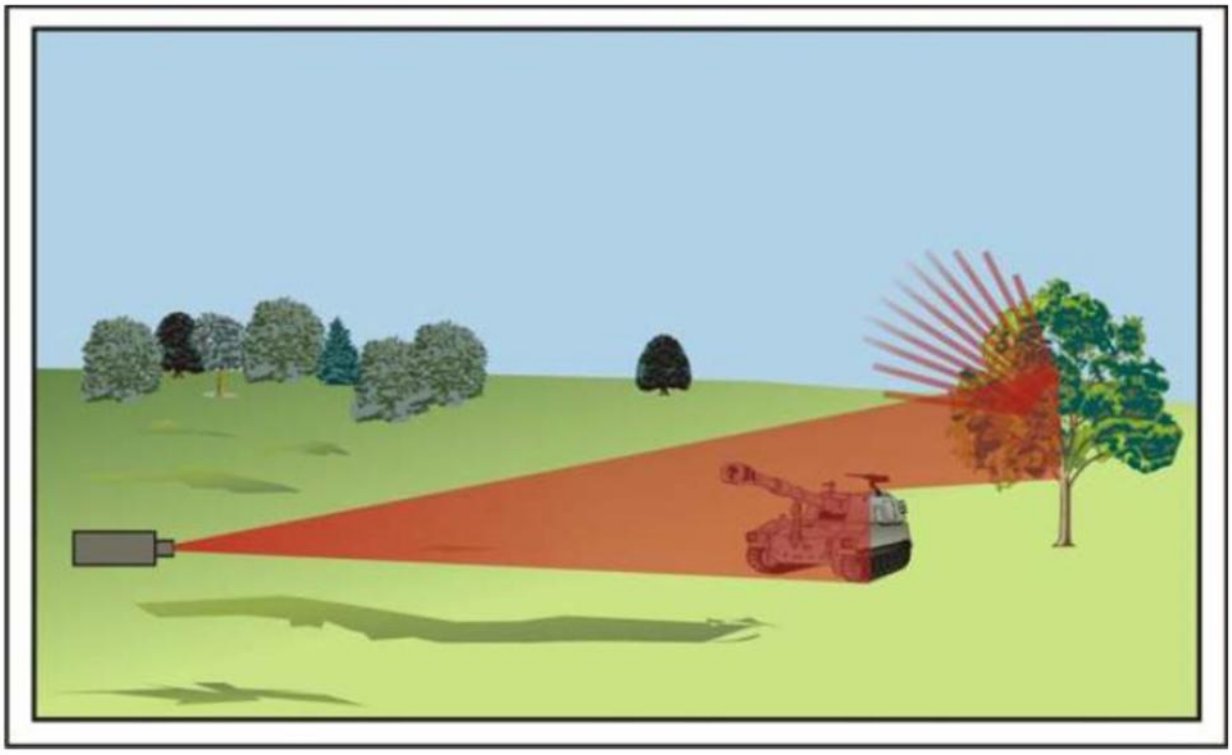


FIGURE 14. Spillover.

(3) Seeker fails to lock-on to anything. This could happen due to a variety of factors:

(a) The laser designator and seeker are not set to the same code.

(b) Poor aiming of designator.

(c) Aircraft seeker position is such that the laser target designator, laser beam, and target are not in the seeker field of view.

(4) Seeker locks on to laser target designator can occur when the designator, rather than the target, is the only return in the seeker field of view. The following precautions should be taken to prevent seeker lock-on errors:

(a) Coordination and synchronization of laser codes between the laser target designator and the laser guided weapon.

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(b) Whenever possible it is essential during planned attacks to avoid placing the designator in the field of view of the seeker. The seeker field of view covers the target but not the area where scatter might be detected by the seeker.

(c) Aircraft attack headings should be close to the laser designator-target line but outside the 20-degree exclusion zone (10 degrees on either side of this line). The following related actions are essential:

(1) Designate and brief approach paths to both the designating and forward air controller personnel and the aircrews prior to conducting the mission.

(2) Plan aircraft approach paths to preclude crossing laser designator beams with the laser seeker.

(3) Ensure the laser seeker intersects the designator beam well forward of the laser firing point, angling toward the target.

(d) It is essential that the pilot of the attacking aircraft has positive knowledge of the location of the designator and the target area before releasing munitions. It is essential that no munitions are launched or released on a heading toward the laser designator.

(e) It is essential that targets are placed to provide enough offset to accommodate the laser operator position so the aerial-, ground-, or naval-delivered munitions do not follow the laser to the operator rather than the target (see figure 15).

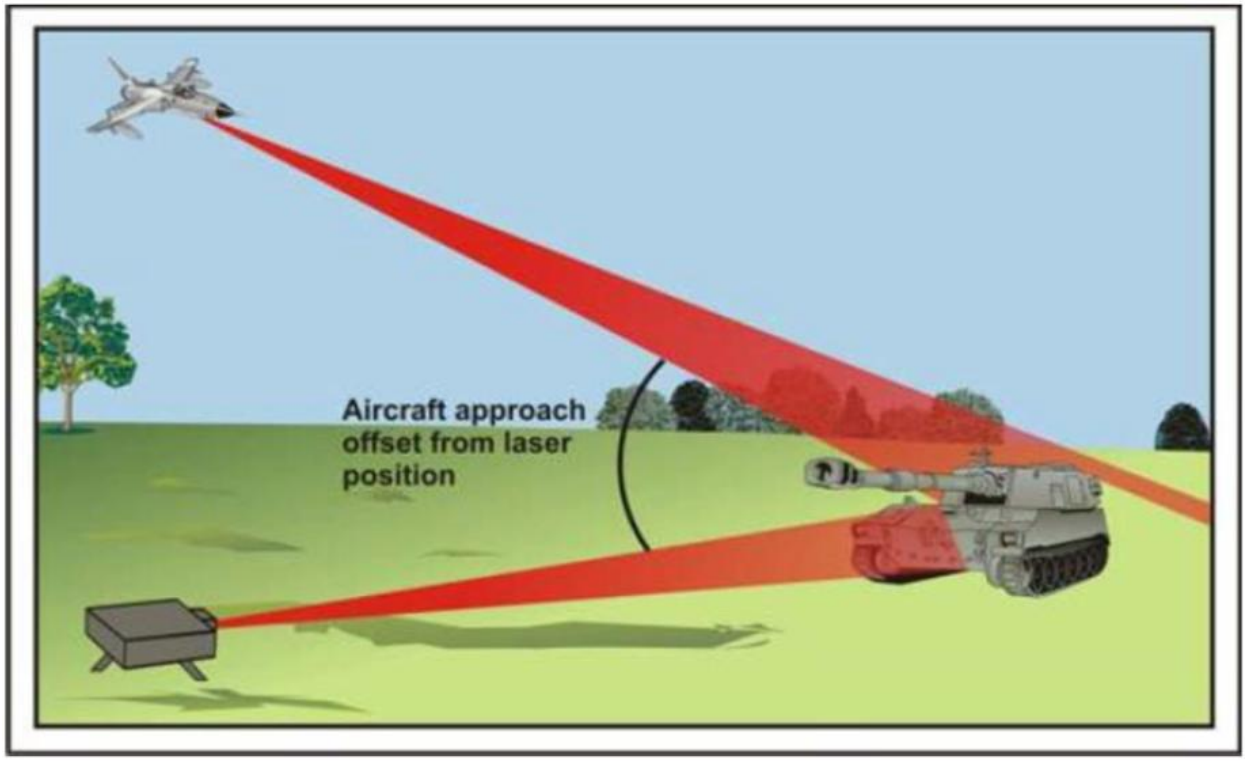


FIGURE 15. Target placement to provide offset.

(f) Ground designator operators are recommended, if possible, to screen the sides of the designator position, from the seeker field of view (out to several meters in front) using vegetation, tarps, and other related materials. WARNING: This does not guarantee that the laser seeker will not lock-on to the laser designator.

e. Laser-guided weapons can erroneously lock–on to the scattered radiation from wingman aircraft laser designators during aerial laser operations. Caution will be taken when using a target-designating laser in conjunction with ordnance delivery aircraft. The potential exists for the on-board laser seeker to lock-on to the designator or its radiated energy (that is, the beam or reflected beam) instead of the target.

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f. Two methods of laser designation can be used with aircraft-released laser guided weapon systems: Lock-On Before Launch (LOBL) and Lock-On After Launch (LOAL). In LOBL, the weapon system acquires the target designation (laser spot) prior to release from the aircraft. In LOAL, the weapon system does not acquire the target designation until after it is released. Even if LOBL is planned, the aircrew should be prepared to employ the weapon in the LOAL mode in case a laser spot is not received before the clearance to launch has been given.

g. Lock-on before launch - In the LOBL mode, the laser-guided weapon line of sight can be displayed in most launch aircraft.

(1) If the line-of-sight cue is above the horizon then the weapon system is probably locked onto an erroneous spot, such as the designator aircraft or atmospheric scatter instead of the desired target spot, and the mission should be aborted.

(2) If the weapon system properly locks onto the target in any LOBL mode, the only risk to the designator would be a midair collision potential if the designator aircraft is operating below the missile trajectory apex. In LOBL modes, the wingman aircraft altitude should remain substantially above the nominal laser-guided weapon apex altitude, keeping in mind that missiles can climb to altitudes well more than their nominal apex values especially if they are tracking a laser designator.

h. Lock-on after launch - In the LOAL mode, no laser-guided weapon line of sight cueing is provided prior to launch of the weapon.

(1) When employed in the LOAL mode, the laser guided weapon will execute a climbing profile searching for a laser coded energy signal prior to tipping over and scanning its field of view along the ground.

(2) The risk to the wingman designator is highest during the initial phase of the laser-guided weapon profile. If it locks onto the designating aircraft, there is a high probability that it will track and impact the laser designator. The dimensions of the instantaneous fields of view of the laser-guided weapons are not absolute and some can detect forward or back scattered radiation at many degrees off boresight.

9. DANGER ZONES AND RISK ANALYSIS

9.1 Deterministic.

a. Danger zones are areas determined by analysis of weapons and laser characteristics combined with historical patterns to define the risk to personnel and equipment within a designated three-dimensional (3D) space. Types of danger zones include LSDZs, Surface Danger Zones (SDZ), and WDZs; for more detail on these see the definitions.

b. A laser footprint is the projection of the laser beam and buffer zone on a surface. A laser footprint serves as the foundation for the worst-case LSDZ calculations. It should be noted that if the laser is not terminated by a surface, no footprint exists and the laser hazard will extend to the NOHD.

(1) An LSDZ only extends as far as the NOHD. If targets are placed beyond the NOHD, at a distance where the laser no longer exceeds current exposure limits, the laser footprint would indicate target placement is in an area outside the LSDZ. The laser footprint may be part of the LSDZ if the laser footprint lies within the NOHD of the laser.

(2) As the laser energy extends beyond the NOHD, the laser footprint accounts for this area. In the case of visible lasers, when the laser footprint extends beyond the NOHD, this can give an indication of an area that may still be of concern for non-injurious transient effects, such as flash blindness, dazzle, disability glare, and startle.

c. Factors affecting an LSDZ based on the hazard potential of the laser system which includes:

(1) Maximum permissible exposure (MPE) limit is the level of laser radiation, up to which, a person may be exposed without hazardous effect or adverse biological changes in the eye or skin. If possible, laser beams and the associated buffer zone should be terminated by backstops, or the radiation level attenuated below the MPE limit within the controlled range or in controlled airspace. If energy below the MPE is allowed to leave the range, the possibility of direct or optically aided viewing by unprotected individuals should be considered in the laser range certification.

(2) NOHD is the distance from an operating laser to the point where the laser irradiance or radiant exposure is not expected to exceed the appropriate MPE. Under normal atmospheric and operational conditions, at distances greater than the NOHD, the laser should be safe for intrabeam viewing. The NOHD for laser systems may be obtained from the Installation/Institutional Laser Range Authority.

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(3) NOHD-M is the NOHD when viewing a laser beam (intrabeam viewing) or reflections through optical instruments, such as binoculars or a scope, can significantly increase the degree of hazard for the eyes and thereby increase the NOHD, which now becomes an NOHD-M. The NOHD-M for optically aided viewing of laser systems may be obtained from the Installation/Institutional Laser Range Authority.

(4) Laser platform stability determines the pointing accuracy of the laser system which, in turn, determines the size of the buffer angle (see table III). The more stable the platform the greater the pointing accuracy; the less stable the platform, the poorer the pointing accuracy.

d. LSDZ consists of a 3D space (volume) extending from the laser system to a backstop, terrain (ground or water), or through NOHD (see figure 16). An LSDZ will always exist if the laser has an NOHD. An LSDZ consists of the following components:

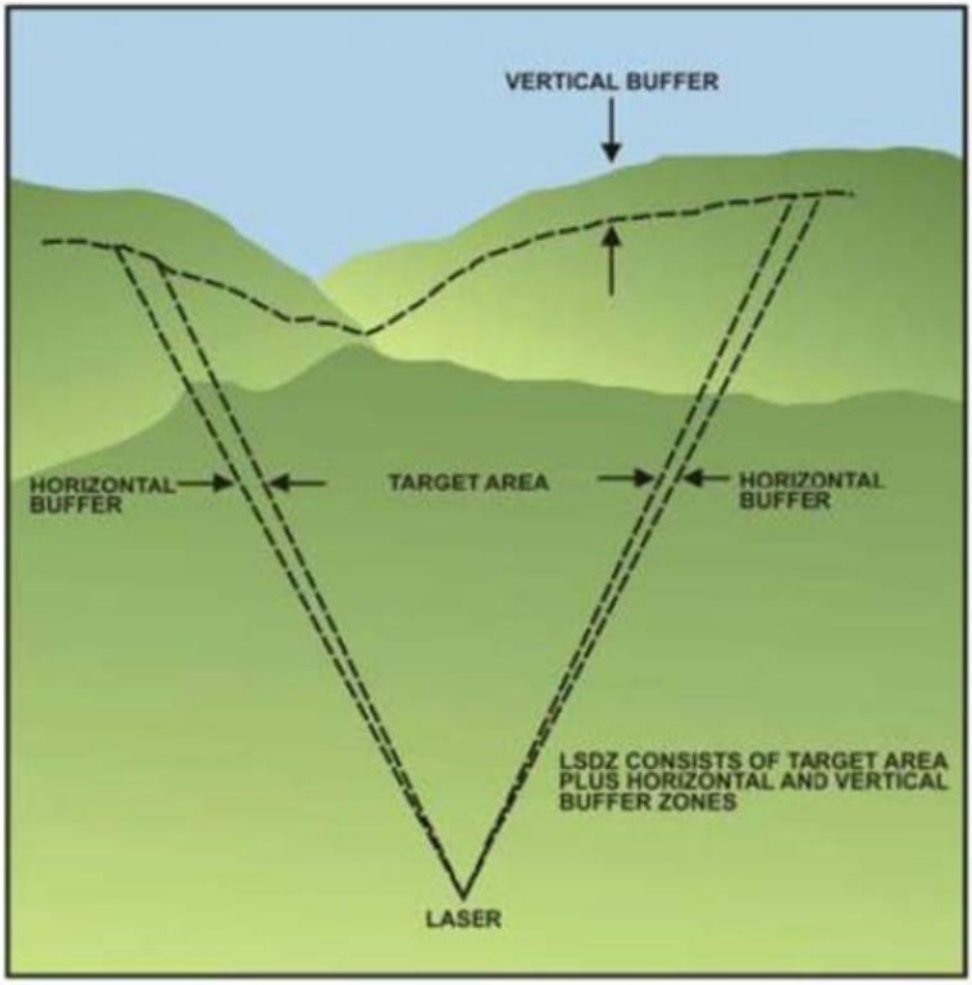


FIGURE 16. Laser surface danger zone (LSDZ).

(1) NOHDs and LSDZs on the ground normally extend out to the NOHD or beam backstop and to the edges of the laser beam buffer zone.

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(2) Use of optics affects the size of the LSDZ. When viewing the laser beam with optics (telescope, binoculars), the hazardous range can be greatly increased. For example, a 10-kilometer NOHD would be increased to 80 kilometers for an individual looking back at the laser from within the beam with 13 power optics (see figure 17). Such large amounts of real estate are difficult to control. The solution is to use a natural backstop behind the target. If no natural backstop is available, the possibility of optically aided viewing by unprotected individuals should be considered in the laser range certification.

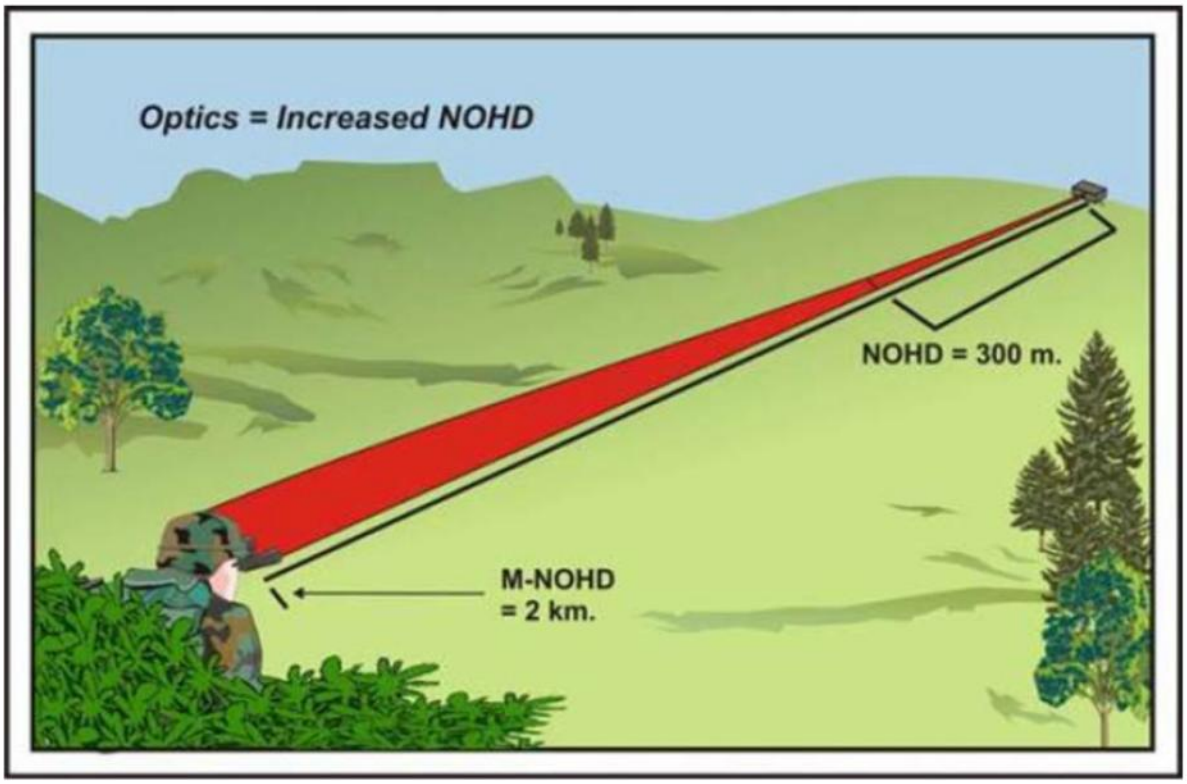


FIGURE 17. NOHD with/without viewing optics.

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(3) Use of backstop - The use of a natural or man-made backstop affects the size of the LSDZ (see figure 18 and figure 19). Laser beams and the associated buffer angles can be terminated by backstops.

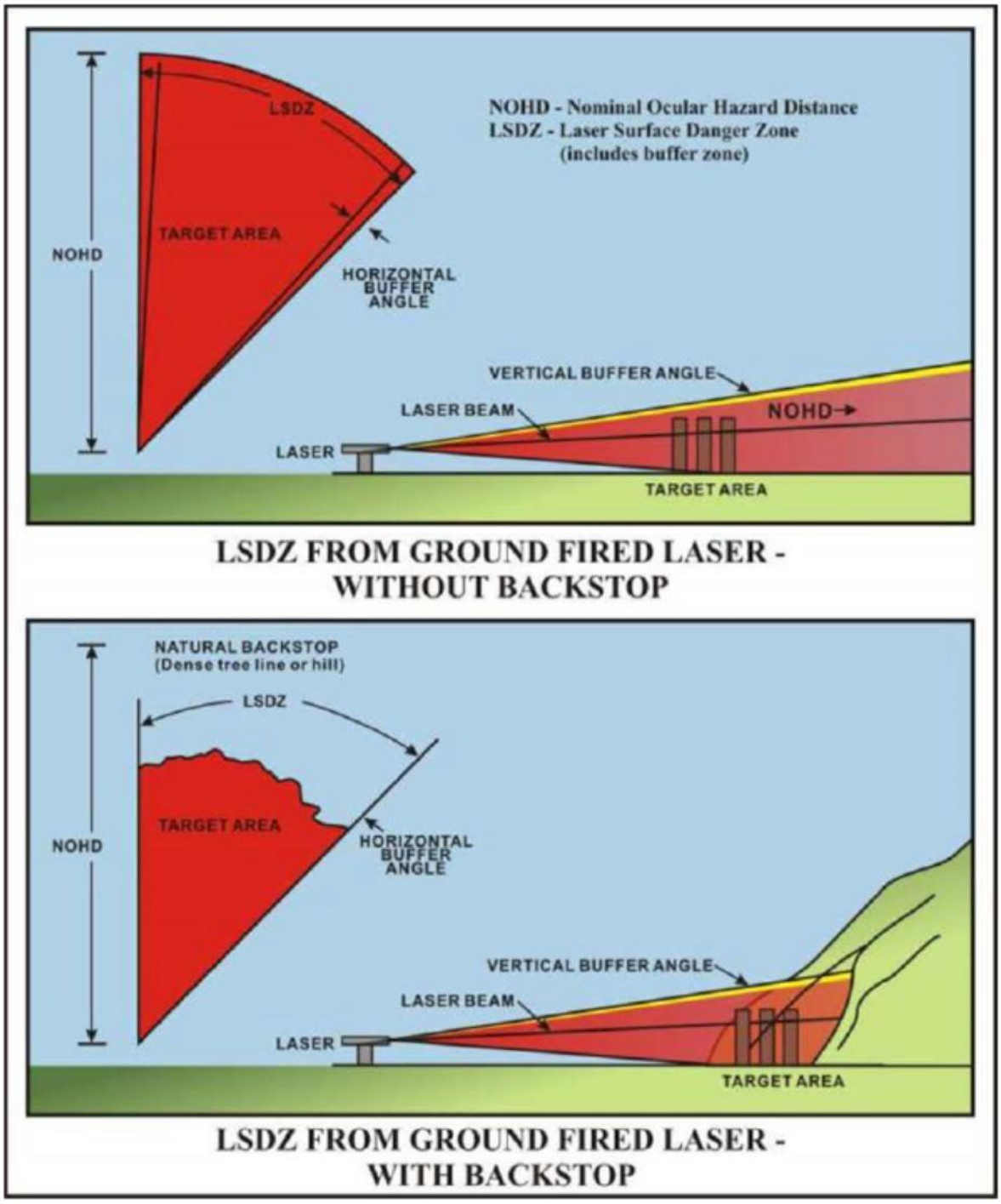


FIGURE 18. LSDZ without and with backstop.

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(a) When the distance to the backstop is less than the NOHD, the backstop determines the absolute hazard distance and the NOHD is a reference value (see figure 19, beam 1).

(b) When a backstop is present, laser energy is prevented from leaving the controlled area or training area.

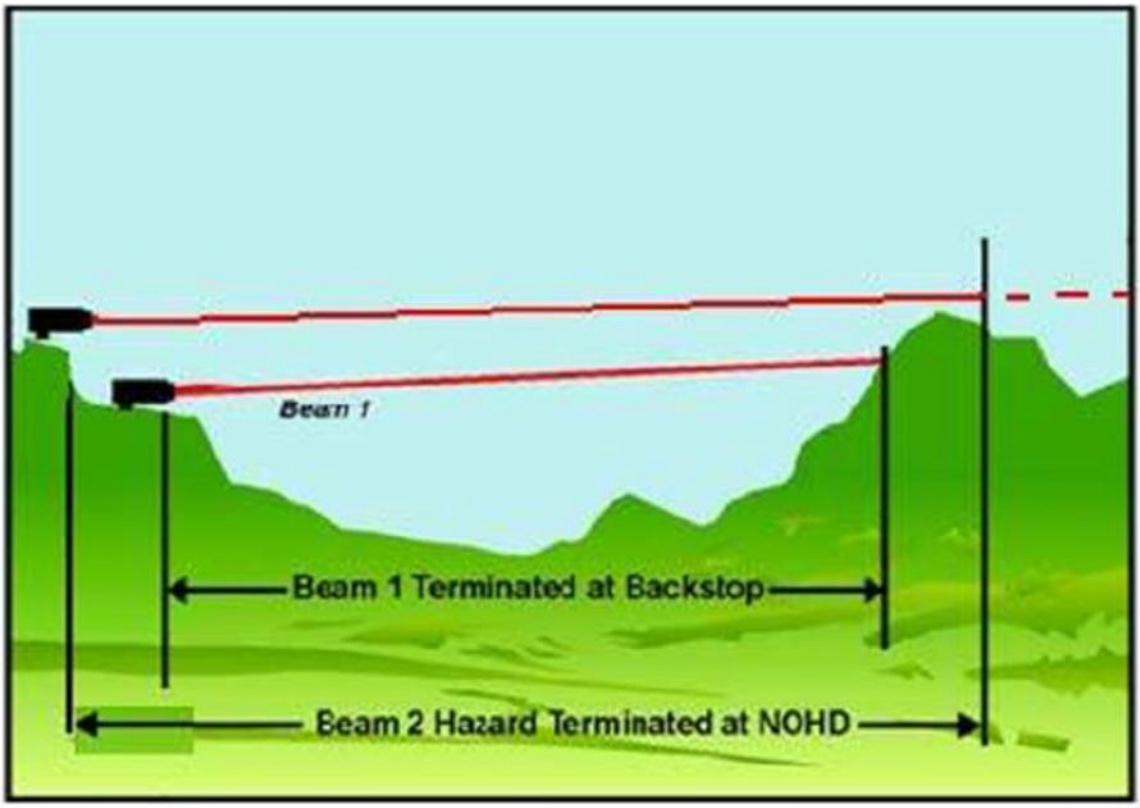


FIGURE 19. Effects of backstops.

(c) In cases where there is no backstop, the LSDZ extends downrange out to the NOHD in the airspace (see figure 18, beam 2). Laser operations at targets on the horizon are permitted if airspace is controlled out to the NOHD. Laser operations at targets above the horizon are permitted if the airspace is controlled out to the NOHD, and the laser operations are approved by the Laser Clearinghouse.

(d) NOHD should be contained within the range boundary or military controlled area.

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(4) Buffer angle is a safety margin added to the beam divergence to establish a laser footprint and is directly proportional to the pointing accuracy of the laser system on its platform. The footprint is then used to assess the viability of potential backstops. Table III lists typical buffer angle values for lasers using an alignment device (optics or sights).

TABLE III. Typical buffer angle values for lasers using an alignment device (optics or sights).

Situation	Buffer Angle	Explanation
Fixed	2 mrad	For targets engaged from a rigid stationary mounting or platform where accurate aiming can be guaranteed.
Inertially Stabilized	5 mrad	For targets engaged from a platform using stabilization (gimbal, gyroscope, or accelerometer).
Supported	10 mrad	For targets engaged by hand-held lasers supported on sand bags or other improvised method.
Unsupported	15 mrad	For targets engaged by unsupported hand-held lasers with an aiming or alignment device.
NOTE: 1. For lasers without alignment devices, the aiming accuracy may require significantly larger buffer angles.		

(5) Factors affecting buffer angle can include the aiming accuracy and platform stability of the laser system; therefore, buffer angles are dependent upon the laser system mounting (that is, a hand-held laser system has a larger buffer angle than a tripod mounted system). The buffer angle increases as the stability of the system decreases. Some laser systems are designed to be used from a variety of mounting configurations.

(6) Determining buffer angle - The buffer angle is measured in milliradians to the left and right of the laser-target line out to the NOHD (see figure 20). (Note: 17.7 milliradians is equal to one degree.)

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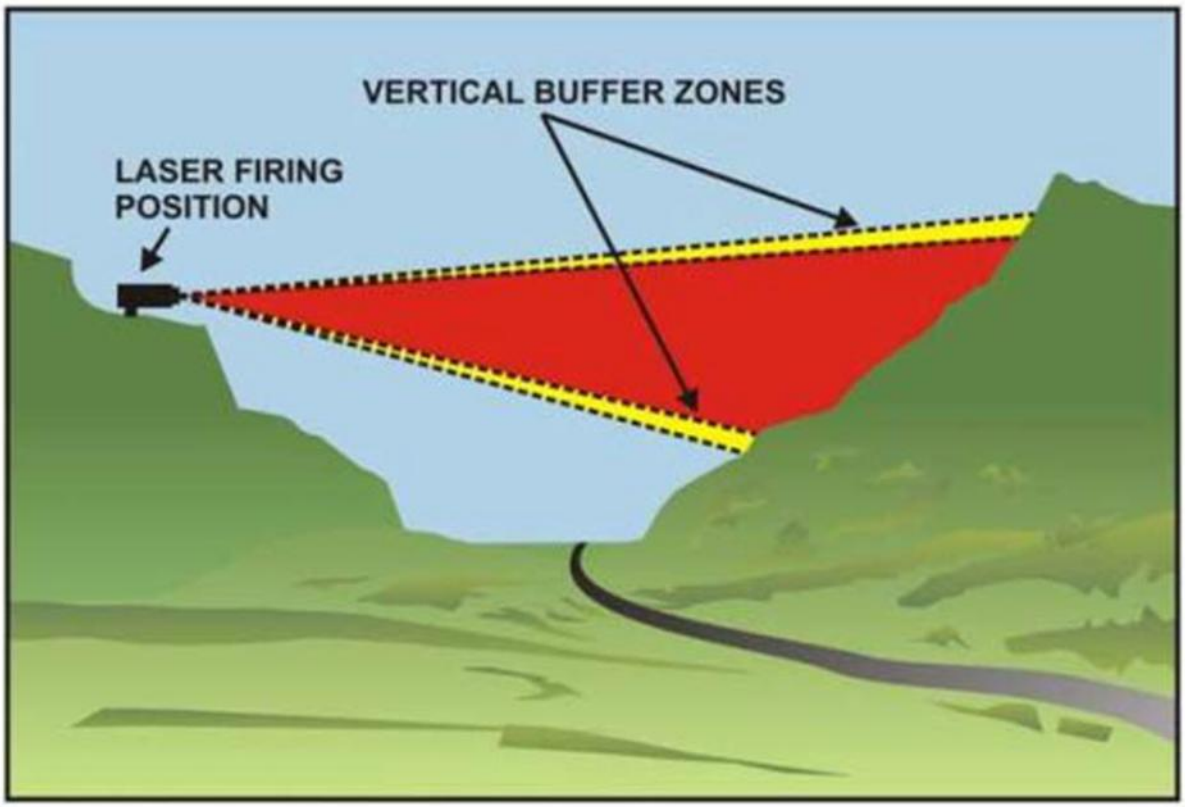


FIGURE 20. Buffer zones with backstop.

(a) If multiple targets are being lased, the buffer angle is applied in milliradians to the left of the left-most target and to the right of the right-most target out to the NOHD.

(b) If the laser is moving the buffer angle is applied to both the initial and final lasing positions for both the left-most and right-most targets/locations.

(c) Minimum buffer angle requirements for currently fielded laser systems under their intended mounting configurations may be obtained from the Installation/Institutional Laser Range Authority. If the laser platform is unknown, the buffer angle will be 15 milliradians on either side of the beam, which may be the worst-case scenario (see table III.)

e. Nominal hazard zone (NHZ) describes the space within which the level of the direct, reflected, or scattered radiation may exceed the applicable MPE during normal operation. Exposure levels outside the NHZ are below the applicable MPE level. The NHZ consists of the target area plus the buffer zones. The NHZ is often confined by the application of backstops. For unterminated direct viewing, this zone would extend to the NOHD. The size, location (distribution), and type of targets to be lased on a range are of primary importance in determining NHZ.

(1) Specular NHZ is created when specular reflectors cannot be removed from the target area due to training requirements. To calculate the specular NHZ, determine the difference between the distance from the laser system to the target and the NOHD distance. This distance is the radius of the volume around the target that constitutes the specular NHZ.

(2) Diffuse NHZ is created from the intersection point of the laser beam and target. If the LSDZ encompasses the target, there is no need to calculate a diffuse NHZ.

f. Airborne LSDZ - To maximize training capabilities for airborne lasing, an LSDZ is a 360-degree generated composite around the target unless dictated by range or training constraints. When generating an airborne LSDZ, the prime consideration is the direction of fire of the laser and not the aircraft heading; it is unlikely that aircraft heading will coincide with the laser-to-target bearing. The delivery aircraft could directly approach the target but

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may also turn away from the target after weapons release before designating with the laser to guide the weapon to the target.

9.2 Probabilistic. This section covers the complex topic of a Range Safety Probabilistic Risk Assessment (RS-PRA) as it applies to outdoor lasing. An RS-PRA becomes necessary for outdoor lasing when the laser hazard is no longer containable such that there becomes an inherent statistical probability of overexposure to people, property, and infrastructure. In this situation, an RS-PRA is necessary to manage overexposure so that the risk of harm is maintained below levels of loss deemed acceptable by competent risk acceptance authority.

a. Currently, there is no RS-PRA capability for outdoor lasing. The notion of intentionally overexposing people to lasing hazard runs counter to current DoD and service level policies. Even the federal regulations and commercial standards for public exposure through laser light shows ensure containment of the hazard above Maximum Permissible Exposure (MPE), so the shift to allowing overexposure while managing the risk of harm will necessarily involve policy and cultural adjustments, in addition to the technical capability of RS-PRA.

b. Nonetheless, it is possible to consider the RS-PRA capability that currently exists for the blunt force trauma hazard of falling debris due to an airborne incident as a template from which generalized requirements and the implementation task perspective can be described. This chapter provides a brief description of PRA, range safety strategies, the four core requirements for an outdoor lasing RS-PRA, the four task groups involved with implementing an outdoor lasing RS-PRA, and the path forward for developing the DoD outdoor lasing RS-PRA capability.

9.2.1 Probabilistic risk assessment (PRA).

a. PRA in simplest terms is defined by the U.S. Environmental Protection Agency (EPA) as “a group of techniques that incorporate variability and uncertainty into the risk assessment process.” It provides estimates of risk as a probabilistic distribution, rather than a single point estimate of a Deterministic Risk Analysis (DRA). Thus, PRA is particularly relevant when understanding the sensitivity, uncertainty, or confidence in the risk estimate is important.

b. Risk analysis based on using point estimates to generate a single-valued risk measure is the traditional basis for deterministic risk analysis. In DoD, the risk assessment code (RAC) is an expression of the risk associated with a hazard that combines the hazard severity and mishap probability into a single numerical value according to enclosure 3 of DoDI 6055.01. This is a classic example of a deterministic risk analysis method. Similarly, the definition of risk in MIL-STD-882, “a combination of the severity of the mishap and the probability that the mishap will occur,” is driven by this same deterministic perspective of risk analysis when severity and probability of occurrence are taken to be point estimates.

c. In the case of range safety, it is useful to consider the operational risk to be a measure of the undesirable effect of uncertainty on safety for a situation involving a hazard, whereby safety is taken to be freedom from harm to a sufficiently high degree of confidence. With this more nuanced understating of risk, the risk model becomes a logical relation of the critical factors characterizing this uncertainty that enables computational evaluation of this measure. The number of factors characterizing the effects of uncertainty within the model reflects the potential degrees of freedom and thus reflect the potential fidelity of the risk model. Higher fidelity computation of risk generally involves consideration of more factors affecting uncertainty. Within this context, the combination of severity and probability of occurrence, i.e., the simplest two-factor risk mode, reflects a minimally low fidelity risk model for which the deterministic approach involving point estimates to create a single value measure is sufficient for some purposes.

d. Confidence, on the other hand, comes from characterizing the factors affecting uncertainty in terms of their variabilities, e.g., probabilistic or statistical distributions, so that the risk measure is no longer a single value, but rather a number with error bars, or a statistical distribution, from which some actual measure of confidence can be derived. If the variabilities are expressed as statistical or probabilistic distributions based on actual studies, then it may be possible to propagate this error mathematically through the risk model. Alternatively, Monte Carlo simulation formed by running the model with the various statistical variances can also reveal the resultant measure of risk as a statistical distribution. In this case, the number of Monte Carlo runs needs to be sufficient to accurately reflect the tails of the risk distribution, so that confidence can be established.

e. Consequently, a PRA in practice typically involves a risk model with sufficient fidelity, e.g., factors affecting uncertainty, combined with the consideration of variation of those factors to gain insight into risk as a measure with sufficient confidence for supporting risk-based decisions. In contrast to the generally simpler and

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quicker calculated deterministic risk, probabilistic risk typically comes at the increased cost associated with the additional data required, the increased level of scholarship involved, i.e., requisite knowledge and expertise, and the increased computational resources and time involved to perform a risk assessment. To understand if the potential benefit outweighs the cost, it is important to understand how a PRA could potentially fit within the context of a particular purpose, which in the case at hand is range safety.

9.2.2 How to apply PRA to range safety. The concept of risk varies according to functional perspective. PRAs have been developed to address specific risk perspectives relevant to the EPA, NRC, and NASA. Developing a PRA for laser range safety starts with an understanding of range safety's risk perspectives and range safety's risk management strategies. Range safety derives its purpose from the Environmental Safety and Occupational Health (ESOH) perspective of operations, e.g., systems use. In accordance with DoDD 4715, it is DoD ESOH policy "to protect the public from risk of death, injury, illness, or property damage because of DoD activities." In this respect, range safety is the responsibility of each range commander to ensure hazards from range operations do not violate established safety, security, or range boundaries (DoDI 3200.18). Harm, which is caused by a hazard or the result of a hazardous situation, requires the existence or occurrence of a hazard, exposure to the hazard, and the right conditions based on specific damage mechanisms to result in some level of casualty or loss. From a range perspective, hazards include thermal, kinetic, radiological, chemical, nuclear, or biological sources, etc., whereby outdoor lasing represents a directed energy form of optical radiation. Optical radiation, in particular, interacts with matter, including skin and eye tissues and optical sensors, with a variety of effects that under the right conditions can easily become damage mechanisms. Such mechanisms may include photo-thermal, photo-mechanical (e.g., photo-ablation), photo-chemical (e.g., photo aging), photo-acoustical, photo-electric, and even photoionization.

a. In general, range safety involves the application of two prominent strategies, namely "hazard containment" and "hazard coexistence." The "hazard containment" range safety strategy is the overall preferred approach and best practice, whereby the hazard is physically contained within some geospatial volume, such that vulnerable people, property, and infrastructure outside of the contained volume have no possibility of exposure to the hazard within. Key to this strategy are physical barriers that can absorb the excess energy of the hazard, range controls that positively restrict access to the hazard in some way, and "safety thresholds" beyond which the hazard no longer poses a threat of harm. In the "hazard containment" strategy, range safety positively controls, e.g., through clearance and surveillance, the hazard volume, such that for harm to occur, there must be a fault, failure, or breach of some barrier or range control. Thus, the concern in "hazard containment" is the residual risk of exposure predicated on such a failure or fault. Additionally, range safety typically requires with this safety strategy that there be at least two barriers (physical or controls) separating a hazard from non-participants, including the public, and at least one barrier, in addition to PPE, separating the hazard from participants.

b. For the particular hazard of outdoor lasing, the established "safety threshold" is the MPE in accordance with DoDI 6055.15, ANSI Z136.1, and IEC 60825-1. MPE is a conservative level of exposure that is at least one or more orders of magnitude below the measured ED50 exposure with respect to Minimal Visual Lesion (MVL), which is the smallest clinically detectable injury. Thus, exposure at or below MPE has no statistical expectation of harm, so that MPE is the basis for the Class 1 Accessible Energy Limitation (AEL), such that the corresponding hazard containment volume is called the Nominal Hazard Zone (NHZ). Outside of the NHZ, there is no expectation of greater than Class 1 exposure. With respect to laser light, Food and Drug Administration (FDA) Center for Devices and Radiological Health (CDRH) Notice to the Laser Industry 40 states that "only Class I levels of laser radiation have no known hazard, and these are the only exposure levels that are considered safe for direct exposure of people." This reflects the federal regulator's perspective with respect to outdoor lasing affecting the public, which is relevant for appreciating the "hazard containment" range safety strategy relevance to outdoor lasing hazard.

c. For participants who may need to be within the NHZ, the "hazard containment" strategy can still be practiced in a different form. Consider that 3R level overexposure (i.e., 5 times the AEL of the next lower applicable class or at most <5 times MPE) is considered generally safe as long as direct viewing of the laser or its specular reflections are restricted. The "R" in 3R refers to the "reduction" of required exposure controls such that situations of direct viewing at the 3R level of exposure must still be restricted, but other exposure situations do not require restrictions. For participants within the NHZ, but not exposed to greater than 5 times MPE, a range control that restricts direct viewing in addition to wearing LEP provides the necessary barrier to harm. For example, communication can ensure that those participants are aware of when the lasing is occurring and know (and can be trusted) not to look toward the lasing source. This would generally not be sufficient for ensuring the protection of non-participants, who cannot be assured to have the information and trusted to behave accordingly, especially overcoming the natural curiosity of the public to look at the interesting military platform. Within the Department of

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the Navy, BUMEDINST 6470.23 specifically states that "Class 3a lasers with red danger labels warning against direct eye exposure to the laser beam require all control measures of a Class 3b laser." Because FDA Class 3a with red danger label corresponds to ANSI/IEC Class 3R, this essentially establishes policy for the Navy that Class 3b controls are required for direct viewing situations of Class 3R level laser exposure. With respect to Class II and Class IIIa, FDA CDRH Notice to the Laser Industry 40 states "these levels are hazardous to the eyes. No direct or reflected beams may be directed into audience areas or used to scan the audience in any way." This reflects the consistent perspective of the federal regulators with regards to viewing restrictions for lasing the public at the levels above class 1, which is consistent with the "hazard containment" philosophy.

d. When "hazard containment" is not possible such that vulnerable people, property, and infrastructure, e.g., the surrounding public, that may be overexposed to some hazard, the strategy of last resorts becomes "hazard coexistence," whereby exposure is statistically possible but the risk of harm from such exposure is managed to below levels deemed acceptable by appropriate authority. In this strategy the NHZ is broken into internal volumes wherein the statistically expected exposure may be above the "safety threshold," but the risk of harm below the level deemed acceptable by appropriate risk acceptance authority. Thus, the risk acceptance authority fully and openly accepts the consequences for the statistically possible bad outcome resulting from exposure to this hazard due to operations by the range. The higher fidelity and confidence necessary for meeting such responsibility introduces the question of PRA for supporting this range safety strategy.

e. In the "hazard coexistence" range safety strategy, it is statistically possible to have harm without any fault or failure. In the event of harm, the question is not "What went wrong?" but rather "Was the harm within the acceptable level of loss?" It is therefore imperative that the risk of harm have relevant units, e.g., expected casualty or loss per range event or per duration of some operation, so that the risk acceptance level represents the quantity or level of casualty or loss that is deemed acceptable.

f. Because appropriate risk acceptance authority may be accepting risk on behalf of a population that is not even informed of the risk, no matter how small, that they are being exposed to, the "risk coexistence" strategy may impact the elements forming the "social contract" that the range has with its surrounding community. This social contract is formed through various means, including the National Environmental Protection Act and the resulting Environmental Impact Statement for the range. For this reason, the risk of harm assessment should have sufficient fidelity and confidence, which is where PRA for range safety becomes especially relevant.

9.2.3 Requirements for a range safety PRA. Assuming range safety PRA (RS-PRA) is used to assess the risk of harm associated with statistically possible overexposure to a hazard, i.e., exposure above the hazard's established "safety threshold," then there are in general four core elements that need to be established with some degree of consensus for managing the risk of intentional overexposure associated with the "hazard coexistence" range strategy. These four elements include (1) casualty standards and thresholds, (2) casualty risk of harm model, (3) risk acceptance criteria for ranges, and (4) physics-based range hazard scenario modeling tools. Within the range safety community of the national test ranges, this is not a new concept. "Hazard coexistence" PRA is a current capability for assessing and managing the risk of blunt force trauma from falling debris hazard of airborne incidents via RCC-321 "Risk Criteria for National Test Ranges." This approach relies on the "Expected Casualty" EC consensus risk of harm model along with various physics-based tools for Monte Carlo simulations of the debris hazard striking the ground that also encompass the variability of factors affecting the uncertainty of harm resulting from this form of kinetic hazard. From this capability, a generalization of the fundamental PRA requirements can be derived to form a roadmap to apply PRA for range safety for other hazards.

a. First requirement: There should be a consensus understanding of what damage, injury, or loss constitutes a casualty and the thresholds, doses, and probabilities associated with such casualty. The military definition of casualty is "any person who is lost to the organization…" (see JP 1-02). In the case of the blunt force trauma kinetic hazard of falling debris, a casualty standard based on some Abbreviated Injury Scale (AIS) level 3 corresponding to an average of 11 ft-lb threshold of blunt force provides a straightforward basis for understanding what constitutes casualty. For other hazards, the standard of casualty, damage, or loss, and the exposure to the particular hazard expected to cause such harm should be understood in consensus.

(1) In the case of outdoor lasing, the question of harm is driven by the broader range of biological effects that relate to impairment or even incapacitation as well as any effects with other materials that relate to compromise or some loss of use, capability, or function. With respect to the eye, biological effects of exposure to optical radiation in the visual and near infrared extend from photo-recovery (i.e., glare with transience), dazzle (i.e., glare with persistence), and MVL, on the lesser side, to enlarged lesion, choroidal hemorrhagic lesion, or vitreous

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hemorrhagic lesion (i.e., blood in the vitreous), on the more severe side. What thresholds on size, types, and possible locations (e.g., peripheral retina vs. macula and fovea) of lesions relate to impairment? Is incapacitation a result of some level of impairment in both eyes? What about skin injury? What about damage thresholds to electro-optical sensors? What is the probability of such harm at a given threshold or dose? The casualty standards, thresholds, and probabilities for a particular hazard are a crucial foundation for developing the associated risk of harm model capable of computing risk in appropriate units. Consensus in this first aspect is therefore crucial for later establishing consensus regarding the risk of harm model.

b. Second requirement: There should be a consensus model for the risk of harm for a given hazard from a range perspective. Furthermore, the risk of harm model need to have the appropriate level of fidelity (i.e., include all the necessary key factors affecting uncertainty), so that with the inclusion of variability sufficient confidence relative to the harm being considered exists to competently support risk-based decisions by appropriate authority.

(1) In the blunt force falling debris case of a kinetic hazard, the RCC-321 "Common Risk Criteria for National Test Ranges" establishes a consensus model for the risk of harm known as the "casualty expectation" EC equation. This consensus risk of harm equation gives "the mean number of casualties predicted to occur as a result of an operation if the operation were to be repeated many times." The formula explicitly relates the key factors characterizing the conditions of exposure to the likelihood of harm from exposure. An equivalent risk of harm model for outdoor laser exposure might be more complex given the different mechanisms of harm, directedness of the energy, and viewing conditions that all contribute to the expectation of harm for some level of exposure. For the kinetic hazard of falling debris, the factors affecting uncertainty include how the force is applied, sheltering factors, etc.

(2) In the case of laser exposure, the apertures, angle of incidence, viewing conditions (e.g., unaided or using aided optics), as well as the irradiance and directedness of the energy (e.g., effective point or extended source origination) are among the factors that may affect the uncertainty of exposure, and be possible elements of a risk of harm equation. If multiple lasers are used, the different characteristics, effects, and corresponding uncertainties of each should be considered. In this sense, the risk of harm model brings the key factors of the potential exposure situation together with the understanding and expectations of the mechanics of harm from the first requirement. This is why this second requirement along with the first requirement are closely intertwined, especially in a consensus sense.

c. Third requirement: There needs to be risk criteria as a basis for acceptable loss due to the hazardous situation or event on the range. Such criteria should include acceptable individual risk, acceptable collective risk, and acceptable cumulative (annual) risk thresholds. Individual risk is the risk of harm that an individual is subject to by virtue of being within the hazard volume (i.e., the expectation of harm to an individual). There may even be different individual risk thresholds for participants, non-participants, neighboring operations, etc. The collective risk, also called the societal risk, is the risk of harm that society is subjected to, such as the number of people expected to suffer harm. Cumulative risk applies when the same population (i.e., a range and its neighboring community) are repeatedly exposed to potential hazard in multiple range missions employing the "hazard coexistence" strategy. Within a given annual time period, the maximum acceptable cumulative risk is like a pie that each range mission employing the "hazard coexistence" strategy consumes a piece of. When the cumulative risk threshold is reached (i.e., the risk pie is consumed), the "hazard coexistence" strategy is no longer acceptable, regardless of the individual and collective risk levels. For each of these risk categories there should be a consensus basis for such criteria based on considerations that are defensible and reasonable from which competent risk acceptance authority can refer and adapt to specific situations as needed. Acceptable loss criteria that are arbitrary or capricious may appear especially negligent in the statistically possible occurrence of harm during a range event.

(1) In the case of the kinetic hazard of falling debris, the basis for loss is casualty or death, so the acceptable risk (or acceptable loss) criteria are stated in quantities of casualties or deaths per mission or per duration of some operation, typically expressed in terms of $1E\hat{} - 7 = 10\hat{} - 7 = 1$ in ten million. For the case of outdoor lasing hazard, the characterization of harm from the first requirement should help to drive the understanding of the level of loss per lasing event or lasing duration that is operationally considered acceptable.

d. Fourth requirement: There needs to be a set of physics-based modeling and simulation (M&S) tools for (1) applying or propagating the particular hazard through a representation of the physical range environment, (2) then apply the risk of harm model to all potential people, property, and infrastructure of concern within the environment to compute the individual, collective, and cumulative risks, and (3) show the comparison with acceptable risks for either representative demographics of people, property, and infrastructure (i.e. mission planning and preparation), or

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in real-time with information of actual people, property, and infrastructure within the range environment (i.e., mission execution support).

(1) For the falling debris kinetic hazard, as an example, this M&S toolset includes aero-modeling tools for characterizing the possible trajectories from a malfunction at any point along the flight path, as well as tools for building the debris catalog associated with a particular type of malfunction and flight body. Then there is a physics-based tool that models all the possible locations' debris may strike the ground to compute the impacting kinetic energy against the casualty threshold to create a contour representation of the likelihood of casualty anywhere within the range environment. Overlaying this with the potential people, property, and infrastructure can then reveal the risks and where risks exceed the acceptable levels of loss.

(2) For the outdoor lasing hazard, a physics-based modeling of the lasing environment is necessary. Such modeling should include relevant aspects of the environment that can scatter, reflect, refract, absorb, and diffract the laser energy as it relates to direct exposure to laser emissions (both primary and secondary emissions, if present), down-range exposure to energy spilling past or through the intended target(s), and to indirect exposure. This also depends on characterizing how well the directedness of the laser energy is preserved throughout the lasing environment. Then the risk of harm model that likely combines aspects of viewing conditions (e.g., aided optics), exposure conditions with respect to potential damage mechanisms, should be applied with respect to people, property, and infrastructure within the lasing environment to assess the expected individual, collective, and cumulative risk from the lasing event.

(3) Because none of these four elements currently exist in a consensus form for the outdoor lasing hazard, the "hazard coexistence" range safety strategy and thus PRA for outdoor lasing remains a capability in development.

9.2.4 Range safety analyst's perspective of PRA. Assuming that the four critical requirements above have been developed and the PRA capability for outdoor lasing supporting the "hazard coexistence" range safety strategy exists, the perspective of the range safety analyst can be understood in terms of four major groups of tasks:

1. Detailed range situation characterization.
2. Detailed characterization of factors within the risk model including variabilities.
3. Simulation of hazard and application of risk model.
4. Establishment of range controls to manage risk of harm to within acceptable losses.

a. First, there is the detailed characterization of the intended range lasing environment. This may involve the measurement of any reflecting surfaces and objects, as well as the mapping of all materials with which the laser energy may interact. Since "hazard coexistence" is generally applicable when "hazard containment" is not practicable, this likely includes higher power laser energies for which even diffuse reflections may be harmful and for which there may be complex interactions, including damage and scattering, with certain materials. The Laser Safety Range Review Assessors (LSRRA), as the range assessment authorities, should be assisting ranges in this effort as part of range assessments supporting range certifications.

b. Second, for a given outdoor lasing situation, the specific characterizations of the laser(s), which should generally be performed by the Laser Safety Review Coordinators (LSRCs), should be understood, so that the use of the laser in-situ can be considered for all aspects of operational variabilities. This may include variation on the lasing platform and target(s) geometries relative to each other, accounting of the pointing error of the laser, lasing timing and duration variation, etc. If lasing platform or target(s) are moving, the uncertainty in path and location becomes part of this consideration. The aspects of the situation along with the variability of each as they relate to the factors in the specific consensus risk of harm model needs to be defined. This sets the stage for the third task group.

c. Third, physics-based modeling tools are employed to combine the characterization of the lasing environment with the specific lasing hazard and with the risk of harm model to enable the propagation of the hazard through the environment to assess the potential risk of harm within the NHZ. By combining this with an expected (or actual) demographic of potentially vulnerable people, property, or infrastructure, and viewing conditions, the individual risk, collective risk, and cumulative risk can be assessed. The physical modeling involved will need to consider how the laser energy as both irradiance and directionality (phase) changes with material and propagating medium interactions to result in specific hazards at any location within the NHZ. Accounting for the various variances of the factors affecting uncertainty may involve a very large number of Monte Carlo runs to assess both risk and the confidence with sufficient fidelity to support safety decisions.

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d. Fourth, the appropriate risk management measures in the form of executable range controls, barriers, and mitigations should be applied to ensure risk everywhere in the lasing environment is maintained below the appropriate acceptable levels. This includes clearance of unacceptable risk areas and surveillance of all volumes under risk management control, which may include restrictions on who, what, and how many may be in a particular volume during a given lasing event. If real-time demographic information is available, the real-time perspective of individual, collective, and cumulative risks may be accessible to manage risk dynamically during mission execution.

9.2.5 Range safety PRA for outdoor lasing path forward.

a. Although the four critical requirements for the outdoor lasing PRA capability supporting "hazard coexistence" are not yet developed, there is the basis for a path forward to establishing such a range safety capability. DoDI 6055.15, dated August 25, 2023, took the first step critical in this direction by placing the onus on the Laser Safety Review Authority (LSRA) community across the military departments (MILDEP) to figure this out: "LSRAs will establish thresholds for acceptable risk of exposure to levels above MPE." The only way "thresholds for acceptable risk" for overexposure can be established, which is essentially the third requirement above, is if first and second requirements have been addressed. Thus, this is a call for the LSRAs to support and help drive the addressing of the first three requirements above.

b. The shift from a laser safety philosophy centered on "safety thresholds" that support the "hazard containment" range safety strategy to thresholds related to harm in a casualty or loss sense, is significant. This includes a shift from studies focused on MVL and ED50 to considerations of harm related to impairment, incapacitation, and tangible loss. Combined with the development of the associated risk of harm model, this will require the involvement of the entire DoD laser safety community in a new consensus building activity with the necessary verification and validation of any standards and models established. This will include the LSRCs across the MILDEP. The subsequent development and validation of the physics-based tools for combining the risk of harm models with the characteristics of the lasing environment will likely involve the LSRRAs as well as direct range input. Much in the way the MIL-HDBK 828 is a product of the DoD LSSWG as part of the Defense Safety Oversight Council (DSOC) governance structure, it would be reasonable to expect that any such PRA capability will also be guided by this working group and governance structure to ensure consistency across DoD.

c. In addition to DoD, NATO has also been working on this problem. Most notably, the NATO SCI-316-RTG has suggested an intermediate path forward by adding further realism into the current deterministic risk assessment approach used for "hazard containment". For the deterministic analysis of reflection hazards, the Reflected Nominal Ocular Hazard Distance (RNOHD) changes significantly with increasing realism of target shape and reflecting properties. More generally, this group has put forward the following recommendations: revised system parameters supported by appropriate engineering controls; revised atmospheric coefficient based on weather conditions; revised exposure times; and revised exposure thresholds. Including actual atmospheric attenuation reduces hazard distances to more realistic values. Considering the actual lasing times as opposed to the 10-second standard for hazard computation might lessen the hazard extent. Taking a closer look at the "safety thresholds" based on more specific laser characteristics and use-case scenarios may reveal where the conservativeness of the current "safety threshold" may be safely relaxed. Sharpening the pencil in this way means developing guidance that may be more context specific and less general, such that the specific conditions and assumptions need to be more carefully considered before applying such guidance.

d. Thus, the intermediate solution is less about changing the paradigm to "hazard coexistence," and more about sharpening the pencil to do "hazard containment" more effectively or efficiently, to result in potentially smaller containment volumes.

e. In addition to the technical challenges, the philosophy of "hazard coexistence" for outdoor lasing represents a policy shift that needs to be formally embraced. DoD outdoor lasing safety involving "hazard containment" above MPE IAW policy has been consistent with ANSI Z136.6 guidance and Laser Light Show guidance of FDA Laser Notice 40 thus far. With respect to laser light shows, FDA policy is that the "Center for Device and Radiological Health (CDRH) will object to any intentional exposure of the public to hazardous levels (i.e., greater than Class 1 levels) of laser or collateral radiation (light)" (FDA CDRH Notice to the Laser Industry 40).

f. Within the Department of the Navy, which includes both the Navy and Marine Corps services, "DON policy is also to ensure that personnel are not exposed to laser radiation in excess of the applicable MPE throughout the life cycle of laser systems." (OPNAVINST 5100.27B) The Department of the Air Force, which includes both the Air Force and Space Force services, "Personnel will not be intentionally exposed to laser radiation in excess of

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the maximum permissible exposure (MPE) levels." (AFI 48-139) Clearly, various policies will likely need updating to allow for "hazard coexistence" for outdoor lasing.

g. More generally, public law requires "wherever and whenever human access to laser radiation levels that exceed the limits of Class I is necessary, these levels shall not exceed the limits of the lowest class necessary to perform the intended function(s) of the product" (21 CFR 1040.10, 1040.11). Thus, the shift to an intentional overexposure along with using a "hazard coexistence" range safety strategy, puts a potentially larger demand on the LSRA community. The LSRAs should continue to manage the application of the military exemption such that exempted products meet "to the extent practicable, the radiation safety provisions of the applicable Federal standard unless adequate alternative controls are to be authorized," and ensure the minimum alternative controls are consistent with overexposure to lowest levels necessary to perform the intended function. The minimum alternative controls are covered in MIL-STD-1425 (series), which applies to ALL military exempted lasers.

h. A condition of the current military exemption (see 76EL-01DOD) granted to DoD by the director, CDRH at the request of the Assistant Secretary of Defense for Readiness ASD(R) as of 2024, is that "Any substantive amendments to the radiation safety procedures found in DoD Instruction 6055.15 or other DoD-established laser safety policy or requirements will be submitted to the FDA for review." Consequently, moving forward any policy changes necessary to support this new "hazard coexistence" strategy for optical radiation may require review if deemed "substantive" by either the CDRH or the ASD(R). As this change in philosophy is navigated, it is important to remember that FDA is the Federal regulator for laser safety and has unilateral authority to revoke or modify the conditions of the "military exemption" (76EL-01DOD) at any time.

i. This all points to a path forward that should be undertaken as a DoD-wide initiative and not at an individual range safety level to develop and establish, even though the result will be an implementation that needs to work and be consistently applicable at the individual ranges.

10. RANGE CERTIFICATION

Prior to any laser range operations, the hazards of laser activities on the range should be identified and incorporated in a laser range certification report. These hazards are identified by a combination of an on-site survey and follow-up analysis of the results. The certification report should include recommendations to mitigate any potential hazards and provide guidance for safe use of the lasers on the range. Laser range certification reports are valid for a period based on DoD service policy. Interim laser certifications may be required if revisions or modifications affecting laser operations occur. The renewal of the laser range certification should be scheduled prior to the expiration of the current certification.

Based on service policy, a laser range certification can be performed for a specific laser system or for a group of lasers with similar characteristics such as Hazard Classes, NOHD, and platform stability. To perform this general certification, the worst-case conditions of all possible systems and missions are used. If these conditions are too restrictive a separate certification for each system should be performed.

10.1 Preparation for certification.

10.1.1 Certification questionnaire. The Institutional Laser Range Authority, in conjunction with the Installation Laser Range Authority, assembles materials needed and reviews this information in to facilitate the certification. The Installation Laser Range Authority should use the applicable service's Laser Range Pre-survey Questionnaire and existing Range Regulations/SOP(s) to provide the Institutional Laser Range Authority with the information necessary to perform the certification. The Appendix B sample questionnaire provides a systematic approach toward preparing for the certification and identifying areas that require additional consideration.

10.1.2 Scheduling. The Installation Laser Range Authority should determine a timeframe for the range evaluation after coordination with the prospective Institutional Laser Range Authority while considering the laser range certification due date. It is suggested that this initial coordination be done a minimum of six months prior to the visit.

10.2 Certification process.

A new laser range may be certified for the first time, or an existing range may be recertified. Once preliminary data has been gathered and reviewed, the Institutional Laser Range Authority performs the certification. The Institutional Laser Range Authority may document the results of collected data on the Laser Range Certification Questionnaire (see Appendix B). As an example, the certification process may proceed as follows:

a. Contact and scheduling

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b. Gather preliminary data

c. Preliminary analysis

d. Perform survey

e. Post survey analysis

f. Report

g. Archiving and distribution

10.3 Gather and review preliminary data. Prior to a site visit, the Institutional Laser Range Authority should gather and review preliminary data provided by the Installation Laser Range Authority or other applicable sources. Preliminary data includes the information in the following items a through f:

a. Range Information

(1) Range owner

(2) Name/identification of each range requiring certification

(3) General description/location of the range to be certified

(4) Baseline range data provided by the range management to include installation SOPs

(5) Detailed maps to include imagery and airspace

(a) Range map - Range maps are essential to show authorized laser firing locations, firing lanes, firing areas, target locations, backstops, populated areas, public roads, and no lase areas. Range maps should include boundaries (lateral and vertical), declination angles, and geographic items such as towers and buildings. The range maps should be assembled using both the Geographical Information System (GIS) and GPS data to ensure accuracy. The datum/projection of the GIS data should be World Geodetic System 1984, Universal Transverse Mercator.

(b) Elevation Data/Models - These usually provide a more accurate representation of elevation compared to range maps due to the increased accuracy of specific elevation data/models. Range maps do not usually have accurate elevation data. As an example, a standard range map is compiled using either 1:50,000 or 1:100,000 base data compared to a United States Geological Survey (USGS) 1:24,000 map. USGS 1:24,000 maps, without an elevation model, can be used when elevation data is of the same accuracy (as the horizontal accuracy). However, elevation data/models, to include GIS models, are preferred to analyze as opposed to using contour intervals and spot heights on a USGS image if the vertical accuracies are equivalent.

(c) Digital Nautical Charts/Navigation Maps - These maps show restricted areas, sea bottom depth, points of interests, and navigation routes on surface and underwater ranges.

(d) Airspace Map - Airspace maps/charts showing restricted airspace, flight paths, laser flight profiles, aircraft orbit points, and no lase areas are required. Controlled airspace is that airspace associated with range-specific characteristics, to include possible non-coincident lateral boundaries and minimum and maximum altitudes. It is important that this controlled airspace, and any other special conditions, are made known. Laser operations are not normally authorized outside the controlled airspace or when other aircraft are between the laser and the target. In addition, if the beam is directed up or above the horizon, additional controls may be necessary, including notification to the FAA and Laser Clearinghouse (LCH).

(e) Recent aerial photos or commercially available satellite imagery are reviewed to look at vegetation, backstops, roads, waterways, and other environmental and man-made features.

(6) Previous laser range certification should be reviewed

(7) Previously determined LSDZs, if available

b. Laser information should be gathered to include:

(1) Description of the lasers authorized to be employed on ranges. Only service-approved lasers are authorized for use on a range.

(2) Engineering control measures

(3) Laser Classification

(4) NOHD (aided and unaided) for laser systems evaluated

(5) Maximum buffer angles by allowable systems to be used on each particular target and range.

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(6) Optical density requirements. Optical density is the amount of protection required in PPE to reduce the incident laser energy to safe eye levels for a particular laser.

(7) Technical orders, technical manuals, SOPs, and reports on the laser system and associated hazards, to include laser systems' parameters and hazardous failure modes that affect laser parameters or beam steering, secondary beams, inadvertent firing, and other potential system problems.

c. Deconfliction of range activities - A description of weapons authorized to be employed on ranges, surface dangers zones, and weapons danger zones per SOP / Installation Laser Range Authority. Determine which Observations Points are used for which ranges, and whether what sites could conflict with lasing operations. This information will be useful for deconfliction / risk mitigation purposes to ensure locations of ground personnel and laser operators are not in conflict with WDZs. Though the laser range certifier is primarily concerned with laser features, hazards, and constraints in the range area, the certifier should understand the effects of changes that the recommendations may make to existing ranges; for example, changing lateral limits to contain laser energy might conflict with other activities that occur on the range. The integration of all the weapon systems and activities occurring on the ranges cannot be overstated.

d. Training information

(1) Intended operational environment for laser use, including types of targets and position, laser firing locations, run-in headings, maximum and minimum firing altitudes and ranges, aircraft positioning and profiles, direction of laser operations, and any other special considerations.

(2) The desired buffer angle for the desired training based on the laser platform.

e. SOP should be comprehensive, up to date, and address all aspects of range safety and operations.

f. Areas of concern. Any areas of concern, either by the Range Manager or the installation commander, need to be addressed, such as changes to targets, target areas, backstops, environmental changes, encroachment issues, or any other areas that may interfere with efficient range operations.

10.4 Perform preliminary data analysis using software/analysis models. Prior to a site visit, a preliminary analysis of the range area should be performed. The purpose of the preliminary analysis is to identify and define potential problem areas and to maximize effectiveness of on-site survey and allocation of resources. The Institutional Laser Range Authority's preliminary analysis may include such items as LSDZ(s), firing areas, target areas, buffer angles, SOPs, requirement for tree lines and backstops. The preliminary analysis can be accomplished manually or with software driven methods.

GIS software is a common medium when depicting LSDZs during the laser range certification process. During preliminary analysis, comprehensive GIS data mining is essential prior to the survey. The certifier should obtain GIS data that includes feature classes, geodatabases, constraint layers, map projections, lasing / target areas, metadata, map datum, digital elevation model accuracy and currency, and file formats as part of the preliminary data. In addition, the data gathering will assist the certifier to identify data shortfalls that may be filled by the Installation GIS analyst prior to the survey analysis.

10.5 On-site survey. To perform a laser range certification, it is essential that a thorough range survey be performed. Range facilities are evaluated in terms of location relative to populated areas, military and civilian industrial sites, water / surface / air traffic, and other constraints, and considers specific environmental factors such as wildlife habitat. The range survey is used to verify geometries of the range boundaries with regards to firing and target positions and verify the validity of SOPs. An on-site inspection is performed to verify these geometries and to inspect the areas in question to verify the lack of specular reflectors and other terrain issues. This inspection might require visual access to the entire LTA. The results of this survey provide the certifying authority the conditions for which laser use is acceptable. It also provides recommendations as to what safety control measures should be implemented.

a. Survey procedures and in-brief - To perform the laser range survey, the certifier should schedule an in-brief with the host installation range control. At a minimum, the in-brief discussion should include a review of the last laser range certification, range personnel requirements for supporting the survey, installation range SOPs, range boundaries, airspace restrictions, lateral limits, laser target areas, range points of interest, magnetic declination, hazard clearances, and new ranges that will require certification since last certification report. In addition, any areas of concern by range control should be identified during this time.

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b. Conduct the survey - During the survey, the certifier should be escorted by an installation range control representative with a comprehensive knowledge of the laser range requirements and the integration of other range activities. During the survey, the certifier will gather data that will be used for the post-survey analysis using mapping-grade accuracy GPS systems, range-finding and azimuth equipment, dispersion-measuring optics (in milliradians), camera(s), and a weather-proof writing tablet for notetaking. At each range the certifier is also looking for man-made and natural backstops, specular reflectors, the presence of laser warning signs, and possible laser hazards. For the laser survey of Infantry Immersion Trainer, Marine Operations Urban Terrain facilities, and shoot houses, many of the same principles, procedures, and observations remain the same with additional attention to specular hazards (such as windows, picture frames, vehicle mirrors), man-made backstops (berms, buildings, walls), and direction of laser fire.

c. Out brief - Once the survey is complete, an Out Brief should be scheduled between the certifier, range control, and any other representatives identified by the Installation Laser Range Authority. Discussion items should include unsafe laser range activities that require immediate attention (if any), changes/deviations from last laser range certification, requested installation GIS data/information to complete the analysis and report, and approximate target date of written report.

10.6 Survey analysis. Including the analysis performed during the Preliminary Analysis (see 10.4), the laser range certifier conducts and consolidates the Survey Analysis using all laser range data and information to create the LSDZs. For the analysis, the certifier will use manual calculations or laser range certification software to create the surface (ground) and aerial LSDZs. Based on the collected survey data, the certifier will input data parameters such as buffer angles, firing points, target points, maneuver areas, azimuths, firing distances, NOHDs, magnetic declination, distances to the target area, lasing altitudes, lateral limits, constraints, tree line size and composition (if required), and target heights (offsets). The LSDZ outputs include a danger zone bounded by firing lines, target lines, lateral limits, the intersection of terrain with the laser energy (or limited by a specific NOHD or an identified constraint), and minimum safe lasing altitudes (for the aerial LSDZs). Once the LSDZ is created, the certifier will verify the LSDZ input parameters for accuracy; once verified, the LSDZ will become part of the laser range certification report.

10.7 Report. Once the LSDZs are completed, the certifier will place the LSDZs in a service-specific layout format and add supporting graphics, data, and text information as required into the report to complete the analysis for a particular range. Including the LSDZ layouts, a report should contain information addressing the date of the survey, range control representative(s) who escorted the laser range certifiers, the date of the last laser range certification, the magnetic declination, date of expiration for new laser range certification, restricted airspace consideration, buffer angles, minimum safe lasing altitudes for aerial ranges, offsets (heights) assigned to the firing and target points, laser systems and associated limitations that may be used on certain ranges depending upon calculated laser range buffers, laser range restrictions, required changes to the installation SOP, and changes from the last laser range certification report. Many of the software tools have report functions that can auto-populate the report template and present information such as firing points, target points, offsets, buffer angles elevation, analysis type, target type, NOHD, ground distances, and minimum safe lasing altitudes. Once the report is complete, it is recommended that the certifier send a draft report to the installation range control that allows a local review to ensure accuracy and completeness.

10.8 Distribution and archiving. Once the report is signed by the Institutional Laser Range Authority, the certifier will distribute the report via an approved distribution list. At a minimum, the report is archived by the Institutional Laser Range Authority. Respective services may use a website to allow access for distribution as approved organizations can download the laser range certification report.

11. LASER ACCIDENT/INCIDENT REPORTING

Report all suspected laser accidents/incidents, regardless of injury, in accordance with AR 385-10, DA PAM 385-10, DA PAM 385-40, TB MED 524, OPNAVINST 5100.27/MCO 5104.1, AR 385-63/MCO 3570.1, BUMEDINST 6470.23, AFI 13-212, and AFI 48-139. Pertinent medical guidance for such emergencies is available from the Walter Reed Army Institute of Research Detachment at Fort Sam Houston or through the DoD Laser Safety Event at 1-800-473-3549. The expeditious examination and treatment of laser eye injuries is critical in minimizing loss of visual acuity.

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12. NOTES

12.1 Intended use. The contents of this handbook are intended to serve as a guide to the safe use of lasers and laser systems used on military reservations and in military controlled areas.

12.2 Subject term (key word) listing.

Direct intrabeam viewing
Flash blindness
Laser guided weapon
Laser operation
Range operations
Reflected intrabeam viewing
Scattered radiation
Skin damage

12.3 References. The following references are not listed within the body of this document; however, these documents play an important role in Laser Systems application and Range certification: JP 3-09.3, AFI 91-401, AR 40-5, MCO 3550.9, and MCO P3550.10.

12.4 International standardization agreement implementation. This handbook implements STANAG 3606, Laser Safety for Military Use and ARSP-4, Laser Safety for Military Use. When changes to, revision, or cancellation of this handbook are proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

12.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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LASER SAFETY SOP

A.1 SCOPE

A.1.1 Scope. This appendix provides information that should be contained within a laser safety SOP and unit responsibilities for compliance with laser safety SOPs.

A.2 INSTALLATION LASER RANGE AUTHORITY SOP RESPONSIBILITIES

A.2.1 Laser range SOP. The following items should be addressed in a laser safety SOP. Installations should prioritize these and other items according to their local requirements and situation. This list is not intended to be all encompassing but should meet the mandates of service orders and directives. Additional or more restrictive conditions may be appropriate. The laser safety SOP should:

a. Provide a schedule for checking target location and environment. Some concerns to be noted are crater holes, dislodged or moved targets from surveyed location, target debris that exposes specular reflectivity, or appropriate backstop for systems employed to contain all laser activity.

b. Delineate Installation Laser Range Authority responsibilities and those of the unit.

c. Provide communication frequencies for training units and cease lasing operations if communications are lost.

d. Coordinate laser codes for multiple users to ensure deconfliction.

e. Require briefings for the unit laser range authorities, including the range description, approved laser systems, targets, firing points, or movement boxes (ground, airborne, or naval).

f. Have provisions for ensuring personnel adjacent to laser activity are aware of the laser activities.

g. Have provisions for ensuring all ranges are marked with appropriate signage with particular attention paid to all access points/areas.

h. PPE maintenance program in place have provisions for designating an assembly/marshaling area that is clear of the laser training area during laser events.

i. Detail the procedures for scheduling laser activity through the common range scheduling system and log all events within it.

j. Specify appropriate PPE for the laser systems being employed.

k. Detail emergency response actions for laser injuries, including medical response and 24-hour hotline (1-800-473-3549 DoD Laser Safety Event Hotline).

l. Have provisions for unit laser safety training.

m. Incorporate any other laser guidance specific to your installation.

n. Target Maintenance Program.

A.3 UNIT SOP RESPONSIBILITIES

A.3.1 Unit laser range SOP. Units should have their own laser safety procedures and brief their personnel prior to conducting laser training. Units are responsible for complying with the installation guidance and directives contained within the laser safety SOP. Responsibilities include the following:

a. All units conducting laser training should provide to the Installation Laser Range Authority an assigned LSO in writing to direct the safe use of lasers, ensure range regulations, the range SOP, and Installation Laser Range Authority directives are complied with. This appointment letter should be provided to Installation Laser Range Authority prior to any laser training.

b. The Unit Laser Range Authority receives a Range Laser Safety Brief from the Installation Laser Range Authority prior to conducting laser operations on the range. This briefing should review laser operations from entry on the range until exit, with specific details for any installation concerns or lessons learned.

c. Ensure only systems approved by the LSRB and reviewed by the installation laser range authority are authorized for use.

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d. Ensure the laser system(s) employed meets the specifications of the range certification, including allowable buffer angle, NOHD, MPE, and platform constraints, activated only from the specified location at the specified target.

e. Provide a log for the laser operations to the Installation Laser Range Authority to be recorded on the common range scheduling system schedule and retained for reference.

f. Conduct a thorough inspection of the range prior to training to ensure the range is clear of equipment and nonessential personnel, and that those present are informed of the laser operations and possess appropriate PPE.

g. Confirm no specular reflectors are associated within the Training Areas/LTA/LSDZ prior to use.

h. Ensure unit assembly/marshaling areas remain clear of the laser training area while lasers are in use.

i. Cease lasing immediately if unauthorized personnel/equipment enter the LSDZ.

j. When lasing, always point the laser down range toward the target/impact area.

k. Ensure personnel forward of the laser are wearing PPE as required; ensure unprotected personnel remain behind the laser operator.

l. Ensure lasers are not activated until the target has been positively identified by the laser operator.

m. Ensure all personnel involved with laser operations know and understand LSDZ and laser activity boundaries.

n. Ensure all safety mechanisms, switches, and covers are used until the laser operator is prepared to engage the target and activate the laser.

o. Cease lasing if communications are lost with the Installation Laser Range Authority, or with any person involved with the laser operations. Ensure lasing does not resume until communications are re-established.

p. Ensure all lasers are safe prior to departing the laser training area using all available measures that could allow the laser to be activated accidentally.

q. Review MIL-HDBK-828 for comprehensive understanding of laser use on outdoor ranges.

r. Ensure personnel are knowledgeable of all laser range restrictions and laser operations on the range.

s. Ensure personnel are briefed on emergency response actions for lasers, including the emergency response hotline 24 hours per day (1-800-473-3549 Laser Safety Event Hotline).

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EXAMPLE LASER RANGE CERTIFICATION QUESTIONNAIRE

B.1 SCOPE

B.1.1 Scope. This appendix provides an example Laser Range Safety Pre-Survey Questionnaire, a Laser Range Certification Survey Questionnaire, and a Laser Range Safety Certification Report. This questionnaire may be used by tailoring or adding items as needed.

B.1.2 Laser range pre-survey questionnaire. The Installation Laser Range Authority should use a Laser Range Pre-Survey Questionnaire (see figure B-1) to provide the Institutional Laser Range Authority with the information necessary to perform the laser range safety certification survey.

B.1.3 Laser range certification questionnaire. The Institutional Laser Range Authority annotates the data gathered during the certification survey on the Laser Range Certification Questionnaire (see figure B-2).

B.1.4 Laser range safety certification report. The Institutional Laser Range Authority compiles the results of the laser range safety certification in a Laser Range Safety Certification Report (see figure B-3).

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Laser Range Pre-Survey Questionnaire

Range/Area Name: _____ Date: _____

Location (Grid Coordinates): _____

Address: _____

Range Owner: _____

Planned Survey Date: _____

Last Survey Date: _____

Phone (DSN) _____ Performed by: _____

Comm: _____ Range POC: _____

User POCs: _____

Description

What is the restricted airspace? _____

What are the range boundaries? _____

What are the range's laser target areas? _____

What are the range's firing points? _____

What are the range's firing lines? _____

What are the range's firing areas? _____

FIGURE B-1. Laser range pre-survey questionnaire.

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What are the range's target lines? _____

_____

_____

What are the range's point targets? _____

_____

_____

_____

What are the ranges points of interest? _____

_____

What is the buffer angle desired by the range? _____

_____

What platforms are approved for ground operations? _____

_____

_____

Data Collection

Documents:

Range SOP: _____

Range Laser Directives: _____

Old Survey Report: _____

Maps:

Range Boundaries: _____

Topography: _____

Restricted Air Space: _____

Target Locations: _____

Laser Operating Locations: _____

FIGURE B-1. Laser range pre-survey questionnaire – Continued.

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Types of Laser Operations:

Airborne: _____

Ground-based: _____

Restricted Air Space: _____

Laser Systems to be used on the range:

_____

_____

_____

	Target Name	Grid Coordinates
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

Laser operator firing positions for Target #	Grid coordinates
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
9.	
10.	

Forward Observer positions for Target # Laser #	Grid coordinates
1.	
2.	
3.	
4.	
5.	

FIGURE B-1. Laser range pre-survey questionnaire – Continued.

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6.		
7.		
8.		
9.		
10.		

1. Does the range have established run-in headings for aircraft? Yes _____ No _____

If yes, what are they?

2. Will more targets be added? Yes _____ No _____

If yes, where (grid coordinates)?

3. Are there manned positions on the range? Yes _____ No _____

If yes, where (grid coordinates)?

4. Describe the surveillance of the range.

5. Are there any conditions off the range that need to be addressed? Yes _____ No _____

If yes, what are they?

6. Any other changes?

FIGURE B-1. Laser range pre-survey questionnaire – Continued.

7. Comments?

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FIGURE B-1. Laser range pre-survey questionnaire – Continued.

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Laser Range Certification Questionnaire

References: a) _____
b) _____

Location (Installation): _____

Facility (Official Range Name): _____

Type of Range: _____

Range Owner: _____

Date of next onsite survey: _____

Date next onsite survey is due: _____

Inspected by:

Last name and initials	Command (Code):	Phone number:

Format of data collected (e.g., WGS 84, NAD 27): _____

Declination angle: _____

Documents referenced for this survey: _____

Laser Range Safety Officer:

Name	Address	Phone

FIGURE B-2. Laser range certification questionnaire.

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Standard Operating Procedure (SOP)			
1. Is there a current, signed SOP for the range?	Yes ____	No ____	N/A ____
2. Are emergency procedures outlined in the SOP pertaining to personnel injury?	Yes ____	No ____	N/A ____
3. Are there procedures outlined in the SOP for the clearing and inspecting of lasers prior to departing the range?	Yes ____	No ____	N/A ____
4. Does the SOP indicate coordinated laser codes for designators and seekers?	Yes ____	No ____	N/A ____
5. Does the SOP indicate permissible aircraft flight profiles and run-in headings for specified targets or target areas?	Yes ____	No ____	N/A ____
6. Does the SOP indicate permissible ship headings and safe firing zones for specified targets or target areas?	Yes ____	No ____	N/A ____
7. Does the SOP indicate permissible ground-based laser operating positions and/or areas for specified targets or target areas?	Yes ____	No ____	N/A ____
8. Does the SOP indicate hazard areas to be cleared of non-operating personnel (road blocks, if required)?	Yes ____	No ____	N/A ____
9. Does the SOP indicate operating personnel locations (including those requiring eye protection)?	Yes ____	No ____	N/A ____
10. Are surveillance procedures delineated in the SOP to ensure a clear range?	Yes ____	No ____	N/A ____
11. Does the SOP indicate radio frequencies for communication?	Yes ____	No ____	N/A ____
12. Does the SOP indicate that laser systems will not be activated until the target has been positively identified?	Yes ____	No ____	N/A ____
13. Does the SOP indicate that Class 3 and 4 lasers shall not be directed above the horizon unless coordination has been completed with DoD components?	Yes ____	No ____	N/A ____
14. Does the SOP indicate that, for ground-based lasers, all unprotected personnel must remain behind the laser safety zone?	Yes	No	N/A

FIGURE B-2. Laser range certification questionnaire – Continued.

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Personnel Responsibilities						
1. Have all personnel authorized to check out a range or training area received the Unit Laser Range Authority class from the Installation Laser Range Authority?	Yes		No		N/A	
2. Has the Installation Laser Range Authority maintained records of qualified Unit Laser Range Authority?	Yes		No		N/A	
3. Are there measures in place to ensure the Unit Laser Range Authority is qualified/certified and knowledgeable of the lasers being fired on the range?	Yes		No		N/A	
4. Is the Unit Laser Range Authority appointed in writing by the commanding officer?	Yes		No		N/A	
5. Does the Installation Laser Range Authority supply the Unit Laser Range Authority with specific check-in and check-out procedures for each range?	Yes		No		N/A	
6. Is it routine policy for the Unit Laser Range Authority to conduct a reading of the range safety rules in detail prior to the range exercise?	Yes		No		N/A	
7. Are there procedures in place to ensure the Unit Laser Range Authority is briefed on changes to policies and procedures since they attended installation laser training?	Yes		No		N/A	
8. Does the Installation Laser Range Authority review all waiver/special event requests that do not comply with installation regulations?	Yes		No		N/A	
9. Are all of the personnel who must be on the range during laser operations equipped with the proper eye protection?	Yes		No		N/A	
10. Have all of the range personnel involved with laser operations had laser safety training?	Yes		No		N/A	

Administrative			
1. Are laser warning signs posted along the perimeter of the LSDZ (in accordance with MIL-HDBK-828D) to warn personnel against unauthorized entry?	Yes	No	N/A

FIGURE B-2. Laser range certification questionnaire – Continued.

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Laser Range Safety Certification Report

Note: This report may require sign-off by the Service Laser Safety Authority.

Range Area/Name: _____

Survey Summary

Date of onsite survey: _____

Date when next on-site survey is due: _____

Applicable Regulations: _____

_____

_____

Range controlled by: _____

Range owner: _____

Survey completed by (Name/Organization): _____

_____

Date of operations for which survey is valid: _____

Format of data collected (e.g., WGS 84, NAD 27): _____

Declination angle: _____

Documents referenced for this survey: _____

_____

_____

_____

_____

Other pertinent information: _____

_____

_____

_____

FIGURE B-3. Laser range safety certification report.

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Survey

Who performed this survey? _____

What are their titles? _____

Who else was involved in this survey? _____

_____

Any adverse conditions? _____

_____

_____

Evaluation

What is the buffer angle desired by the range? _____

_____

Survey Results

1. Degree of compliance with applicable regulations:

_____

_____

_____

_____

_____

_____

2. Safety deficiencies that must be corrected before approving range for laser use:

_____

_____

_____

_____

_____

_____

3. What are the range's desired run-in headings for aircraft?

_____

_____

_____

_____

_____

_____

FIGURE B-3. Laser range safety certification report – Continued.

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Recommended Actions

1. Corrective actions for existing deficiencies:

2. Ground Laser Restrictions - Description of Laser Surface Danger Zones (LSDZ):

3. Aircraft Mounted Lasers - Description of Laser Surface Danger Zones (LSDZ):

4. Recommended operating procedures/range regulations:

5. Recommended laser eye protection:

FIGURE B-3. Laser range safety certification report – Continued.

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6. Controls for protection from reflected laser beams:

7. Recommended training:

8. Recommended pre-briefs for:
   a. Laser users:

   b. Laser Range Personnel

FIGURE B-3. Laser range safety certification report – Continued.

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Laser Range Certification

1. _____ (range/range training facility) and attendant laser hazard analysis have been designated to accommodate the following, under the supervision of a qualified laser range officer in charge and Laser Range Safety Officer, per the range SOP:

   a. The use of _____ laser(s).

   b. The use of lasers with the following NOHD:

   _____

   c. The use of lasers with the buffer angles:

   _____

   d. Appropriate _____ wavelength PPE:

2. A risk assessment of all "no" answers on certification questionnaires has been completed and steps taken to mitigate the risk. A summary of that risk assessment and risk mitigation steps is enclosed.

3. _____ (range/range training facility) meets the criteria established in service directives and is certified effective as of the date of this letter.

Signature: _____ (Date)_____

Copy to: Institutional Laser Range Authority
Service Range Safety Authority
Installation Laser Range Authority

FIGURE B-3. Laser range safety certification report – Continued.

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METHODOLOGY FOR RANGE LASER HAZARD EVALUATION

C.1 SCOPE

C.1.1 Scope. The goal of laser safety evaluations on ranges is to determine what is required to keep the laser beam plus and its associated buffer within the confines of the designated constraint boundary of the range. This appendix provides a methodology for performing range laser hazard evaluations consistent with Section 10.

C.1.2 Methodology. The information provided in this appendix may be used in addition to the service-specific range laser evaluation techniques. The equations are the means to determine minimum laser altitude which will satisfy the safety constraints for use of an airborne laser system on a particular range and at a specified distance from the target. Equations are provided to determine positions of ground based lasers that will satisfy the safety constraints on a given range.

a. The Range Safety Officer needs to ensure that all laser ranges are assessed according to the service specified schedule. A pre-survey questionnaire should be provided and an example of questions that will be asked is provided in this appendix. It is important for the Range Manager to ensure all stakeholders are available when the Institutional Laser Range Authority (ILRA) performs the survey. This includes Range Safety, the LSO, range users, range support staff, and the Command representative. With the survey completed, the RCA will prepare a Range Laser Safety Certification (RLSC) Report to be provided at a later date. The Command is not obligated to allow all scenarios covered in the report. Examples of the information the report covers are provided. The Range Manager needs to be familiar with the information and its presentation. Sample calculations are also provided to give some insight into how the report is prepared.

b. This appendix provides examples of what to expect from an RLSC Report and how to interpret the report. An RLSC report displays results of the range evaluation. A range evaluation requires performance of a site survey and analysis for every scenario where Class 3B and Class 4 lasers will be used. This appendix also provides information on various hazards the range operator needs to be aware of and how to best work with the RCA.

c. Sample equations in this document are informative and not intended to teach the reader how to perform an evaluation. Each service has professionals that perform evaluations to create these certification reports, and those professionals should be used when completing a RLSC. These sample calculations focus on air-to-ground and ground-to-ground geometry. The process of calculating other lasing scenarios is similar.

C.2 BACKGROUND

Lasers are used between five domains. These are subsurface (water), surface (water) ground, air and space. Each combination of lasing and target location has its own considerations. There are 25 categories listed in table C-1. It is important for the Range Manager to understand that while surface to surface, surface to air, air to air, and air to surface have been the four dominant scenarios based on laser-guided munitions, targeting lasers, and rangefinders, other lasing conditions are becoming common as space and subsurface uses for lasers are developed.

Ranges require certification before initial use and recertification on a schedule determined by each service, or when changes occur that could affect safe usage. The RLSC team will collect data pertaining to the intended location of the lasers, the intended location of the targets, and all applicable user and geographic data for laser energy containment purposes. Containment data may include topography as well as range or installation boundaries, backstops, and other items of interest. This data is typically modeled as points, lines, and polygons. While collecting or verifying this data on the range, the RLSC team will inspect the range for specular reflectors, areas of concern, and other considerations that may not be found on maps.

Analysis of the data mentioned is performed and the results are provided in an RLSC Report. As each range has unique characteristics, each range requires an individualized evaluation. The following sections provide a few examples of common scenarios, considerations, and mitigations performed or provided in the RLSC Report.

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TABLE C-I. Scenarios.

Target → / Laser ↓	Subsurface	Surface	Ground	Air	Space
Subsurface	Subsurface to Subsurface	Subsurface to Surface	Subsurface to Ground	Subsurface to Air	Subsurface to Space
Surface	Surface to Subsurface	Surface to Surface	Surface to Ground	Surface to Air	Surface to Space
Ground	Ground to Subsurface	Ground to Surface	Ground to Ground	Ground to Air	Ground to Space
Air	Air to Subsurface	Air to Surface	Air to Ground	Air to Air	Air to Space
Space	Space to Subsurface	Space to Surface	Space to Ground	Space to Air	Space to Space

C.2.1 Laser interaction with water surface. Laser interaction with the water surface can create safety hazards in two ways: reflections off the surface from above and transmission through the water-air boundary. Reflections off the water need to be considered when the water surface is used at a backstop. The worst scenario is still water, such as in a flooded field where surrounding grass keeps the surface nearly flat by providing a windbreak. This surface has the potential to act as a flat specular reflector, as does black ice. Water in a bay can be very still, but flat water is rare in the ocean. As the power of laser systems becomes greater, the danger of reflections off water with waves becomes greater. The RCA has resources to analyze the associated hazards. Lasers can also reflect off wetted surfaces such as ships or tank hulls from rain or wave action, and this will be accounted for by the RCA.

C.2.2 Underwater use of lasers. The use of lasers is expanding to include underwater applications. Lasers pointed into the abyss can be the safest application, but there are potential hazards to be considered. These include divers and reflective objects that could redirect the beam. Visible lasers and the tow bodies containing the laser can attract marine mammals such as dolphins. All environmental issues are addressed by the Environmental Impact Assessment and by the Environmental Review Board. The RLSC Report will not address environmental issues.

a. The stability of the underwater lasing platform also should be considered. Depth switches and inclinometer switches are often used on systems. Laser platforms such as unmanned autonomous vehicles (UAVs) and tow bodies should be tested for hydrodynamic stability prior to using the laser.

b. Lasing above the horizontal from underwater presents the risk the laser energy will exit the water surface into the air above. This is a high probability even when the angle of incidence is low with the water surface because waves can have very high slopes. The water surface will tend to increase the divergence of the laser. The RCA can work with subject matter experts (SMEs) to analyze this hazard.

C.2.3 Lasers directed above the horizon. Lasers directed above the horizon include direct skyward lasing from the air or ground. It also includes subsurface lasers directed towards the surface with or without the intent of reaching the atmosphere. Even low angle of incidence subsurface lasing can interact with waves that have significant slope allowing energy to exit the water into the air. Lasers can also interact with specular and diffuse reflectors, which redirect the energy skyward. Laser energy directed (or redirected) skyward can be a hazard to aviation and to satellites. The RCA can analyze the hazard to aviation. The United States Air Force Laser Clearinghouse should be consulted to determine the steps needed to avoid damage to satellites. Generally, the hazard of concern here is damage to satellite optical systems, so the laser system need not be very powerful. Some lasers will be determined not a hazard to satellites, while others will require coordination with the Laser Clearinghouse to ensure the laser is aimed at clear skies.

C.3 INTERACTION WITH THE RCA

C.3.1 Pre-survey questionnaire. The RCA will send a pre-survey questionnaire. The following represents the type of information they will need prior to the visit:

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a. Prior range laser certification reports.

b. Changes made to the lasing scenarios since the last certification.

c. Range regulations and SOPs.

d. Current maps, charts, images, and other important information of the region.

e. A brief description of the area to be surveyed.

f. Special arrangements required for access to the area. This can include permission or coordination and transportation requirements.

g. Availability of transportation to all of the areas to be surveyed, including target areas.

h. Scenarios involving the F-35 Joint Strike Fighter.

i. Scenarios involving High-Energy Lasers (HEL).

j. Scenarios involving above the horizon lasing.

k. Information on the surrounding landowners and the activities that take place in proximity to the range areas.

l. The magnetic declination of the areas to be surveyed.

m. The presence of restricted or SUA.

n. The points of interest in the area (buildings, towers).

o. Target types and composition.

p. List of all scenario types and areas that are to take place. This involves the combination of lasing from air, surface or subsurface-to-air, surface or subsurface.

q. Firing to and from point, line, or areas.

r. Laser platform, vessel type, and, if aircraft, fixed or rotary wing.

C.3.2 Range certification authority visit.

It is important for the Range Manager to ensure all stakeholders are available when the Range Certification Authority (RCA) performs the survey. This includes Range Safety, the LSO, range users, range support staff, and the Command representative.

There should be a briefing by the RCA. All concerns with the present range and considerations for future uses should be addressed at this time. The RCA will endeavor to establish the range to meet all operational needs addressed at the briefing. The Range Command should be ready to provide transportation and coordinate access to all laser firing areas and laser target areas. Targets will be inspected.

The RCA will provide guidance on how to read and apply the Range Certification Report. They will also provide guidance on those factors that increase or decrease safety, and when the Range Manager should contact the RCA for follow-up.

C.3.3 RLSC report results. An RLSC Report is provided to inform Range/Installation Commanders where the hazards exist, and provide guidance on how the lasers can be used safely (mitigations, limits, etc.). The report consists of models of the hazards of each scenario and special instructions to maintain a safe laser range. Key components of these reports include the flight profiles, the images or maps that show how the models are applied, and specific instructions. The RLSC Report images and format will vary by Service. Examples of images that represent the variations in style one can expect are included in this appendix.

C.3.4 Post-survey/report action items. The RLSC report may come with a list of action items that need to be addressed before one or more ranges at a facility are used. Issues that may need to be addressed include:

a. Targets (e.g., despecularization).

b. Training of range users and operators.

c. Communications on range.

d. Limitations of ranges.

e. Range clearance procedures.

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f. Lasers and platforms authorization list.

g. LEP maintenance procedures.

h. Review of SOPs.

i. Range user community review of RLSC Report.

j. Laser Safety Review board approval for all lasers used.

k. Training of range personnel and anticipated turnover.

l. Contacts.

While this is not a complete list, it does cover the types of action items most commonly seen. The RCA will also discuss the timeline of expectations for the final report and for action items. The RCA should use the briefing of the Range Manager and with the range users to provide education on issues of concern. Special-case laser systems, which present additional hazards, should be discussed. A number of these systems are in service or are under development. The RCA will also discuss the types of systems that may be seen on the range in the future.

C.3.5 Range changes and events that necessitate consultation with the range certification authority. There are a number of events and changes to the range that will trigger the need to obtain input from the RCA. Both the Range Manager and the range users should be aware of these events. Often the RCA can resolve an issue quickly. In some cases, a recertification of the range will be necessary to accommodate a change in needs or to address an event that exposed shortcomings in the range certification. These changes and events include the following:

a. Changes to firing positions.

b. Changes in target positions.

c. Changes in targets.

d. Changes in tree lines (clear cutting, fire, blight).

e. Changes to range boundaries.

f. Requests for changes to flight profiles.

g. Requests for changes to maximum or minimum altitudes.

h. Changes to the Range SOP.

i. Questions on interpretation of the SOP or the Range Certification Report.

j. Mishaps, injury, property damage, or other events that expose potential shortcomings in the safety plan.

k. Questions on LEP.

l. Changes to tactics or training procedures.

m. Requests for the use of high-energy lasers.

n. Requests for training with the F-35 not previously approved.

C.4 OVERVIEW OF THE CALCULATIONS PERFORMED BY THE RCA.

The following sections and graphics describe the basics of the calculations performed by the RCA. The outputs presented do not represent the form that all services use. Part of the job of each service RCA is to present to the Range Manager the format of the output that they will receive. The RCA should ensure that the Range Manager is familiar with how to interpret the graphics.

C.4.1 Laser spot, buffer, footprint, and angle of incidence (basic geometry). A uniform circular Gaussian beam will project a circular spot at a distance if the laser is orthogonal to the surface upon which it is being projected. A buffer is applied to the laser system to accommodate for the system's instability. On figure C-1, the footprint is the area where the laser spot and buffer intersect perpendicular to the surface. The angle of incidence can affect the shape of the beam and buffer. If the laser is not orthogonal to the surface, the spot and buffer projection will be ellipsoidal in shape. Figure C-2 displays the footprint of the same laser with an angle of incidence less than 90 degrees (not perpendicular to the surface).

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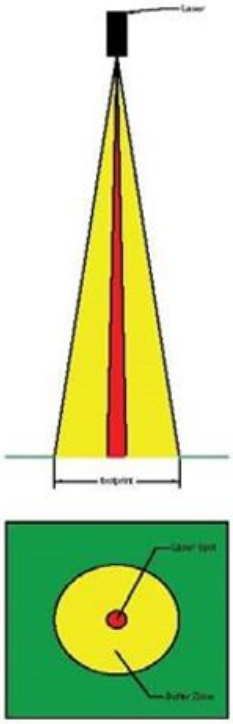


FIGURE C-1. Laser footprint – side view and top view.

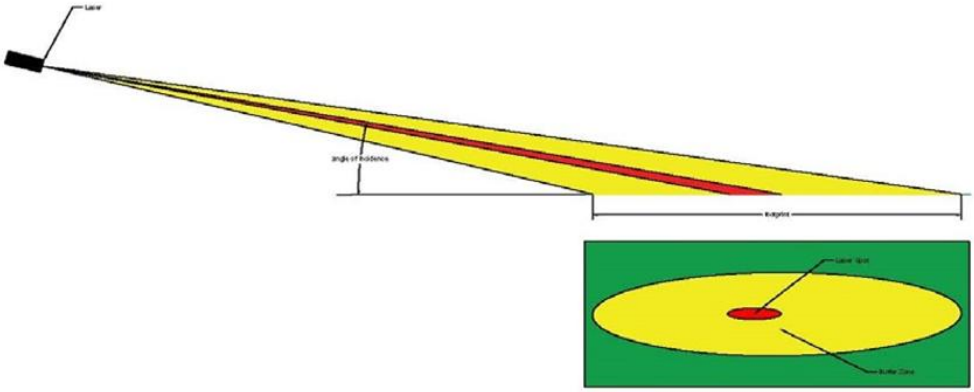


FIGURE C-2. Laser footprint – side view and top view with an angle of incidence less than 90 degrees.

Figure C-1 is the shape of the laser spot and the buffer zone if a laser is perpendicular to a flat surface. Figure C-2 shows the same laser at the same distance at a low angle of incidence. The shape depicted on Figure C-2 or derivations of it are more likely found on ranges from undulating terrain, varying elevations, and offset angles.

C.5 BASIC METHODS FOR LSDZ CREATION

This evaluation should be done for each laser heading and should account for any slope of the terrain.

C.5.1 Single laser heading. Provided that the laser target and surrounding area are clear of specular reflectors, the mathematical model used to evaluate range safety assures that the laser beam and its associated buffered footprint fall within the prescribed boundaries of the controlled or restricted ground space. C.5.2 through C.5.5 describe the equations used for this model. Figure C-1 shows an aircraft laser illuminating a small target area with the associated buffer zones fore and aft. Figure C-2 shows an airborne laser illuminating a large target area with

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near and far buffer zones assigned as if the laser were always aimed at the nearest and farthest targets. The plan view of the footprint is shown on figure C-3. In either case, the hazard from the actual laser and the buffer zone is considered in a range evaluation.

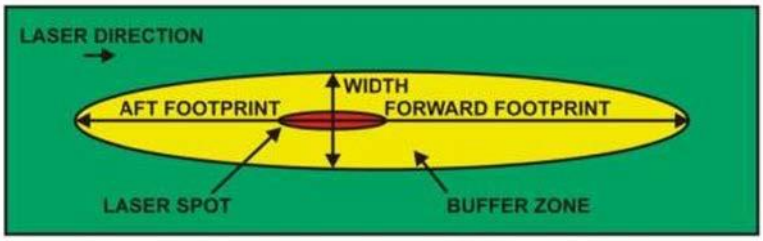


FIGURE C-3. Laser footprint – top view.

C.5.2 Multiple laser headings. If the laser energy is from several bearings (for example 70 to 110 degrees), the LSDZ will be a summation of all possible buffered footprints as shown on figure C-4. If the bearings are not specified from any direction, the LSDZ will be a circle with a radius equal to the longest forward or aft buffered footprint dimension for the possible altitudes or slant ranges (see figure C-5).

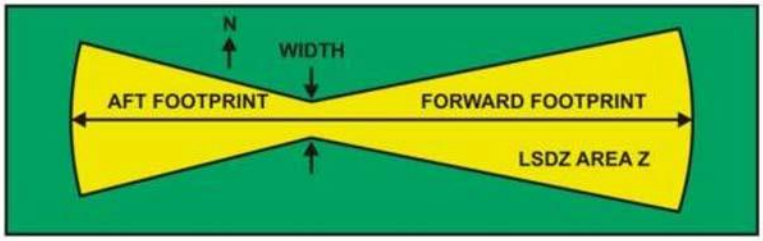


FIGURE C-4. Integrated footprint – bearing 70 to 110 degrees.

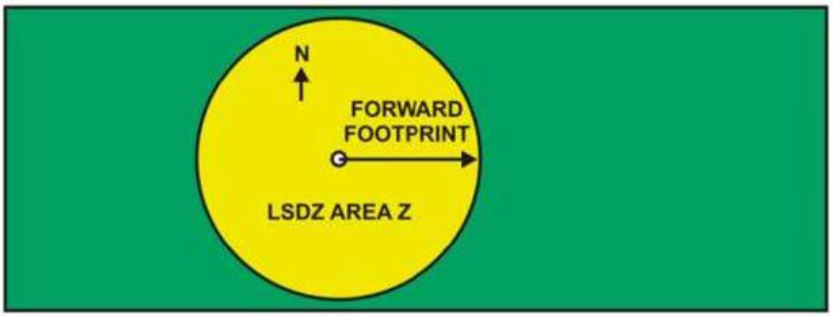


FIGURE C-5. LSDZ – attack from any direction.

C.5.3 Unlevel terrain. Although actual procedures vary on a case-by-case, the conditions described in C.5.4 through C.5.13 are presented as common.

C.5.4 Target on rising terrain or hills behind target (natural backstop). The condition of targets on rising terrain sometimes lengthens the near boundary and makes the far boundary less restrictive than the level ground condition. Hills behind the targets can act as natural backstops and reduce the size of the forward footprint as rising terrain did (see figure C-6, figure C-7, and figure C-8).

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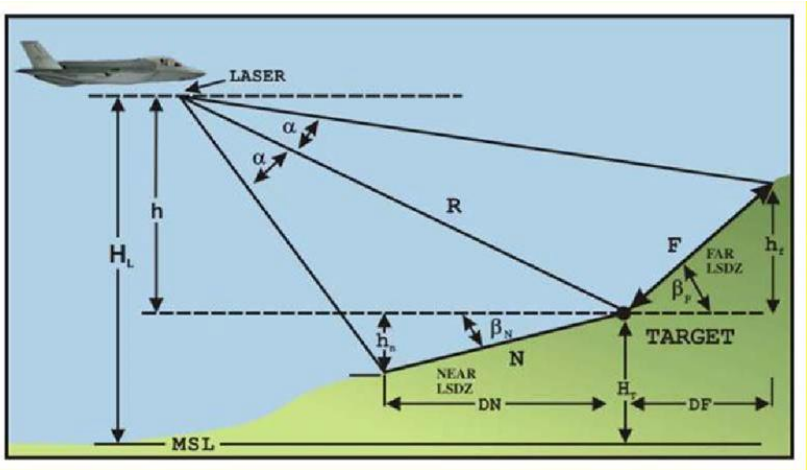


FIGURE C-6. LSDZ with rising terrain.

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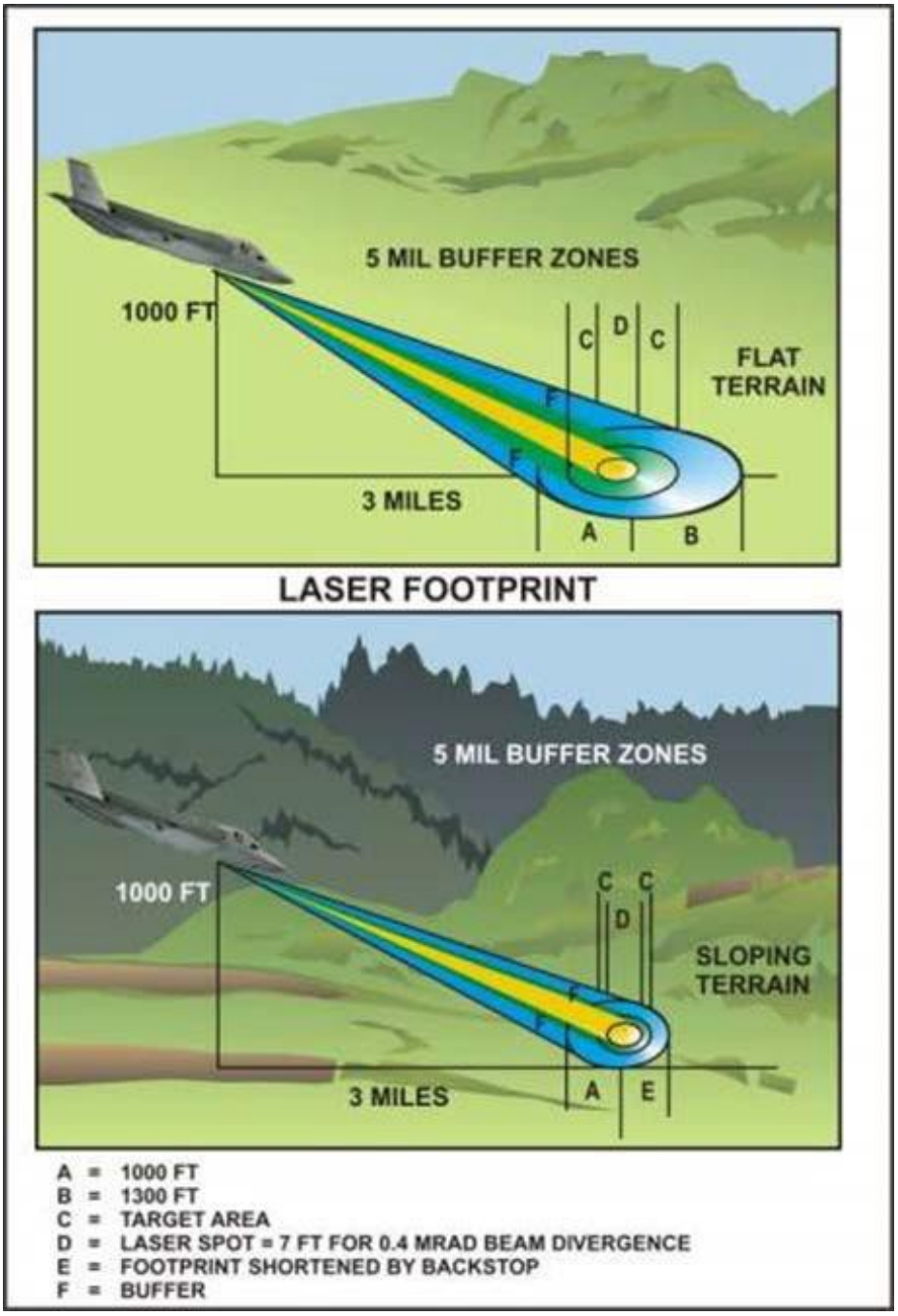


FIGURE C-7. Natural backstops to control laser beam.

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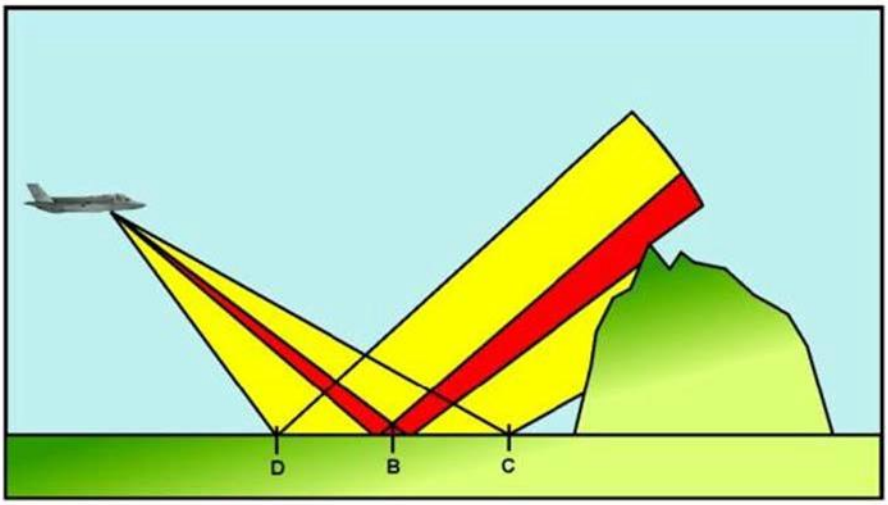


FIGURE C-8. Insufficient backstop to control laser beam.

C.5.5 Falling terrain in target area or hills in foreground. This condition will result in longer forward buffered footprints and more restrictive conditions. The height, Mean Sea Level, or Above Ground Level (AGL) of the laser in reference to the target should be determined for all distances between the laser and target. The downward sloping ground beyond the target can greatly extend the forward footprint as shown on figure C-9 and figure C-10. If flight profiles are not limited, the forward footprint could be as long as the NOHD.

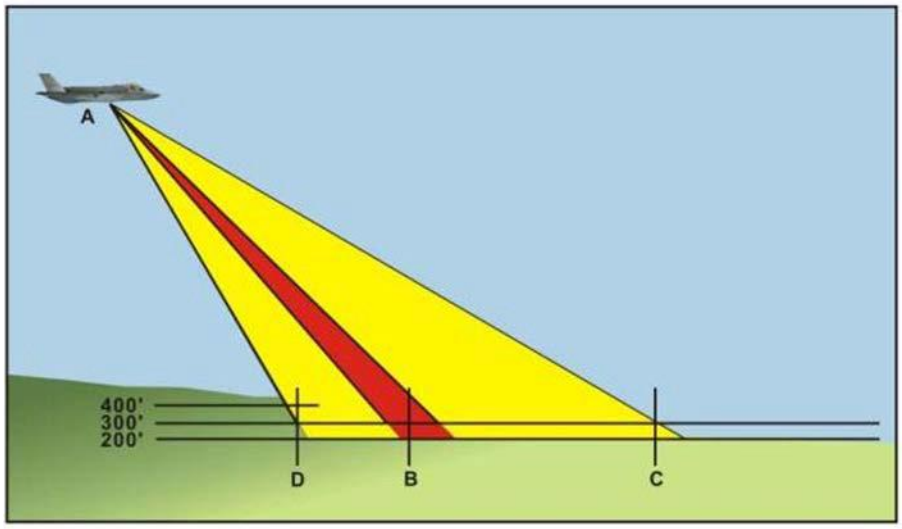


FIGURE C-9. LSDZ with terrain sloping down when range is less than NOHD.

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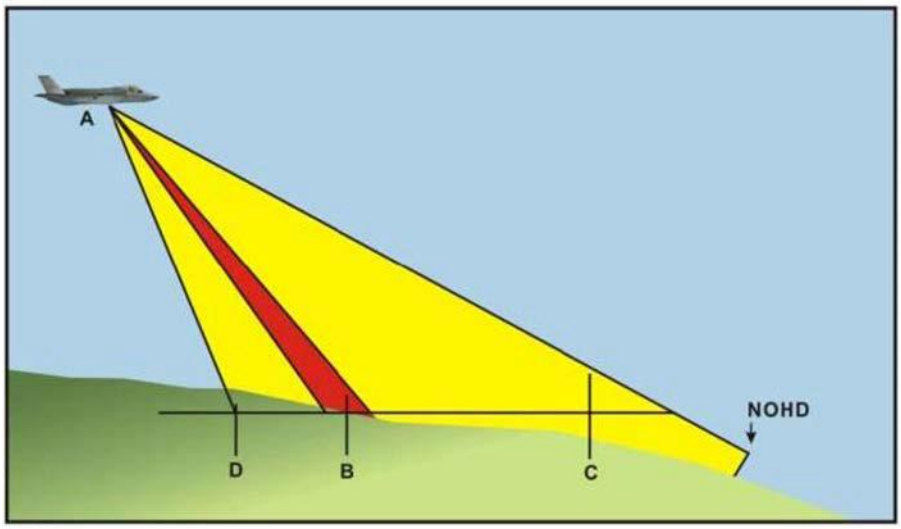


FIGURE C-10. LSDZ with terrain sloping down when range is greater than NOHD.

C.5.6 Vertical buffer far boundary. Addressing the far boundary constraint first, figure C-11 illustrates the geometry of the problem. First, determine the available buffer above and below the target out to the edge of the backstop where:

α = buffer angle plus beam divergence on either side of the laser line of sight (LOS). The beam divergence is extremely small compared to the buffer angle, so the beam divergence may be ignored.

δ = available vertical buffer angle between laser LOS to target and laser LOS to backstop.

h = altitude of laser.

a1 = altitude of far target.

bl = altitude of far boundary.

dl = horizontal distance on surface from laser to furthest target.

A = distance from target to far boundary of LSDZ (backstop).

The angle δ may be calculated from:

$\delta = \arctan((b1 - h)/(d1 + A)) + \arctan((h - a1)/(d1))$.

As long as the angle δ remains greater than angle α, the beam is safely contained vertically within the designated LSDZ.

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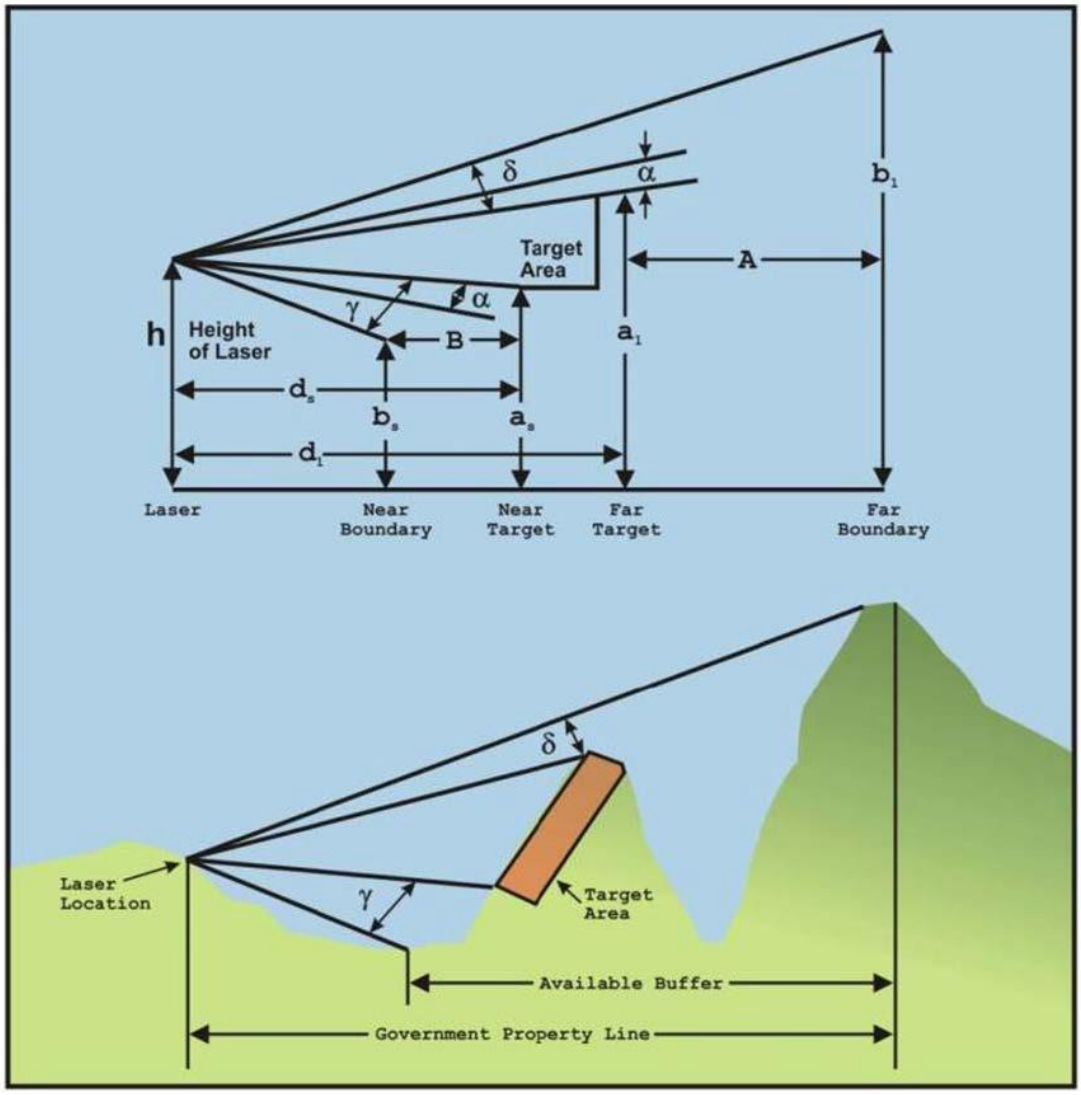


FIGURE C-11. LSDZ geometry and vertical buffer.

C.5.7 Vertical buffer near boundary. Similarly for the near boundary:

α = buffer angle plus beam divergence on either side of the laser LOS. The beam divergence is extremely small compared to the buffer angle so the beam divergence may be ignored.

γ = vertical angle from either side of the laser, LOS to the near edge of LSDZ (backstop) between the laser and the target.

h = altitude of laser.

as = altitude of nearest target.

bs = altitude of near boundary.

ds = horizontal distance on surface from laser to nearest target.

B = distance from target to near boundary of LSDZ (backstop).

The vertical angle γ may be calculated from:

$\gamma = \arctan((h-bs)/(ds-B)) + \arctan((as-h)/ds))$.

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As long as the angle γ remains greater than the angle α, the beam is safely contained vertically within the designated LSDZ.

C.5.8 Horizontal buffer. Available buffer to the left and the right of the target out to the backstop may be calculated as follows (see figure C-12):

AB = arctan((FPN-EBN)/(FPE-EBE)) - arctan((FPN-TN)/(FPE-TE)) where:

AB = available buffer angle in radians left and right of target out to the backstop.

FPN = laser firing position north coordinate in meters.

EBN = edge of backstop north coordinate in meters.

FPE = laser firing position east coordinate in meters.

EBE = edge of backstop east coordinate in meters.

TN = edge of target north coordinate in meters.

TE = edge of target east coordinate in meters.

If the angle AB is greater than angle α and is negative for the right edge of the backstop and positive for the left edge of the backstop, the beam is safely contained horizontally within the designated LSDZ.

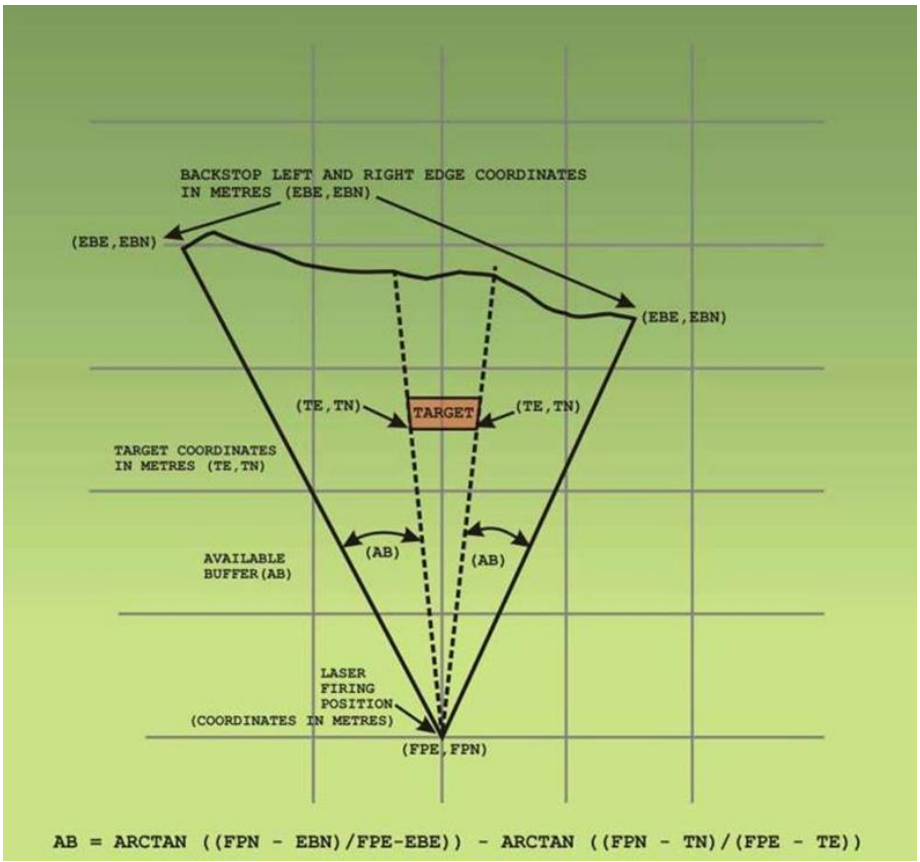


FIGURE C-12. Calculation of available buffer versus allowed buffer.

C.5.9 Minimum safe lasing altitude (MSLA). MSLA can be determined from the following equations:

The equations below yield the minimum safe lasing altitude AGL for aircraft lasing. For the derivation of MSLA for the far boundary, the variable H below represents the altitude relative to the top of the target or the far edge of the target area. For the derivation of MSLA for the near boundary, H represents the altitude relative to the ground at the target or the near edge of the target area. In both cases, R represents horizontal (ground) distance from firing (lasing) point to target, not slant range.

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The equation for the far boundary does not consider possible laser backstops between the target and the far boundary. However, the equation may be applied for the backstop locations to possibly reduce the MSLA. Figure C-13 depicts the geometry for the determination of the far boundary and near boundary MSLA for level terrain, figure C-14 depicts the geometry for the determination of the far boundary and near boundary MSLA for rising terrain, and figure C-15 depicts the geometry for the determination of the far boundary and near boundary MSLA for falling terrain.

The MSLA equation for the far boundary is:

$$MSLA_{AGL} = E_T - E_P + T + \frac{P\tan(b) + D_F - \sqrt{(P\tan(b) + D_F)^2 - 4\tan(b)(R^2\tan(b) + RD_F\tan(b) - RP)}}{2\tan(b)}$$

A simpler form is to use the distance from the firing point to far boundary, QF:

$$MSLA_{AGL} = E_T - E_P + T + \frac{P\tan(b) + D_F - \sqrt{(P\tan(b) + D_F)^2 - 4\,\mathrm{R}\tan(b)(\mathrm{Q_F}\tan(b) - P)}}{2\tan(b)}$$

For the near boundary, T = 0. The MSLA equation for the near boundary is:

$$MSLA_{AGL} = E_T - E_P + \frac{P\tan(b) + D_N - \sqrt{(P\tan(b) + D_N)^2 - 4\tan(b)(R^2\tan(b) - RD_N\tan(b) + RP)}}{2\tan(b)}$$

A simpler form using the distance from the firing point to the near boundary, QN:

$$MSLA_{AGL} = E_T - E_P + \frac{P\tan(b) + D_N - \sqrt{(P\tan(b) + D_N)^2 - 4R\tan(b)(\mathrm{Q_N}\tan(b) + P)}}{2\tan(b)}$$

Diagram notations:

FP = firing point location

EP = elevation of FP

TGT = target location

ET = elevation of target

R = distance from FP to TGT

b = buffer angle

P = elevation of FB or NB minus elevation of TGT

(note P can be positive or negative)

for near, P = EN – ET; for far, P = EF – ET – T

FB = far boundary location

EF = elevation of far boundary location

DF = distance from TGT to FB

QF = distance from FP to FB = R + DF

NB = near boundary location

EN = elevation of near boundary location

DN = distance from TGT to NB

QN = distance from FP to NB = R – DN

T = height of target

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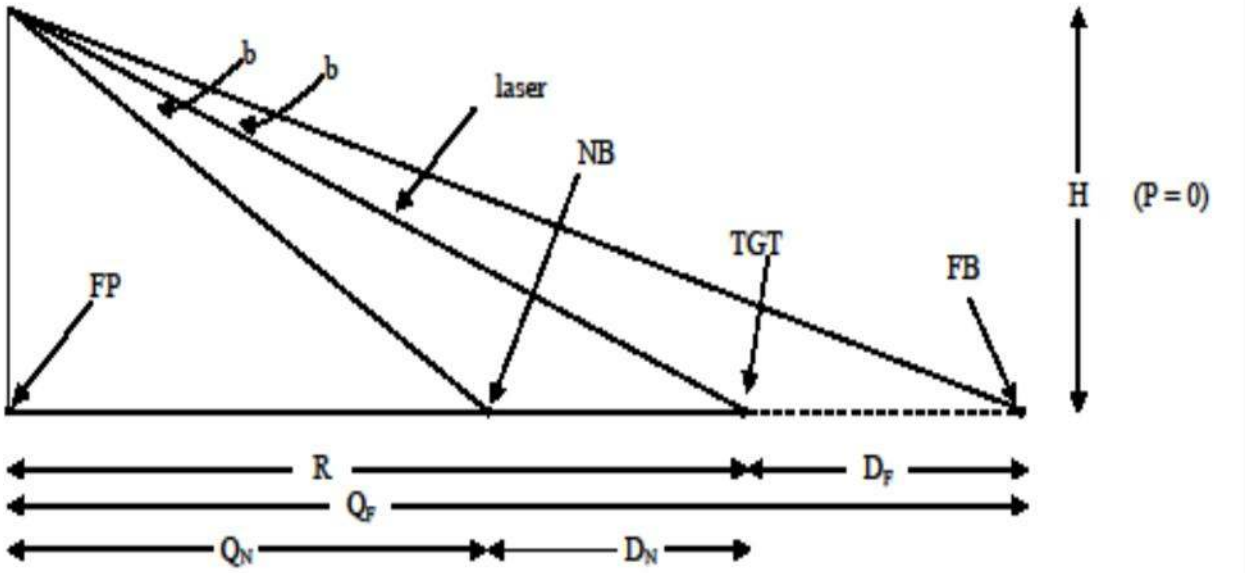


FIGURE C-13. Case 1: FB or NB level with TGT.

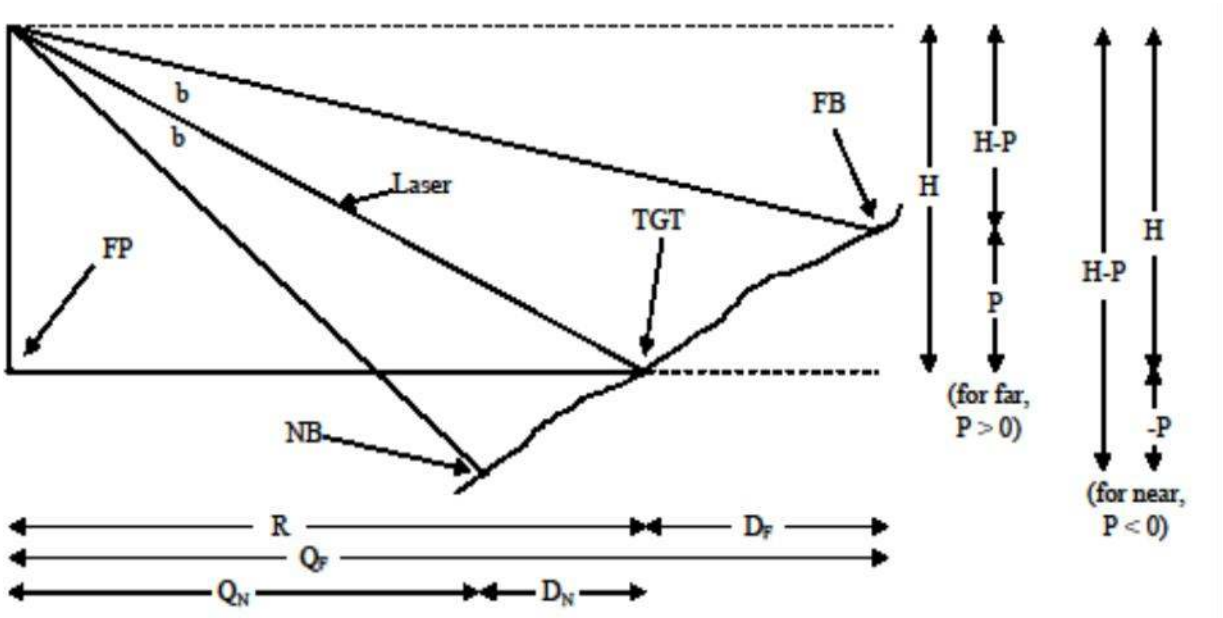


FIGURE C-14. Case 2: rising terrain.

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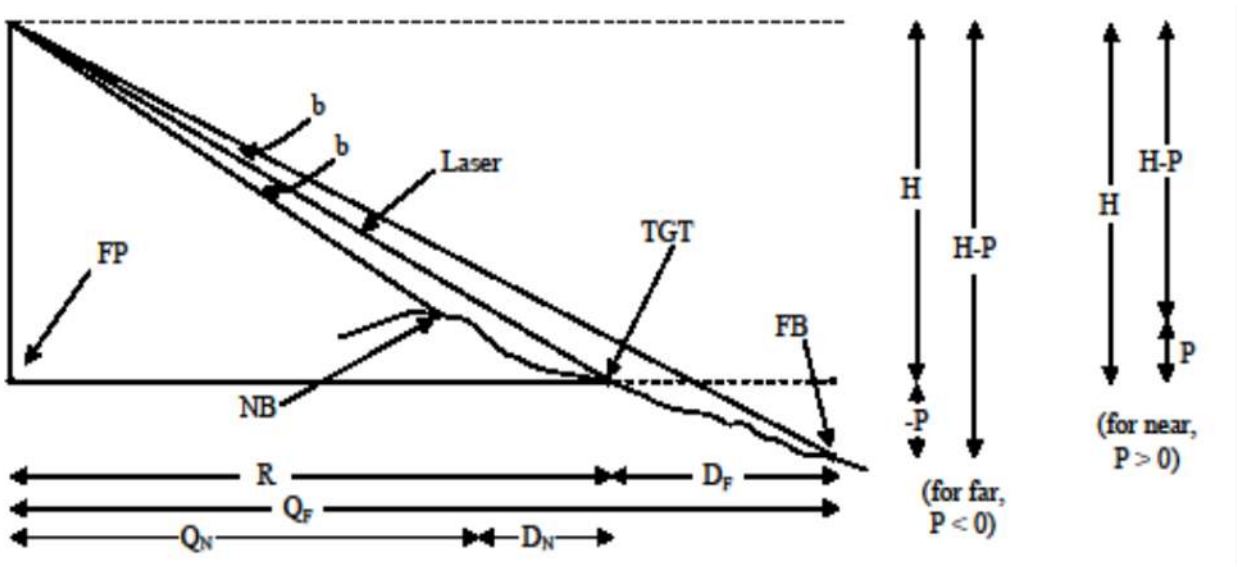


FIGURE C-15. Case 3: falling terrain.

The point identified as FP is the horizontal location; the vertical location of the firing point is not shown as the elevation of the FP may be higher or lower than the TGT. The point identified as the target represents the near edge of the target area, far edge of the target area, or the top of the target for a single-point three-dimensional target.

C.5.10 Airborne laser with target on level ground. For airborne laser buffer geometry at ground level (see figure C-16).

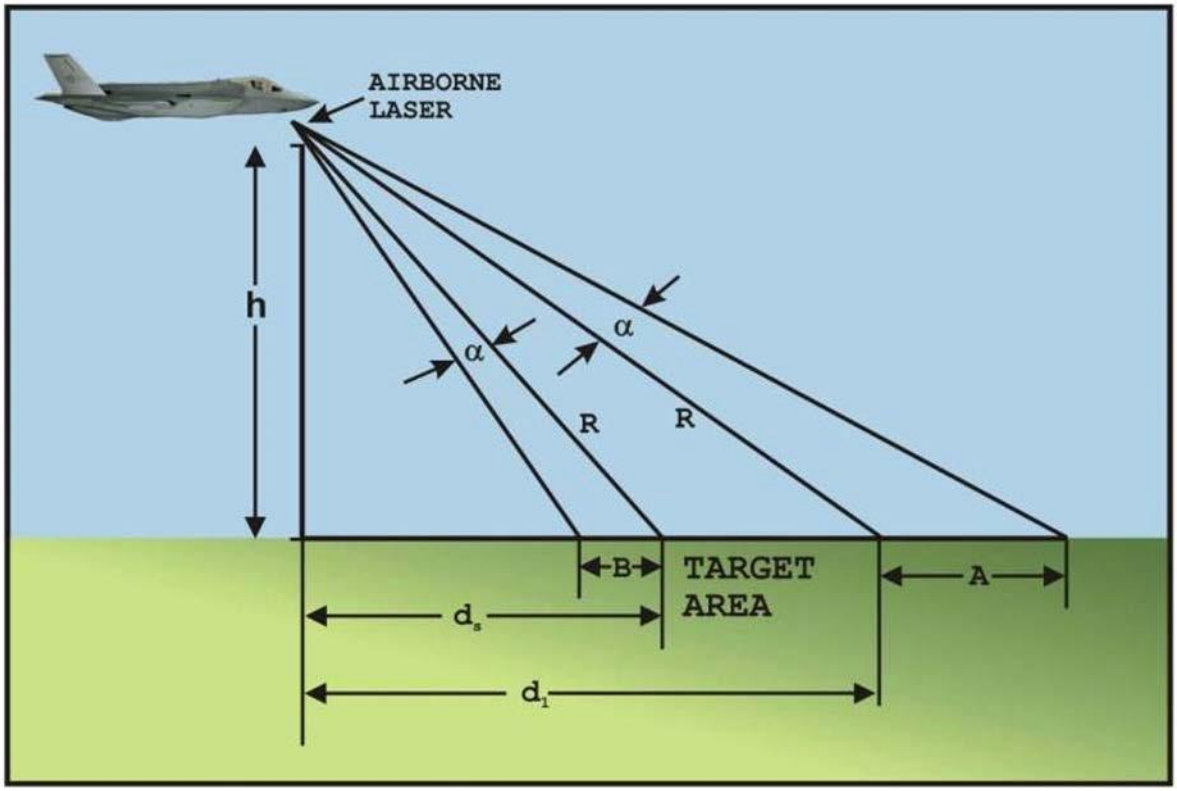


FIGURE C-16. Airborne laser buffer geometry – level ground.

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C.5.11 Aircraft minimum altitude. The minimum laser altitude (h) relative to the target to keep the buffered laser footprint within the far boundary when at slant range (R) from target is:

$h = R\sin(\arcsin((R/A)\sin(\alpha)) + \alpha)$.

The minimum altitude relative to target to keep the buffered laser footprint within the near boundary when at slant range R from target is:

$h = R\sin(\arcsin((R/B)\sin(\alpha)) - \alpha)$ where:

R = slant range from laser to target.

α = buffer angle plus beam divergence either side of laser LOS. The beam divergence is extremely small compared to the buffer angle and hence, the beam divergence may be ignored.

A = distance from target to far boundary of LSDZ.

B = distance from target to near boundary of LSDZ.

h = altitude of laser relative to target surface.

HL = altitude of laser above Mean Sea Level.

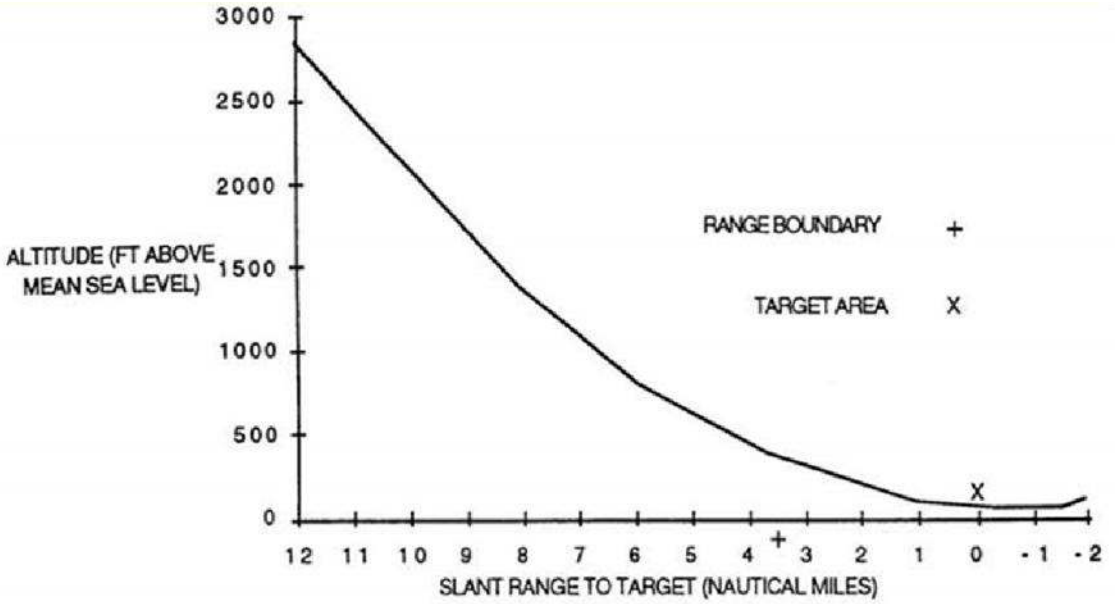
HT = height of target above Mean Sea Level.

Choose whichever h is the higher number and assign it as the safe altitude for lasing at range R. If altitude is altitude above mean sea level, then the required laser altitude is:

$HL = h + HT$.

Repeat this calculation for every nautical mile (or fraction of a mile depending on the risk) starting at about 12 nautical miles up to and beyond the target then plot the results. Remember, as you pass over the target, that the far and near boundary definitions reverse. A typical flight profile is plotted on figure C-17.

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SLANT RANGE TO TARGET nmi	MINIMUM LASING ALTITUDE ft MSL	SLANT RANGE FROM TARGET nmi	MINIMUM LASING ALTITUDE ft MSL
12	2850	0	10
11	2500	- 1	50
10	2100	- 2	112
9	1750		
8	1400		
7	1100		
6	850		
5	650		
4	450		
3	300		
2	150		
1	100		

FIGURE C-17. Example laser aircraft flight profile.

C.5.12 Left and right hand LSDZ. The width of the right hand and left hand LSDZ are calculated as:

$s = R \times \sin(\alpha)$.

s = left hand LSDZ width or right hand LSDZ width.

R = slant range from laser to target.

α = assigned buffer angle plus beam divergence on either side of the laser LOS. The beam divergence is small compared to the buffer angle and may be ignored.

Generally, the “small angle approximation” can be utilized for this equation since the buffer angle is usually small. This simplifies the equation to:

$s = R \times \alpha$.

Note: At 15 milliradian, this introduces an error of 0.004 percent. At larger buffer angles the error grows rapidly. If calculations require accuracy greater than this introduced error, do not use this small angle approximation.

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C.5.13 Airborne laser with target on sloping ground. Altitudes to keep buffered laser footprint within near or far boundary LSDZ can be calculated as:

Buffered Footprint (see figure C-18).

HT = altitude of target above mean sea level.

h = altitude of laser above target.

HL = altitude of laser above mean sea level = h + HT.

hn = height of near boundary above or below target.

hf = height of far boundary above or below target.

DN = horizontal distance from target to near boundary.

DF = horizontal distance from target to far boundary.

N = slant range distance from near edge of near target to edge of near boundary = square root of the sum of the squares of hn and DN.

F = slant range distance from far edge of far target to edge of far boundary = square root of the sum of the squares of hf and DF.

βF = declination or elevation angle from horizontal between edge of far target and edge of far

boundary = arctan(hf/DF) (positive number for far boundary higher than the target and negative number for far boundary lower than target).

βN = declination or elevation angle from horizontal between edge of near target and edge of near boundary = arctan(hn/DN) (positive number for near boundary lower than target and negative number for near boundary higher than target).

R = slant range from laser to target.

α = assigned buffer angle plus beam divergence.

For Far Target:

$h = R\sin(\arcsin((R/F)\sin(\alpha)) - \beta F + \alpha)$ and

$HL = R\sin(\arcsin((R/F)\sin(\alpha)) - \beta F + \alpha) + HT$.

For Near Target:

$h = R\sin(\arcsin((R/N)\sin(\alpha)) - \beta N - \alpha)$ and

$HL = R\sin(\arcsin((R/N)\sin(\alpha)) - \beta N - \alpha) + HT$.

Choose whichever h is the higher number and assign it as the safe altitude for lasing at range R. Repeat this calculation for every nautical mile (or fraction of a mile depending on the risk) starting at about 12 nautical miles up to and beyond the target then plot the results. Remember, as you pass over the target, that the far and near boundary definitions reverse. A typical flight profile is plotted on figure C-17.

Left and Right Hand LSDZ. If the buffer angle is small, the width of the right hand and left hand LSDZ can be calculated as:

$s = R \times \alpha$.

s = left hand LSDZ width or right hand LSDZ width.

R = slant range from laser to target.

α = assigned buffer angle plus beam divergence on either side of the laser LOS. The beam divergence is small compared to the buffer angle and may be ignored.

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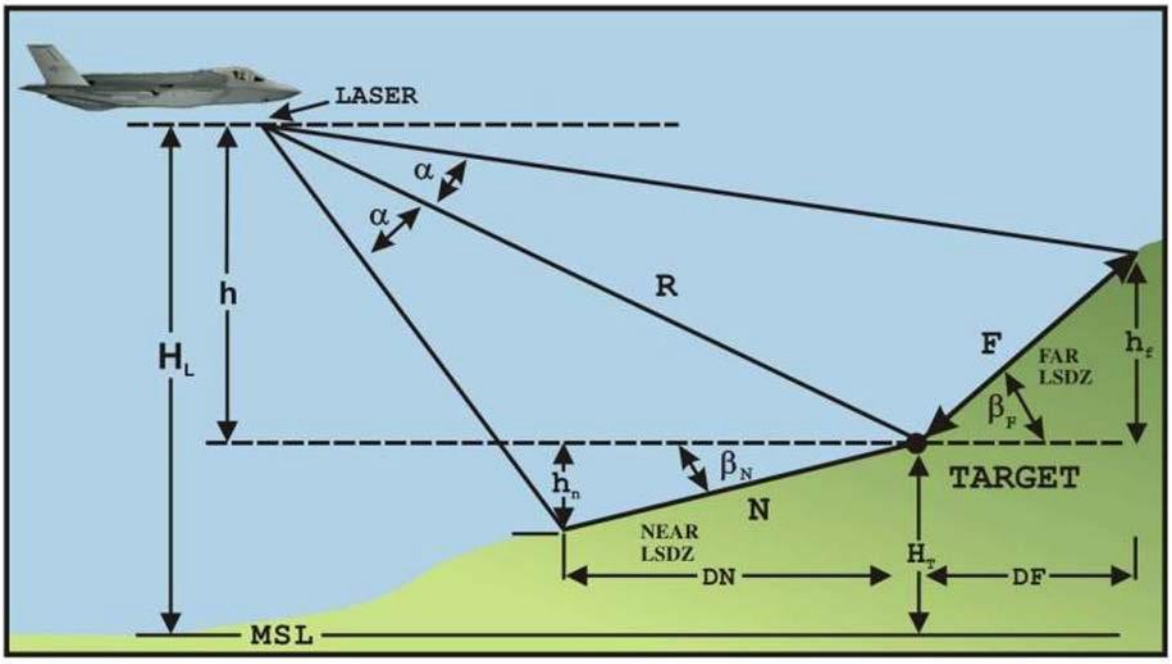


FIGURE C-18. Laser target on sloping terrain.

C.6 MITIGATIONS

There are two ways to mitigate hazardous energy coming from a laser system. The use of a backstop is desired. This is a physical barrier that will prevent energy from leaving the area of control by absorbing the energy. The other method requires a case-by-case comparison of the nominal ocular hazard distances (NOHDs) of each system to ensure the laser is safe for all anticipated viewing conditions beyond the area of control.

To mitigate specular reflectors either remove the reflector, cover it with non-reflective material, paint it with flat black, or disturb the reflective surface. Disturbing the surface can be accomplished by sanding or scuffing up objects or low flights with rotary wing aircraft over water to cause turbulence in the water's surface.

C.7 RLSC REPORTS

C.7.1 Typical considerations. The following sections contain examples of considerations, mitigations, content, and equations that are used to generate an RLSC Report:

a. Elevation – Typically considered in aircraft and sometimes in range design, this is the elevation from where the user would like to use the laser.

b. Direction or heading – Will there be limits established for the direction of laser use?

c. Standoff – How far away from the target is the laser turned on and off?

d. Platform stability – These are normally described as:

(1) 5 mrad for gyro-stabilized systems.

(2) 10 mrad for stabilized systems (monopod, bipod, tripod, sandbag, etc.).

(3) 15 mrad handheld or hand carried systems with alignment devices.

e. Target location – Target location considers proximity to the range or installation boundary, if the target can be grouped with nearby targets, distance to the target, terrain, line of sight, and if energy can be contained.

f. Target elevation – The elevation of where the target is and the height of the target.

g. Target type or condition – Some target materials are either reflective or can become reflective.

h. Terrain mitigated – Can the energy be physically contained on the range?

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i. Vegetation – Is the vegetation deciduous (loses leaves) or evergreens (do not lose leaves)?

j. NOHD – The calculated hazard distance of a particular system.

k. Maximum allowable NOHD – The maximum allowable distance that a system can be used on a range. If provided this value, it will need to be compared to the NOHD of every system being used on a range to determine if the NOHD of the system exceeds the maximum allowable NOHD, in which case the systems NOHD cannot be contained on the range and should not be used.

l. Control zone – Range/installation boundary and applicable restricted airspace or warning areas.

m. Buddy lasing versus self-lasing – Is a unit or person lasing for another range user, or is the person lasing also going to deliver the laser guided munition?

n. Maneuverability – Will the laser or target move during the laser on cycle?

o. Specular reflector – A mirror-like reflection (mirrors, polished alloy, still standing water, flat glass, etc.).

p. Special case considerations – These systems are normally indicated by the service lead responsible for verifying the parameters of a laser system. The systems have anomalies such as stray laser energy, a system with an output that will be difficult to contain under normal circumstances, and systems mounted to platforms that may exceed the platform stability norms.

q. Using the following chart and graph, a pilot can quickly determine the required elevation above the target and various standoff distances (see figure C-19 and table C-II).

AIR-SURFACE – W-151 LTA TO W-151 FIXED-WING TA SAFE LASING FLIGHT PROFILE			
Lateral Limits	000°-360° (unrestricted)	**Max Buffer Angle**	5 mrad(gyro-stabilized)
Air Safe Lasing LHZ graphic page #			

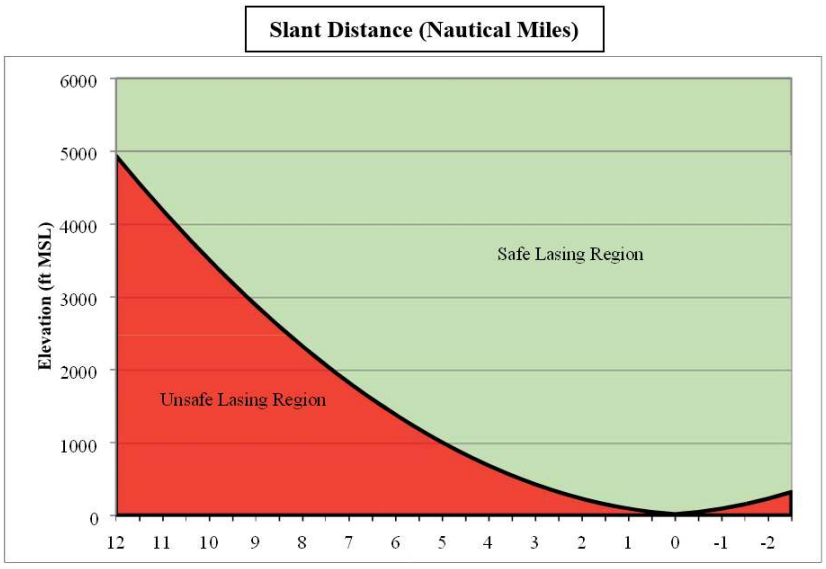


FIGURE C-19. Example of safe lasing minimum altitude graph.

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TABLE C-II. Example of safe lasing minimum altitude table.

Slant Distance (nmi)		Altitude (feet MSL)	Slant Distance (nmi)		Altitude (feet MSL)
12.0	-	4933	4.5	-	834
11.5	-	4554	4.0	-	683
11.0	-	4189	3.5	-	546
10.5	-	3840	3.0	-	425
10.0	-	3506	2.5	-	318
9.5	-	3187	2.0	-	227
9.0	-	2884	1.5	-	151
8.5	-	2595	1.0	-	91
8.0	-	2322	0.5	-	45
7.5	-	2064	0.0	-	15
7.0	-	1821	-0.5	-	45
6.5	-	1593	-1.0	-	91
6.0	-	1381	-1.5	-	151
5.5	-	1183	-2.0	-	227
5.0	-	1001	-2.5	-	318

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r. The following (figure C-20) is a graphic of a scenario where a pilot can engage targets anywhere within the green target area, and, if the previous flight profile is followed (see figure C-19 and table C-II), the hazardous energy is contained in the red area depicted on figure C-20.

1	**AIR-SURFACE–W-151 LTA TO W-151 FIXED-WING TA**			
	Lateral Limits	000° to 360° Magnetic (unrestricted)		
	Maximum Buffer	5 mrad (gyro-stabilized)	**NOHD**	N/A
	Air Safe Lasing Flight Profile page #			

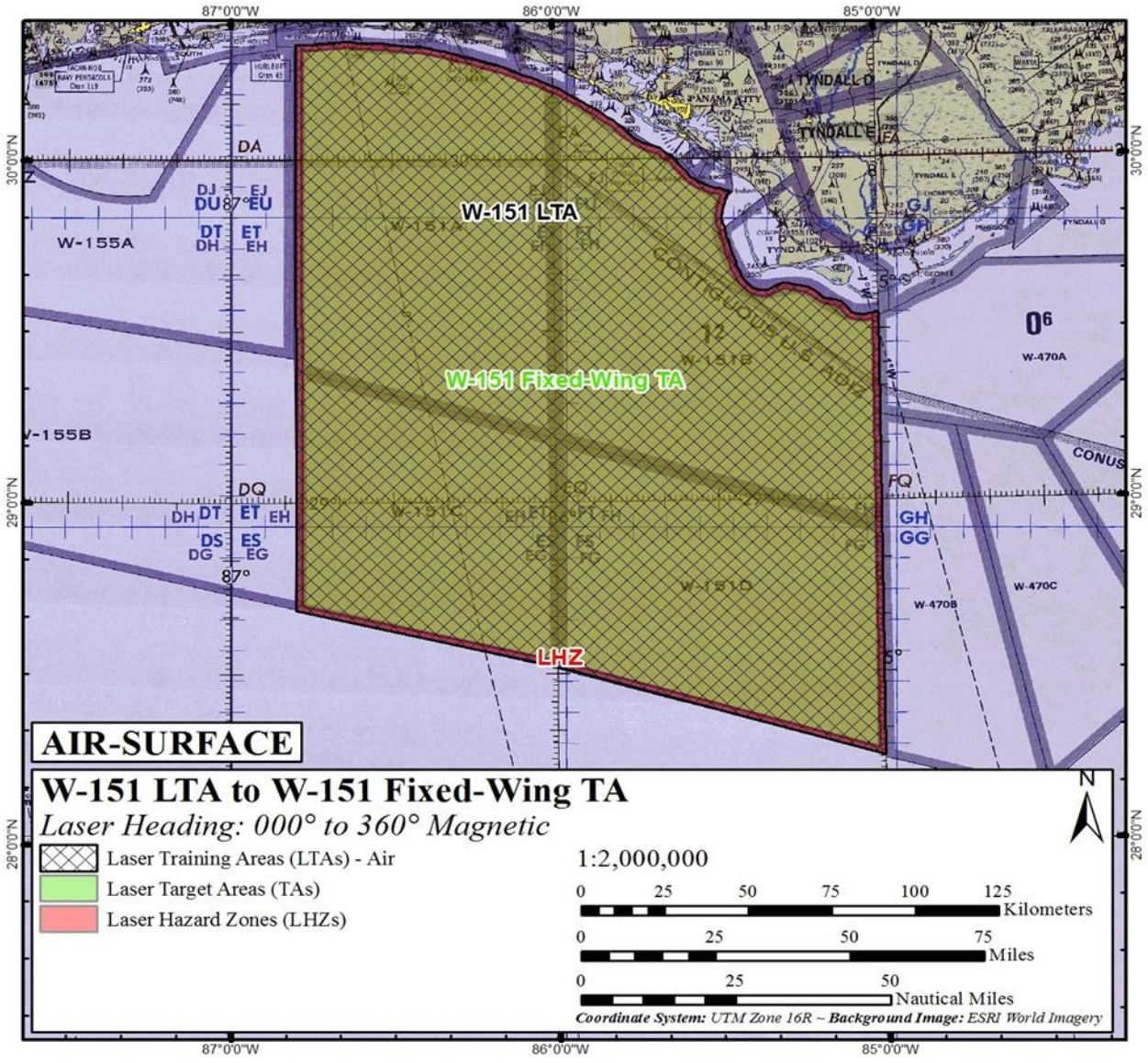


FIGURE C-20. Example of large area laser range.

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s. Every surface scenario is different. The following is a surface scenario example that can contain most surface use lasers laser systems fired from point Echo at surface targets between an azimuth 269°–283° magnetic. Lasers fired from point Echo at ground targets between an azimuth of 262°–269° magnetic should not exceed an NOHD of 4800 m (see figure C-21).

74	SURFACE-SURFACE–OP 1 ECHO FP TO ADVERSARY AIRFIELD TA			
	Lateral Limits	NOHD: 262° clockwise to 269°/Mitigated: 269° clockwise to 283°		
	Maximum Buffer	15 mrad (un-stabilized)	**NOHD**	4800 m*

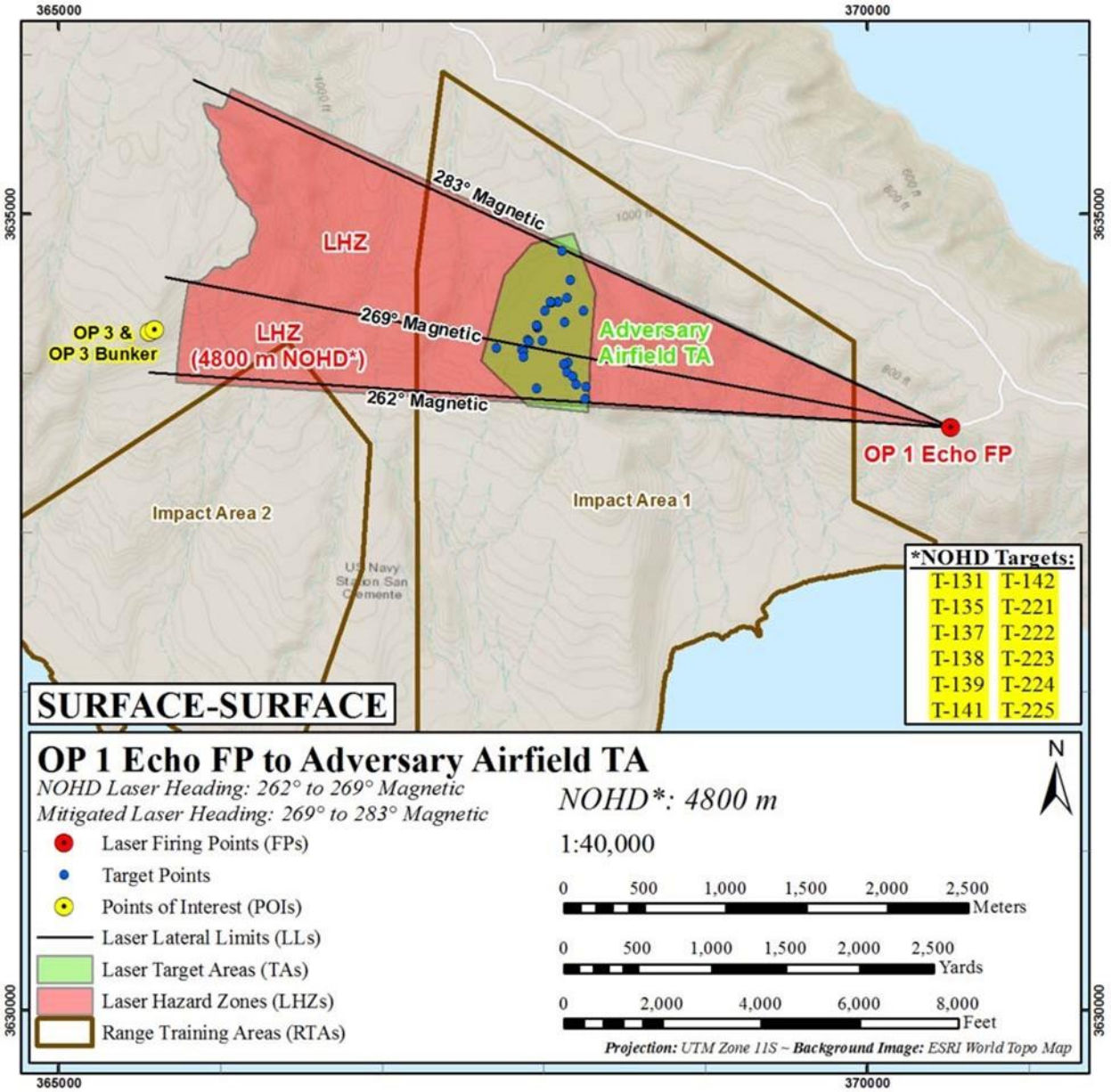


FIGURE C-21. Example of range with backstop limited and NOHD limited areas.

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t. Figure C-22 is similar in function to figure C-19 and table C-II. It was generated using different software tools.

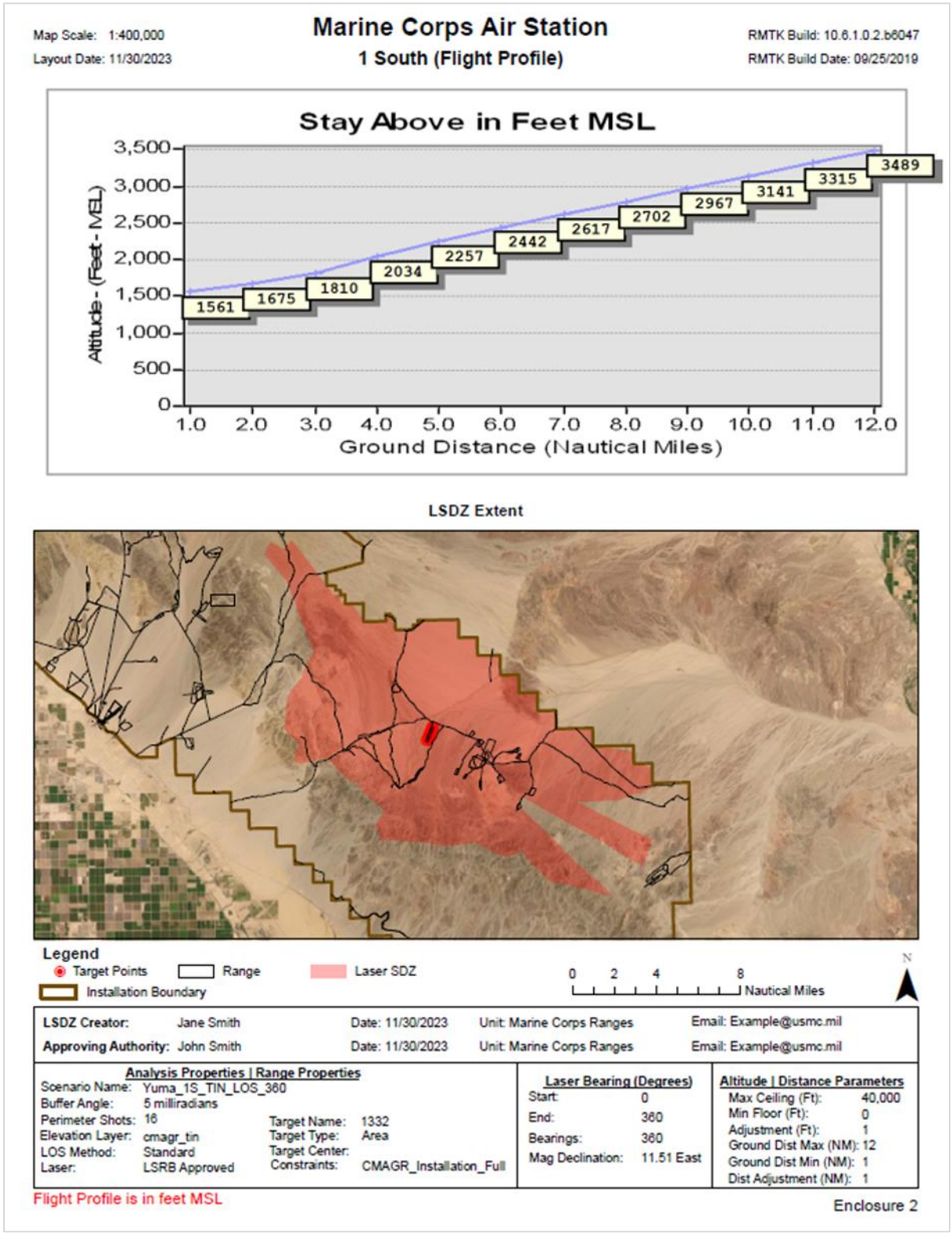


FIGURE C-22. Example of safe lasing area and flight profile for over land range.

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u. Figure C-23 upper graphic represents minimum lasing altitude with color codes. The lower graphic shows the Surface Danger Zone for lasing operations on range.

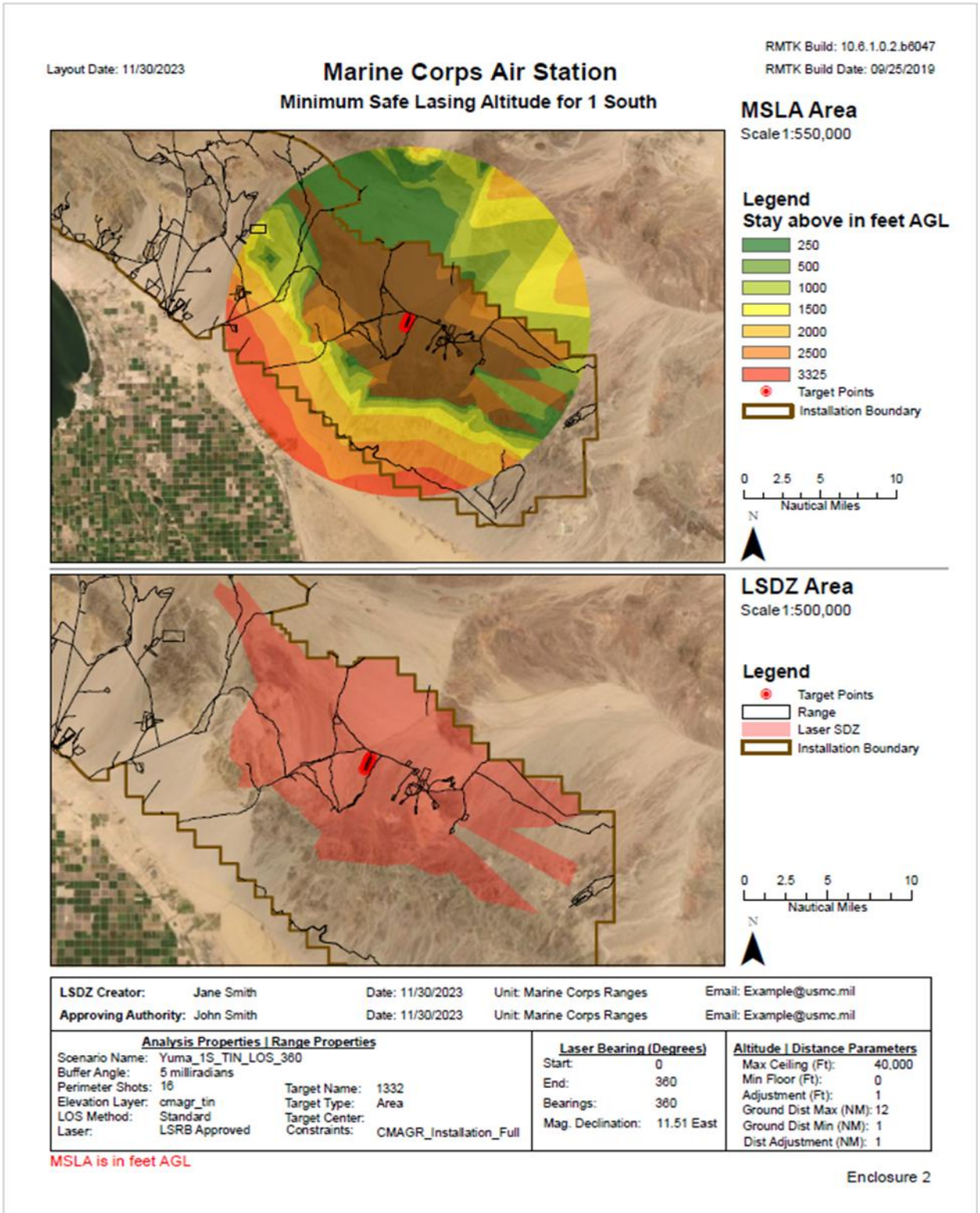


FIGURE C-23. Example of safe lasing altitude and surface danger zone graphics.

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CONCLUDING MATERIAL

Custodians:
Army – SAF
Navy – MC
Air Force – 10

Review activities:
Navy – EC
OT – SE

Preparing activity:
Navy – MC
(Project SAFT-2025-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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